

To my family.

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Chapter 1

Introduction

1.1 Topic and motivation

Generative diffusion models have become the dominant paradigm for synthesising high-dimensional data: they power state-of-the-art systems for images, audio, video, and molecular structures (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Karras et al., 2022). The same construction underlies the image and video generators deployed at scale by the large technology companies, from latent diffusion (Rombach et al., 2022) onwards, and reaches beyond media: structure prediction in AlphaFold 3 places atomic coordinates with a diffusion module (Abramson et al., 2024). Their central object is the *score function*, the gradient of the log-density of progressively noised data,

$$S(x, t) = \nabla_x \log p_t(x), \tag{1.1}$$

which is the one object a model must learn in order to reverse a noising process: given the forward process, its schedule and its terminal law, the score determines the reverse drift (Anderson, 1982). In practice the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011); the network is a black box, and the structure of the function it has to learn remains poorly understood.

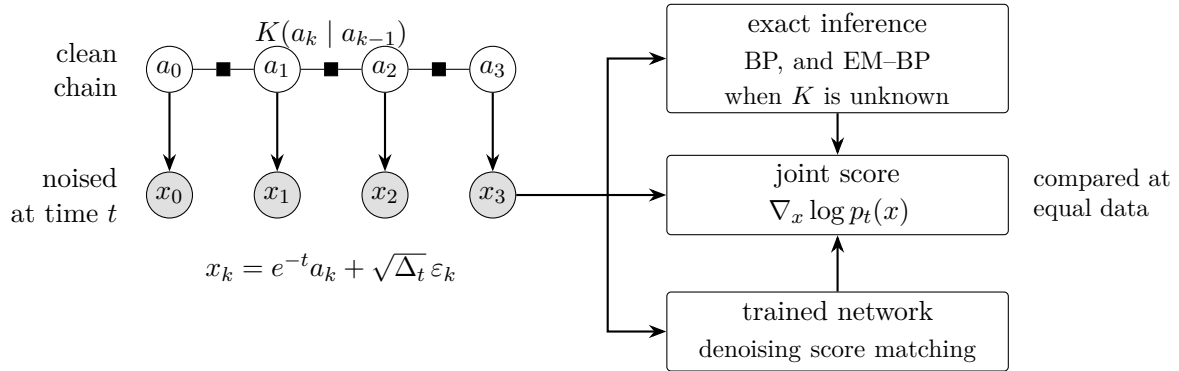


Figure 1.1: The object of the thesis and the two ways it is obtained. A Markov chain generates a clean trajectory; the Ornstein–Uhlenbeck channel corrupts every coordinate independently, so each observation attaches to the latent chain on its own and the posterior stays a chain. That is what makes the joint score exactly computable by message passing — with the transition kernel known (Chapters 5 and 6), or estimated from noised sequences alone when it is not (Chapter 7). The same score is what a denoising-score-matching network is trained to approximate without being given the chain structure. Chapter 8 measures what supplying that structure is worth, at equal training data, and where it stops paying.

A complementary research programme, initiated in the statistical-physics community, replaces the black box with *exactly solvable models*: data distributions simple enough that the score can be computed in closed form, yet rich enough to exhibit the phenomena observed in real systems, such as speciation, memorisation, and dynamical phase transitions along the generation trajectory (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024; Achilli et al., 2026). Exact scores act as a microscope: every regime, transition time, and failure mode can be traced back to explicit formulas.

This thesis works inside that programme, in a direction that is natural for data with *temporal or sequential structure*: the data are not a cloud of independent points but a *trajectory* $a_0, a_1, \dots, a_{L-1}$ of a Markov process, the prototype of a video, a time series, or any dynamic object. Each frame of the trajectory is corrupted independently by an Ornstein–Uhlenbeck (OU) channel, and the object of study is the *joint* score of the whole noisy sequence. Two classical bodies of theory then meet the modern one. First, the joint score of a Gaussian Markov chain is governed by the algebra of structured matrices: Toeplitz covariances, tridiagonal precisions, and their common eigenbasis. Second, by an identity that holds for any data distribution, computing a score is solving a Bayesian

inference problem, and for chain-structured data that problem lives on a tree-shaped factor graph, the domain of *belief propagation* (BP) and of the Kalman filter, together with the Gaussian message closures of statistical physics (TAP, AMP) (Pearl, 1988; Mézard and Montanari, 2009; Kalman, 1960; Donoho et al., 2009). The thesis develops both routes in full, proves where they coincide, and measures what is lost when they are forced apart.

1.2 Positioning and gap

The literature this work builds on can be organised in three strands; Chapters 2 and 3 review them; the extended treatment is in the companion compendium.

Statistical physics of learning machines. Machine learning and statistical mechanics share a genealogy that runs from spin glasses through the Hopfield network and the Boltzmann machine to today’s generative models, a lineage recognised by the 2024 Nobel Prize in Physics (Hopfield, 1982; Ackley et al., 1985; The Royal Swedish Academy of Sciences, 2024). Its modern branch produced closed-form analyses of diffusion generation for Gaussian mixtures, ferromagnets, and high-dimensional asymptotics: the regimes of the backward dynamics and their transitions (Biroli and Mézard, 2023; Biroli et al., 2024), the general theory of speciation (Achilli et al., 2026; Sclocchi et al., 2025), the dynamics of memorisation and generalisation in training (Bonnaire et al., 2025; Garnier-Brun et al., 2026), and, on the training side, the demonstration that energy-based models learn through a cascade of phase transitions (Bachtis et al., 2024). These works treat the data as unstructured collections of samples; the time axis they study is the diffusion (or training) time. The internal time of a *sequence* — a video frame being correlated with its neighbours through a dynamical rule — is absent from these analyses.

Inference on chains. Exact posterior inference on Gauss–Markov chains is classical: the Kalman filter and smoother (Kalman, 1960; Rauch et al., 1965), hidden Markov models (Rabiner, 1989), and their unifying formulation as sum–product message passing on factor graphs (Kschischang et al., 2001; Mézard and Montanari, 2009). Continuous, non-

Gaussian state spaces have a long literature of their own, and it is worth saying precisely how this work sits inside it, because the machinery is familiar enough to look like a reapplication. Kalman–RTS is exact only for linear-Gaussian models. Assumed-density filtering and expectation propagation keep tractability by projecting each message back into a parametric family, so the approximation is in the messages themselves. Gaussian-sum filters widen that family without removing the restriction. Particle filters and smoothers place no parametric restriction but substitute Monte Carlo error, which can be quantified but adds a second error term to every comparison, so that what an approximation costs and what the reference costs have to be disentangled before either can be read. Grid-based filtering for continuous-state HMMs is the closest relative of what is done here, and is normally treated as an approximation to be tolerated rather than as an object whose error is itself the measurement.

Two things distinguish the present use. Deterministic quadrature is adopted precisely so that the reference carries *no* sampling error and its discretisation error can be bounded separately and reported, which is what makes it usable as a reference rather than as another competitor. And the target is not filtering: it is the diffusion score at every noise level, obtained from the same recursion that learns the kernel from noised sequences. What the classical literature does not ask is the question generative modelling forces — how the exact posterior, and therefore the exact score, deforms as a function of the diffusion time, and what that deformation implies for anything, algorithm or network, that must compute it at every level.

Message passing with continuous variables. BP is exact on trees for discrete variables, where messages are finite probability vectors. For continuous variables the messages become probability densities, and practical algorithms must close them within a parametric family, almost always the Gaussian one; this is the step that connects BP to AMP and to the TAP equations of statistical physics (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016; Genovese and Piana, 2026). The classical justification of the Gaussian closure is a central-limit argument valid when every node receives many

messages. On a chain each node receives exactly two: the central-limit argument fails by construction, and we are not aware of a quantitative assessment of the Gaussian message closure on chain-structured diffusion posteriors.

Amortising inference into a network. The three strands above concern how the score can be *computed*. A diffusion model does not compute it; it trains a network to reproduce it, and the distinction is one of kind rather than of accuracy. Sum-product on the posterior chain determines a whole distribution over the latent sequence, by making a free energy stationary over a family rich enough — because the graph is a tree — that the minimiser is exact rather than approximate. A score network is a fixed-depth feed-forward map: it spends the same computation on every input, whatever that input demands. The distinction is not that such a map cannot represent the *result* of the inference — a sufficiently expressive network can approximate the input-output function of an iterative algorithm, and Chapter 3 discusses architectures argued to do exactly that on tree-structured data. It is that the network does not *perform* instance-wise inference: it produces the answer directly rather than by a search whose length adapts to the instance. Reversal is iterative, over a horizon and a discretisation the sampler fixes, but each step’s score still comes from the fixed-compute map, so the network stands to the inference as an *amortisation* (LeCun et al., 2006).

Whether that amortisation is cheap or expensive is, in general, unmeasurable: it would require the very object the network was trained to stand in for. On the family of this thesis that object is available, so the question becomes an experiment. The works above do not measure it, because none of them combines a correlated non-Gaussian target computable across the whole noise schedule with an estimator that learns the transition from noised data and a baseline trained on the same data. What the measurement shows is reported in Chapter 8, including the respect in which it is *not* a simple function of the sample size: the error ratio is not monotone in the training-set size, and no trend in it is claimed here.

The gap, in one sentence: there is no end-to-end, exactly solvable account of the joint diffusion score of a structured sequence, covering its matrix structure, its message-passing

computation, the price of the Gaussian closure when the sequence is not Gaussian, and the price of amortising the whole computation into a network.

1.3 Research questions

The thesis addresses six questions. The first four concern scores under *known* dynamics and the last two concern learning dynamics that are unknown.

- RQ1.** For a stationary Gaussian AR(1) chain observed through coordinatewise Ornstein–Uhlenbeck noise, what is the joint score, and how do its spatial couplings vary with the diffusion time?
- RQ2.** How can the same Gaussian-chain score be computed by belief propagation, and how does the resulting recursion relate to Kalman filtering and smoothing?
- RQ3.** For selected non-Gaussian innovation models, how accurate is a Gaussian-message approximation relative to a validated numerical belief-propagation reference, over the parameter ranges studied?
- RQ4.** In the Gaussian model, how rapidly does the influence of remote observations on the score decay, and what limited guidance does this provide for local approximations of the score?
- RQ5.** Under what assumptions can an *unknown* local transition law be recovered from noised sequences alone, and which numerical, optimisation and statistical checks are required before such a recovery claim is reliable?
- RQ6.** On a family where the Markov assumption holds exactly, what is supplying that assumption worth, measured against denoising-score-matching networks trained on identical data and given none of it?

These questions are deliberately confined to the models named in them. No claim is made about general Markov processes, about universal properties of learned diffusion

scores, or about practical neural architectures. In particular RQ6 is a question about the value of a correct structural assumption on one family, not a comparison of learning algorithms in general: the two arms are asymmetric in supplied information by construction, which is what makes the measurement meaningful and also what bounds it.

Chapter 9 is a self-contained case study rather than a seventh question. It asks what joint observations can identify that no single-frame marginal can, on a rotating-ring model whose state space and estimand both differ from the chain's, and it is placed after the main line rather than inside it.

1.4 Contributions

This thesis makes four principal contributions, stated here in the order the body develops them.

1. **Exact scores for a noised Gaussian Markov chain.** For the stationary Gaussian AR(1) chain under coordinatewise OU corruption we derive the joint score in closed form, diagonalise it for all diffusion times in a single eigenbasis, and characterise how its coupling structure changes with t : a band-fill law $(Q_t)_{i,i+d} = (-1)^{d-1}(2t)^{d-1}(Q_0^d)_{i,i+d} + O(t^d)$ at small t , a return to isotropy at rate e^{-2t} , and exact bulk formulas for the posterior variance and for the geometric decay rate $q(\alpha, t)$ of posterior correlations. We show that belief propagation on the chain's factor graph is exactly the Kalman filter followed by Rauch–Tung–Striebel smoothing, update by update, and that it reproduces the matrix score to double-precision rounding. We prove that the influence of evidence at distance d decays as $q(\alpha, t)^d$, and separately *measure* that the radius- r windowed estimator's error decays at the same rate over $r = 2, \dots, 13$ — separately, because windowing changes the boundary conditions of the recursion in a way the influence-coefficient argument does not cover (Chapter 5).
2. **A controlled non-Gaussian reference, and an exact reading of Gaussian message closure.** For Laplace innovations, functional belief propagation remains

exact on the chain, but its messages no longer close in a finite-dimensional family. We construct a numerically validated quadrature reference and prove that moment-projected Gaussian messages return exactly the LMMSE estimator of the covariance-matched Gaussian model, and the same estimator for every innovation law sharing the first two moments (Theorem 6.6). The measured gap is therefore the price of discarding everything beyond second moments, not the error of one implementation: a median relative score error of 0.24 at $t = 0.05$, below 10^{-2} by $t \approx 1$ and below 10^{-3} by $t \approx 2.1$, with direction recovered more accurately than magnitude (Chapter 6).

3. **Learning an unknown transition from noised sequences.** We drop the assumption that the transition law is known. Fisher’s identity turns one forward–backward pass into the exact gradient of the marginal log-likelihood, with no differentiation through the recursion; the E-step is structurally exact for any innovation law, while the implementation uses controlled grid quadrature and a generalised M-step. The convergence experiments separate the mixture label’s own latent-variable difficulty from the additional missing information the observation channel introduces, and measure the consequence: the fitted innovation shape settles at a median of 229 updates against 80 for the correlation, so a stopping rule read off the correlation trace certifies a kernel that has not converged in the coordinate every non-Gaussian claim depends on (Chapter 7).
4. **The measured value and limits of supplied Markov structure.** At equal training-set size, EM–BP attains $7.0\text{--}17.6\times$ lower pointwise denoising error than an unstructured network across 8 sizes, and $2.3\text{--}6.1\times$ lower error than a weight-shared window head selected by a screen over three architectures on disjoint seeds. Controlled experiments then delimit that advantage under finite-grid mixture capacity and under two departures from the Markov assumption. All of these conclusions are restricted to the stated numerical and statistical protocols (Chapter 8).

Two further results sit alongside the main line rather than inside it. The rotating-ring model of Chapter 9 extends the state at each site from a scalar to a point on a circle and

gives an exactly interpretable example in which a dynamical parameter is absent from every one-frame marginal yet identifiable from the trajectory. The per-node AMP/TAP closure on the Gaussian chain, developed in Appendix B, returns the exact score but a different posterior variance, with an existence boundary in closed form, a breakdown time $t_c(\alpha)$, and a critical coupling $\alpha_c = \sqrt{2} - 1$ (statement of results in Section 10.1).

1.5 Methodology

The methodology is deliberate: closed forms first, numerics as audit. Analytical results lead, and every approximation is measured against a reference computed by an independent method rather than assumed small. Networks appear in exactly one role — as the baseline RQ6 measures the structured estimator against — and never as an instrument for establishing anything about the model itself. Claims in this thesis are established in three ways, and the difference is not cosmetic. Most are proved by explicit algebra (with the longer passages isolated in numbered *toolboxes*). Others are verified by deterministic numerical checks against a reference computed by an independent method, in the spirit of the exactly solvable programme of Biroli and Mézard (2023). Where an approximation is introduced — the Gaussian closure of BP messages, local truncations — its error is *measured* against a reference computed either by matrix algebra or by dense grid quadrature validated on the solvable case. The latter is a numerically validated reference, not an exact one, and is used only inside the resolution regime where that validation holds. The learning chapters (7 and 8) are neither: they are stochastic estimation experiments, reported with paired uncertainty over seeds, and their conclusions are statistical rather than certain. All numbers in the thesis are regenerated by deterministic scripts from frozen outputs, documented in Appendix C; we regard this reproducibility as a strength of the method, not as a contribution in itself.

The status a result then carries, in five labels. How a claim is established is not quite the same as what it may afterwards be asserted to be, and “exact” is used for

several different things in this literature. The thesis therefore fixes one vocabulary of five labels, finer than the three modes above, and each research chapter closes with a section classifying its own results in it.

Exact, by derivation. An identity of the continuum model, proved algebraically. It holds for the mathematical object, whatever a computer subsequently does with it.

Exact for the discretised model. A statement true of the finite grid after a stated discretisation — BP on the resulting chain-shaped factor model, for instance, is exact up to floating point. This does *not* claim the discretisation reproduces the continuum answer; that is the next category. Which discretised object is meant has to be said, since more than one is available: Section 7.4 states the convention used here.

Numerical, in a validated regime. A measurement whose error is bounded against an independently computed reference, over the stated grids, tolerances and parameter ranges, and not beyond them.

Statistical, over seeds. An estimate from a stochastic experiment, reported with paired uncertainty across replicates. It can be resolved, unresolved, or resolved against a predeclared region, and none of those is the same as proved. The learning chapters are almost entirely of this kind.

Interpretation. A reading of the measurements that is consistent with them but not established by them. Kept where it is useful and labelled wherever it appears.

The distinction between the first two matters most in Chapters 6 and 7: sum-product on the posterior chain is exact by derivation as a recursion over functional messages, while what is computed is a grid approximation to it whose error is measured separately. Both statements are true, and they are not the same statement.

1.6 Scope and limitations

The exact results concern one-dimensional state per frame, linear AR(1) dynamics, and stationary initial conditions. The non-Gaussian results are numerical, valid within the validated resolution regime of the reference solver and the tested parameter ranges. The locality results are proved for the Gaussian bulk only.

The learning results (RQ5, RQ6) carry their own limits, and they are tighter than the analytical ones. The estimator is given the Markov factorisation, the autoregressive form, homogeneity across sites, and an innovation family flexible enough to represent the truth closely; nothing here speaks to a setting where those do not hold. The last of these is a representation rather than an exact containment: a C -component Gaussian mixture does not contain a Laplace density for any finite C . What maximum likelihood approaches is therefore the best approximation within the fitted family, and the recovery statements of Chapter 8 concern specific summaries — α , the innovation variance, its excess kurtosis — rather than the transition density as a whole. The comparison in RQ6 is thus a measurement of what a correct structural assumption buys on one family. It is not evidence that likelihood learning through belief propagation is a better learning algorithm than score matching — a distinction the thesis maintains wherever the numbers appear. Any implication for trained score networks in practice remains a hypothesis suggested by a controlled model, not an established property of practical systems.

One line of experiment is reported as unresolved rather than as a finding: the reverse-generation study of Section 8.4, whose fits used a fixed iteration budget far shorter than the innovation shape needs to settle, so that it scores converged estimators against unconverged ones on the one coordinate it measures. It supports no conclusion in Chapter 10. A second question — how much innovation-mixture capacity is worth having — was unresolved for the same reason in an earlier and much smaller design; it was rerun to a convergence criterion at sixteen seeds and is answered in Section 8.3, in the opposite direction to the one the earlier design suggested.

1.7 Structure of the thesis

Chapters 2 and 3 contain the background and literature review; Chapters 4–8 contain the research. Throughout, each concept is developed once, in the chapter where it naturally belongs, and then used without repetition.

Chapter 2 fixes the four objects the research chapters manipulate — the Markov sequence that generates the data, the Ornstein–Uhlenbeck channel that corrupts it, the time reversal that makes generation possible, and the score–posterior identity that turns the reversal’s unknown term into a posterior expectation — and then states what the surrounding literature settles and what it leaves open. It is deliberately compressed to what the body uses. The statistical mechanics from which these models descend, the lineage that carries it into machine learning, and the step-by-step derivations behind each result quoted there are in the companion compendium, at full length. Chapter 3 builds the inference machinery this thesis reasons with: Markov random fields, factor graphs, the sum–product algorithm and the conditions under which it is exact, and the architectural mirror of locality in convolutional networks and U-Nets. The multivariate Gaussian and its precision-matrix reading, the spectral theorem, the identification of Gaussian BP with the Kalman filter, and the TAP/AMP closures are developed where they are used — in Chapter 5 and Appendix B — rather than stockpiled here.

Chapter 4 settles the object of study: why the joint score of a trajectory rather than a per-frame score, and why the stationary AR(1) chain under coordinatewise OU noise is the minimal model in which that object is nontrivial yet exactly computable. Chapter 5 solves the Gaussian chain exactly (RQ1, RQ2, and the Gaussian half of RQ4), with the spectral form of the precision matrix as the organising object and belief propagation identified with the Kalman smoother. Chapter 6 leaves Gaussianity in the most controlled way possible, Laplace innovations, and measures what the Gaussian message family costs when the data are no longer Gaussian (RQ3). Chapters 7–8 leave the innovation law itself unknown: an EM procedure built on the exact chain inference of Chapter 5 estimates it from noised data alone, and the structured estimator is compared against trained neural

denoisers — unstructured and structure-aware — along axes of model capacity and of misspecification of the Markov assumption itself. That closes the main line: one continuous argument from the Gaussian chain, through a controlled non-Gaussian departure, to a learned transition and its measured value. Chapter 9 then extends the state at each site from a scalar to a point on a rotating ring, a nonlinear dynamic object, and shows that a Cartesian surrogate is exactly solvable while the model as drawn is solvable numerically to a measured accuracy, both admitting the same structural description of the joint score. It is placed after the main results because it is an independent extension and nothing in them depends on it. Chapter 10 states findings per research question, limitations, and the questions that follow. The appendices contain proofs, numerical checks, aggregation analyses and reproducibility material that directly support the thesis’s original results: the Gaussian identities with proofs, the AMP/TAP analysis with its exact phase boundary, the linear-algebra and rotating-ring derivations, the reproducibility protocol, and the robustness of both reported ratios to how they were aggregated. Extended historical background, literature notes, alternative derivations and the development record are collected in the companion compendium.

Chapter 2

Background and Positioning

This chapter fixes the four objects the rest of the thesis manipulates: the Markov sequence that generates the data, the Ornstein–Uhlenbeck channel that corrupts it, the time reversal that makes generation possible, and the identity that turns the reversal’s one unknown term into a posterior expectation. It closes by stating what the surrounding literature settles and what it leaves open.

The treatment is deliberately compressed. Every result below is standard, and each is derived in full in the companion compendium: the variational calculus and the statistical mechanics from which these models descend, the integrating-factor solution of the Ornstein–Uhlenbeck equation, the Fokker–Planck route to the reverse drift, and an extended review of the solvable-diffusion literature. Nothing in the body depends on those derivations. The body depends only on the statements collected here.

2.1 Markov sequences

A *stochastic process* is a collection of random variables indexed by time, $(X_k)_{k \geq 0}$ in discrete time or $(X_t)_{t \geq 0}$ in continuous time, defined on a common probability space. Two notions of time homogeneity are used constantly below and are worth separating precisely (Brockwell and Davis, 1991).

Definition 2.1 (Strict and weak stationarity). A process (X_k) is *strictly stationary* if all its finite-dimensional laws are invariant under time shifts: for every n , every set of indices $k_1, \dots, k_n$, and every shift h ,

$$(X_{k_1}, \dots, X_{k_n}) \stackrel{d}{=} (X_{k_1+h}, \dots, X_{k_n+h}).$$

It is *weakly (second-order) stationary* if its mean is constant and its autocovariance depends only on the lag, $\text{Cov}(X_k, X_{k+d}) = \gamma(d)$.

Strict stationarity implies weak stationarity whenever second moments exist, and for *Gaussian* processes the two coincide, because a Gaussian law is determined by its first two moments. The distinction matters here because Chapter 6 replaces the Gaussian innovation with a Laplace one, at which point the two notions separate and only the strict one is preserved by the construction. Stationarity also has a matrix consequence the research chapters exploit: the covariance of L consecutive observations of a weakly stationary process, $(\Sigma)_{ij} = \gamma(i - j)$, is constant along diagonals, a symmetric *Toeplitz* matrix. Wherever a covariance in this thesis is Toeplitz, stationarity is the reason.

Definition 2.2 (Markov chain). A sequence of random variables $(X_k)_{k \geq 0}$ with values in a state space $\mathcal{X}$ is a *Markov chain* if for every k and every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{P}(X_{k+1} \in A \mid X_k, X_{k-1}, \dots, X_0) = \mathbb{P}(X_{k+1} \in A \mid X_k). \quad (2.1)$$

The conditional law is encoded by a *transition kernel* $K(x' \mid x)$: a density, or a matrix for discrete $\mathcal{X}$, with $\int K(x' \mid x) dx' = 1$ for every x .

Example 2.3 (The Gaussian AR(1) chain). The running example of this thesis is the scalar Gaussian autoregression of order one,

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad k = 0, \dots, L-2, \quad (2.2)$$

with transition kernel $K(a' \mid a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$. Taking $\sigma_\eta^2 = 1 - \alpha^2$ and $a_0 \sim \mathcal{N}(0, 1)$

makes it stationary with unit marginal variance and autocovariance $\gamma(d) = \alpha^{|d|}$, so its covariance is Toeplitz for the reason just given. Every chapter from Chapter 5 onwards concerns (2.2), one of its non-Gaussian variants, or what an estimator can recover of it through noise.

The Markov property (2.1) is exactly a statement about *factorisation*: the joint law of the first L states is

$$p(x_0, x_1, \dots, x_{L-1}) = p_0(x_0) \prod_{k=0}^{L-2} K(x_{k+1} | x_k). \quad (2.3)$$

This is the most heavily used equation of the thesis. It says the joint distribution of a Markov trajectory is a product of *local* terms, each touching two consecutive variables: the structure that makes the chain a tree in the factor-graph sense of Chapter 3, that makes its precision matrix tridiagonal in the Gaussian case of Chapter 5, and that makes belief propagation exact. Marginals evolve by integrating the kernel, $p_{k+1}(x') = \int K(x'|x) p_k(x) dx$, and a law π with $\pi = K\pi$ is *stationary*: a chain started in π remains in it, and its trajectory is then strictly stationary in the sense of Definition 2.1.

One change of perspective sets up the research problem. Markov chains are classically studied as *algorithms*, where the stationary law is the target and the trajectory a means to reach it. From Chapter 5 onwards the chain (2.2) plays the opposite role: it is the *data-generating process*, one trajectory $(a_0, \dots, a_{L-1})$ is one data point, and the object of interest is the law of the whole trajectory after every coordinate has been independently corrupted. The factorisation (2.3) then stops being a computational convenience and becomes the structure to be recovered.

2.2 The Ornstein–Uhlenbeck corruption channel

The noising process used by diffusion models, and by every result of this thesis, is the *Ornstein–Uhlenbeck process* (Uhlenbeck and Ornstein, 1930): Brownian motion pulled

back to the origin by a linear spring,

$$dX_t = -X_t dt + \sqrt{2} dW_t, \quad X_0 = a. \quad (2.4)$$

The drift and diffusion constants are the variance-preserving normalisation of the diffusion literature (Song et al., 2021); any other choice rescales time. The process is the unique Gaussian, Markov, stationary process in continuous time, and it is exactly solvable. Solving (2.4) by integrating factor — the derivation is routine and is carried out in the companion compendium — gives a Gaussian transition kernel, $p(x_t | X_0 = a) = \mathcal{N}(x_t; ae^{-t}, \Delta_t)$ with $\Delta_t = 1 - e^{-2t}$. Corrupting a data point a for a time t therefore produces

$$x = ae^{-t} + \sqrt{\Delta_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1) : \quad (2.5)$$

the signal is *contracted* by e^{-t} while isotropic noise of variance Δ_t fills in. At $t = 0$ the channel is the identity; as $t \rightarrow \infty$, $x \sim \mathcal{N}(0, 1)$ regardless of a . The signal-to-noise ratio e^{-2t}/Δ_t sweeps continuously from ∞ to 0 and diverges as $1/2t$ for small t , a factor that reappears throughout the thesis as an error amplifier. The channel interpolates between the data distribution and a standard Gaussian, which is what a diffusion model needs.

Two structural remarks are used repeatedly later. First, the kernel is Gaussian *in a as well as in x* : read as a function of the clean variable, $\mathcal{N}(x; ae^{-t}, \Delta_t) \propto \exp[-(e^{-2t}/2\Delta_t)(a - xe^t)^2]$, a Gaussian likelihood factor of precision e^{-2t}/Δ_t centred at the deconvolved observation $e^t x$. This is what makes the posteriors of Chapter 5 jointly Gaussian and the belief-propagation messages closable in a finite-dimensional family. Second, if $a \sim \mathcal{N}(\mu, \Sigma_0)$ is pushed through (2.5) coordinate-wise with *independent* noises, the output is Gaussian with

$$\mu_t = e^{-t} \mu, \quad \Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I, \quad (2.6)$$

since means and covariances transform linearly and the added noise is white. Equation (2.6) — shrink the covariance, add a multiple of the identity — is the entire forward process of this thesis in one line.

2.3 Reverse diffusion

Suppose data $X_0 \sim p_0$ are corrupted by a forward SDE

$$dX_t = f(X_t, t) dt + g(t) dW_t \quad (2.7)$$

up to time T , producing marginals p_t ; the Ornstein–Uhlenbeck channel (2.4) is the case $f = -x$, $g = \sqrt{2}$. Can the corruption be undone — is there an SDE which, run from $X_T \sim p_T$ backwards, reproduces the same marginals? The answer (Anderson, 1982) is yes, and the reverse drift needs exactly one object that the forward process does not supply:

$$dX_t = \left[f(X_t, t) - g(t)^2 \nabla_x \log p_t(X_t) \right] dt + g(t) d\bar{W}_t, \quad (2.8)$$

where time runs backwards from T to 0 and $\bar{W}$ is a reverse-time Wiener process. The correction to the drift is the *score* $\nabla_x \log p_t$ of the noised marginal. The forward process and the terminal law p_T are known by construction; the score is not, and estimating it is the entire modelling effort of a diffusion model (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021). A Fokker–Planck argument making (2.8) plausible, together with the probability-flow ODE it produces, is given in the companion compendium; the rigorous statement is Anderson (1982). A recent book-length survey of the model family is Lai et al. (2025).

2.4 The score as a posterior expectation

Every result of this thesis passes through one rearrangement of the score. Writing the noised marginal as the convolution of p_0 with the Gaussian channel and differentiating under the integral (Vincent, 2011):

$$\boxed{\nabla_x \log p_t(x) = \frac{e^{-t} \mathbb{E}[a | x] - x}{\Delta_t} \iff \mathbb{E}[a | x] = e^t \left(x + \Delta_t \nabla_x \log p_t(x) \right).} \quad (2.9)$$

We refer to (2.9) throughout as the *score-posterior identity*; in the empirical-Bayes literature it is Tweedie’s formula (Robbins, 1956; Miyasawa, 1961; Efron, 2011). The score of the noisy marginal and the Bayes-optimal denoiser $\mathbb{E}[a|x]$ determine each other exactly, with no assumption on p_0 . Two readings recur. The score points from x towards the contracted image of the posterior mean, so denoising direction and score direction coincide; and estimating the score *is* solving a Bayesian inference problem, not merely an analogue of one. For sequence data with a Markov prior that inference problem lives on a tree, which is why it is computable by message passing — the programme made precise in Chapter 3 and executed in Chapter 5.

For *vector-valued* data $a \in \mathbb{R}^L$ noised coordinate-wise, (2.9) holds componentwise with the *full* posterior:

$$\left(\nabla_x \log p_t(x)\right)_k = \frac{e^{-t} \mathbb{E}[a_k | x_0, \dots, x_{L-1}] - x_k}{\Delta_t}. \quad (2.10)$$

The conditioning is on *all* coordinates. Even though the noise is independent across frames, the posterior mean of frame k , and hence the k -th score component, depends on every observation, because the prior couples the frames. Equation (2.10) is the precise reason the *joint* score differs from the per-frame marginal score (Chapter 4), and Chapter 5 quantifies how much coupling the conditioning induces as a function of t .

2.5 Positioning

Five lines of work bound the problem this thesis addresses. Each is summarised at the level of what it settles; the extended discussion, paper by paper, is in the companion compendium.

Solvable diffusion models and phase transitions. Treating the reverse process as a stochastic dynamics on a high-dimensional energy landscape has produced an exact picture for tractable priors: Gaussian mixtures and random-energy targets exhibit sharp times at which the backward trajectory commits to a class (Biroli and Mézard, 2023; Biroli

et al., 2024; Raya and Ambrogioni, 2023; Achilli et al., 2026), and the sampling problem itself admits a hardness analysis (Ghio et al., 2024; Cui et al., 2025). These analyses fix the prior to a form with no internal sequential structure, so the noised law factorises over coordinates or couples them only through a low-dimensional latent.

Memorisation and generalisation. A second strand asks when a diffusion model reproduces its training set rather than the law behind it, and locates the transition in the ratio of data to capacity (Achilli et al., 2024; Bonnaire et al., 2025; Kadkhodaie et al., 2024; Favero et al., 2025b). The mechanism is again characterised for priors without a time axis internal to the data point.

Filtering and smoothing in state-space models. Recovering a latent Markov trajectory from noisy observations is classical: the Kalman filter (Kalman, 1960) and the Rauch–Tung–Striebel smoother (Rauch et al., 1965) solve it exactly in the linear-Gaussian case, and expectation–maximisation (Dempster et al., 1977) learns the transition when it is unknown. This literature fixes the observation noise at a single level. The diffusion setting instead requires the whole family indexed by t , and it is the deformation across t , not the solution at any one t , that carries the information a score model needs.

Exact inference on trees, and what happens off them. Belief propagation is exact on tree-shaped factor graphs (Pearl, 1988; Mézard and Montanari, 2009). On a chain with continuous variables the messages are functions rather than vectors, and they close in a finite-dimensional family only for special priors; moment projection (Minka, 2001) is the standard response. What the projection costs, for a specific non-Gaussian chain, is not generally known in closed form.

Networks as amortised inference. Score networks are trained to approximate a posterior expectation, which invites reading their architecture as an inference procedure, and cascade-style analyses show structure being acquired in stages (Bachtis et al., 2024; Sclocchi et al., 2025; Favero et al., 2025a; Garnier-Brun et al., 2026; Zdeborová and Krzakala,

2016). The reading is suggestive rather than established, and the comparison between a network and the exact inference procedure for the same prior is rarely available, because the exact procedure is rarely computable.

Real sequential data — video, audio, trajectories — have exactly the structure the solvable programme leaves out: an internal frame time with a Markovian dependence along it. For such data the score (2.10) is a smoothing posterior on a chain, an object with seventy years of inference theory behind it. Yet the questions a diffusion modeller asks about it are answered neither by the classical literature, which fixes the noise level, nor by the diffusion literature, which sets internal structure aside: how does the coupling structure of $\nabla \log p_t$ deform with t ; can it be computed locally; what a Gaussian parametrisation of the messages costs when the data are not Gaussian; can the transition itself be learned through the channel; and how does the resulting estimator compare with a network trained on the same data? Those five questions are the subject of Chapters 5 to 8, with the inference tools prepared in Chapter 3 and the modelling choices that fixed the formulation recorded in Chapter 4.

Chapter 3

Graphical Models and Message Passing

This chapter introduces the representation the thesis computes in: a distribution written as a product of local factors, and the recursion that returns its marginals exactly when those factors form a tree. It is short by design. The multivariate Gaussian in both parametrisations, the spectral and matrix-inversion tools, the Gaussian special case of the recursion, and its relation to TAP and AMP are all in Appendix D; what is kept here is the structure, because it is the structure and not the algebra that decides what is computable.

3.1 Markov random fields

Appendix D assigns structure to matrices, through the zeros of a precision. Graphical models assign it to distributions instead, for arbitrary and not necessarily Gaussian variables; the two turn out to be the same assignment.

Definition 3.1 (Markov random field). Let $G = (V, \mathcal{E})$ be an undirected graph with one random variable x_i per node. The collection $x = (x_i)_{i \in V}$ is a *Markov random field* (MRF)

if it satisfies the local Markov property: for every node i ,

$$p(x_i | x_{V \setminus \{i\}}) = p(x_i | x_{\partial i}), \quad (3.1)$$

where ∂i is the set of neighbours of i in G .

The Hammersley–Clifford theorem connects conditional independence to an energy representation (proved in Besag, 1974): for a strictly positive density, the Markov property with respect to G is equivalent to factorisation over the *cliques* of G ,

$$p(x) = \frac{1}{Z} \prod_{C \in \text{cliques}(G)} \psi_C(x_C) = \frac{1}{Z} \exp \left[- \sum_C E_C(x_C) \right]. \quad (3.2)$$

Conditional independence, graphical locality, and additive local energies are one and the same thing. The Ising model is the textbook example. The decisive example for us is one-dimensional: the chain factorisation (2.3) makes a Markov chain an MRF on a line, with pairwise nearest-neighbour energies. For *Gaussian* MRFs the correspondence becomes matrix-concrete: the energy is the quadratic form $\frac{1}{2}a^\top J a - h^\top a$, its pairwise terms are the entries J_{ij} , and the graph of the MRF is exactly the sparsity pattern of the precision matrix. The dual reading of zeros in Appendix D.1 and the Hammersley–Clifford theorem are the same statement, once in linear algebra and once in probability. A Gaussian Markov chain therefore has a tridiagonal precision, a fact Chapter 5 derives explicitly and exploits throughout.

One more structural remark that the research chapters rely on: if each variable of an MRF is observed through its own independent noisy channel, the likelihood factors touch one variable each, so no clique is enlarged. *The posterior of an MRF under per-node observations is an MRF on the same graph.* Observation changes the potentials, never the graph.

3.2 Factor graphs

MRFs record *which* variables interact but not *how* the joint factorises: a triangle of pairwise terms and a single triple term have the same graph. The representation that makes the factorisation itself graphical, and on which message passing is most naturally defined, is the *factor graph* (Kschischang et al., 2001; Koller and Friedman, 2009).

Definition 3.2 (Factor graph). Given a factorisation $p(x) = \frac{1}{Z} \prod_a \psi_a(x_{\partial a})$, its factor graph is the bipartite graph with a *variable node* for each x_i , a *factor node* for each ψ_a , and an edge (i, a) whenever $x_i \in x_{\partial a}$.

A connected factor graph with one fewer edge than nodes is a *tree*: it contains no cycles, so any edge separates it into two disjoint components. Trees are the tractable case of inference: on them, marginals and partition functions are computable exactly by the recursion of the next section, whatever the alphabet of the variables. The factor graph of a Markov chain observed through per-node noise is a tree of a particularly simple shape (a spine of variables and transition factors, with one evidence leaf per variable), as Chapter 5 verifies by counting.

3.3 The sum-product algorithm

On a factor graph, sum-product belief propagation (BP) passes messages along every edge (Pearl, 1988; Kschischang et al., 2001; Mézard and Montanari, 2009). Variable-to-factor messages multiply the incoming factor messages from all *other* neighbours, and factor-to-variable messages integrate the factor against the incoming variable messages:

$$\nu_{i \rightarrow a}(x_i) \propto \prod_{b \in \partial i \setminus a} \hat{\nu}_{b \rightarrow i}(x_i), \quad \hat{\nu}_{a \rightarrow i}(x_i) \propto \int \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} \nu_{j \rightarrow a}(x_j) dx_{\partial a \setminus i}, \quad (3.3)$$

with the belief at node i proportional to $\prod_{a \in \partial i} \hat{\nu}_{a \rightarrow i}(x_i)$. The “exclude the recipient” rule is the heart of the algorithm: the message $i \rightarrow a$ summarises everything node i knows *except* what it learned from a , so that no piece of evidence is ever counted twice along a

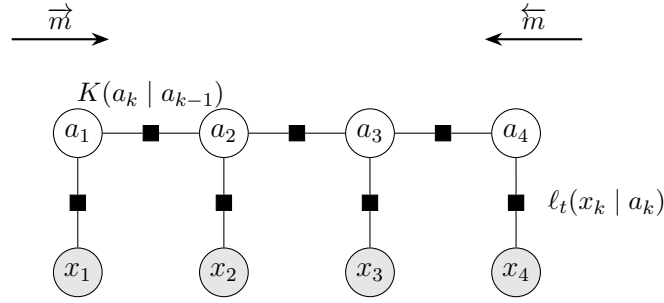


Figure 3.1: The posterior factor graph of a Markov chain observed through the coordinatewise channel. Open circles are latent variables, shaded circles observations, squares factors. Each transition factor $K(a_k | a_{k-1})$ joins two adjacent latent variables; each observation enters through a *unary* factor attached to a single latent variable, because the channel corrupts every coordinate independently. Conditioning therefore reweights node potentials without creating any edge among the a_k , so the posterior graph is still a chain and Theorem 3.3 applies to it. Arrows show the two message sweeps of (3.3). The construction fails as soon as the forward noise is correlated across coordinates: the likelihood then touches several a_k at once and the graph acquires edges.

path.

Theorem 3.3 (Tree exactness of sum-product). *On a tree-shaped factor graph, one inward and one outward pass of (3.3) compute the exact node and factor marginals of p (Mézard and Montanari, 2009, Theorem 14.1). The resulting beliefs are unique; the individual unnormalised messages are defined only up to positive multiplicative constants.*

The reason is the separation property of trees: each edge splits the graph into two disjoint halves, and the message crossing it summarises exactly the half behind it. On graphs with cycles the same updates, iterated, define *loopy* BP, which is extraordinarily useful but approximate (Weiss and Freeman, 2001; Malioutov et al., 2006); none of that is needed in this thesis, because our graphs are trees.

The representation problem. For *discrete* variables each message is a finite probability vector and each update in (3.3) is finite arithmetic. For *continuous* variables each message is a probability *density*, an infinite-dimensional object, and the update integrals must be carried out in function space. Tree exactness (a property of the graph) says nothing about representability of the messages (a property of the model’s function class);

keeping the two apart is essential, and the distinction is where all the difficulty of the non-Gaussian research chapter lives. There are three ways to make continuous BP computable: restrict to models whose messages close in a finite-dimensional family (the Gaussian route, next section); represent messages numerically on a grid (the route our non-Gaussian reference solver takes, Chapter 6); or project each message onto a tractable family and accept the error (the route whose cost this thesis measures).

3.4 Locality in learned architectures: convolutions, receptive fields, U-Nets

The message-passing story has an architectural mirror, and it supplies the vocabulary for the locality questions of the research chapters. A convolutional network processes a signal by applying the same local filter at every position, stacking such layers so that depth enlarges the *receptive field*: the window of input coordinates that can influence a given output coordinate (LeCun et al., 2015). Convolutional architectures powered the breakthrough of deep learning in vision (Krizhevsky et al., 2012), and their central inductive bias is precisely locality plus translation invariance: the value of an output pixel is computed from a bounded neighbourhood, identically across the image. The standard denoising backbone of diffusion models, the U-Net (Ronneberger et al., 2015; Ho et al., 2020), is a convolutional encoder–decoder with skip connections, whose receptive field grows along the contracting path and whose skip connections reinject fine-scale local information along the expanding path.

Why local architectures should be *sufficient* for score estimation is a theoretical question, and message passing has begun to answer it. Mei (2024) shows that for generative hierarchical models (tree-structured graphical models over image-like data), the denoising function computed by belief propagation can be implemented efficiently by a U-Net, with the network’s layers mirroring the sweeps of the message-passing schedule; convolutional layers are efficient implementations of the local BP updates. This gives the U-Net an inference-algorithm reading: the architecture is good at denoising because denoising, on

structured data, *is* message passing, and message passing is local.

The correspondence is not confined to convolutional architectures. Garnier-Brun et al. (2024) train transformers on data generated by a hierarchical model in which the exact posterior is again computable by belief propagation, and find that the trained attention maps come to reproduce the message-passing schedule. Read together with Mei (2024), this is *cross-architecture evidence* for a particular reading: that graphical inference is a useful description of what a trained denoiser ends up computing, and that the depth it needs is set by the structure of the data rather than by the architecture chosen. It is evidence across two architecture families and two model families, not a theorem about every denoiser — a network can approximate the same posterior map without reproducing the algorithm internally, and neither result rules that out. This is precisely why the chain matters here. A chain is a degenerate tree, of depth equal to its length and width two, so the depth these results would predict is not the logarithmic depth of a balanced hierarchy but a depth linear in the correlation length — a sharply different architectural prediction, on a model where it can be checked.

The present thesis works one step below this correspondence, on a model small enough that the locality of the score is not an architectural hypothesis but a computable quantity. For the Gaussian chain, the exact score at frame k depends on the observation at frame $k \pm d$ with a weight that decays geometrically in d , at a rate $q(\alpha, t)$ computed in closed form in Chapter 5. The error of the radius- r windowed estimator is *measured* to decay at that same rate over the tested range — a separate and weaker statement. Comparing truncation of the inference against truncation of the score matrix then quantifies, in one solvable case, how a local computation should be organised. These results describe the function a network would have to learn rather than the dynamics of learning it; the trained baselines of Chapter 8 supply the other half of that comparison.

Chapter 4

The Object of Study, and the Minimal Model

Before any of the analysis, two choices have to be justified: that the quantity worth studying is the *joint* score of a whole trajectory rather than a per-frame score, and that the stationary Gaussian AR(1) chain under coordinatewise OU noise is the right model in which to study it. This short chapter settles both. The exploratory models that led here, and the sequence of formulations they ruled out, are recorded in the companion compendium rather than in this document.

4.1 Why the joint score

Standard diffusion models generate *static* objects; the data of interest here are *dynamic*, a data point being a sequence of frames indexed by an internal time $u = 0, 1, \dots, L - 1$ — a video, a rotating shape, a stochastic trajectory. Two times therefore appear from the outset, and keeping them apart is a permanent discipline of this thesis: the *internal* time u of the sequence, and the *diffusion* time t of the noising channel, which acts independently on every frame.

It is tempting to treat such a trajectory as L separate samples, computing each frame's noisy marginal score independently. In the Gaussian model that carries most of this thesis,

that is not merely lossy — it is empty.

Proposition 4.1 (One-frame marginals in the stationary Gaussian chain). *Let the clean chain be stationary Gaussian AR(1) with unit marginal variance, put through the variance-preserving channel $x = e^{-t}a + \sqrt{1 - e^{-2t}}z$, with $z \sim \mathcal{N}(0, 1)$ independent of a . Then every noisy frame is marginally $\mathcal{N}(0, 1)$ at every diffusion time, its per-frame marginal score is exactly $-x_k$, and the collection of one-frame marginals carries zero Fisher information about the coupling α .*

This is immediate: variance preservation gives $\text{Var}(x) = e^{-2t} \text{Var}(a) + (1 - e^{-2t}) = 1$, and a Gaussian a makes x Gaussian, so the marginal law is $\mathcal{N}(0, 1)$ whatever α is and cannot depend on it.

Remark 4.2 (The same statement is false off the Gaussian model). Variance preservation alone gives only $\text{Var}(x) = 1$; it does not give Gaussianity, and the distinction matters because the non-Gaussian chains of Chapters 6–8 are the ones the rest of this thesis measures. Writing φ for characteristic functions,

$$\varphi_x(u) = \varphi_a(e^{-t}u) \exp\left[-\frac{1-e^{-2t}}{2}u^2\right], \quad (4.1)$$

so if x were exactly $\mathcal{N}(0, 1)$ at some finite t , dividing by the Gaussian factor would force $\varphi_a(e^{-t}u) = \exp[-\frac{1}{2}e^{-2t}u^2]$ — that is, a Gaussian. A genuinely non-Gaussian frame therefore stays non-Gaussian under any finite amount of Gaussian corruption, and its one-frame score stays nonlinear; Section 6.2 computes that score for a Laplace prior and shows it approaching $-x$ only as $t \rightarrow \infty$.

So the correct general statement is narrower than “per-frame is empty”, and it is about *dependence* rather than about information as such. One-frame marginals can see the innovation law, and can see a parameter that changes the marginals; what no collection of them can recover is how consecutive frames are coupled, since coupling is a property of the joint law and of no marginal of it. In the Gaussian case those two statements collapse into one, because the marginals are α -free by Proposition 4.1; off it they come apart, and

Chapter 9 proves the sharper version for one specific parameter — that the rotation rate of the ring model has exactly zero Fisher information in every one-frame marginal, under that model’s rotational symmetry.

A sequence is *one* sample from a joint law, and the quantity that drives the reverse dynamics is the joint score of the whole noisy trajectory, the gradient of $\log P_t(x_0, \dots, x_{L-1})$, whose k -th component is a conditional expectation over the *full* posterior of the clean frames, as in (2.10). That is the object in which the temporal structure lives. Chapter 5 exhibits the contrast in closed form at $L = 2$.

This also settles the shape of the structural question. One might hope that consecutive frame-blocks of the score are related by a fixed map, a “propagator”,

$$S_{:,u}(x, t) \approx \mathcal{P}_t(S_{:,u-1}(x, t)), \quad (4.2)$$

so that a learning system could hold one block and one map instead of L blocks. What the joint view shows is that the structure worth studying is not a map between per-frame scores at all, but the coupling pattern of the joint score — a statement about a matrix in the Gaussian case, and about a posterior in general. That is the object the rest of this thesis analyses. Figure 4.1 shows it in the smallest case where it exists: the diagonal tilt of the joint score field is the whole of the coupling, and it is exactly what a per-frame treatment throws away.

4.2 The minimal model

Among the candidate dynamic objects — rotations, coupled phases, bounded trajectories with reflections, curved data manifolds — the stationary Gaussian AR(1) chain under coordinatewise OU noise is the minimal one in which the joint score is nontrivial yet exactly computable: one coupling parameter α for the internal dynamics against one noise parameter t for the corruption, linear algebra available at every step, and every structural question (coupling range, locality, message passing) posable in closed form.

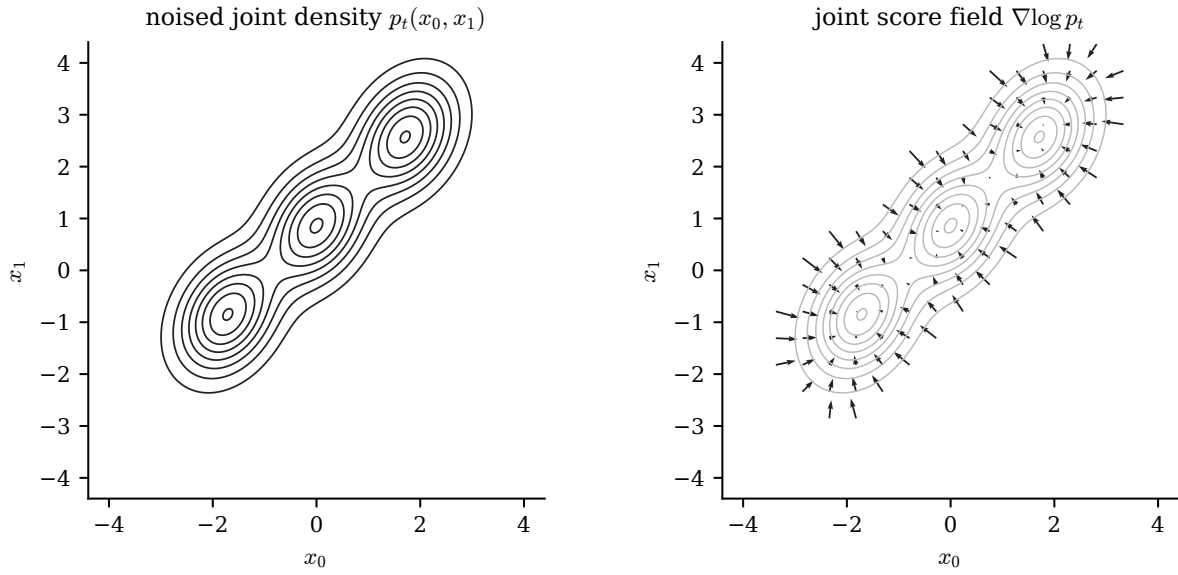


Figure 4.1: The coupling made visible, in the smallest case where it exists: $x_1 = x_0 + c + \eta$ with a trimodal Gaussian-mixture prior on x_0 , drift $c = 1$, innovation standard deviation $\sigma = 0.45$, after diffusion time $t = 0.15$. Left: the noised joint density $p_t(x_0, x_1)$. Right: the joint score field $\nabla \log p_t$ (arrows) over the same density (contours). Both panels are computed from the closed-form mixture expressions on a grid. The arrows point towards the modes and tilt along the diagonal, and that tilt is the entire effect of the coupling between frames — it is exactly what any per-frame treatment discards. Note that this prior is *not* Gaussian, so by Remark 4.2 its noised one-frame marginals are Gaussian mixtures with nonlinear scores rather than $\mathcal{N}(0, 1)$ with score $-x_k$; what those marginals cannot see is the coupling, which is the point the figure makes.

The requirement doing the work here is *exact computability*. The whole programme of this thesis is to measure what approximations cost — Gaussian message closure in Chapter 6, an estimated transition kernel in Chapters 7 and 8, a finite receptive field in Chapter 5 — and every one of those measurements needs a reference that is known rather than approximated. A model in which the answer is available in closed form is not a simplification of the question; it is what makes the question answerable at all.

The nonlinear models are set aside at that point deliberately, as targets for after the solvable case is understood. Chapter 9 takes up the first of them.

Chapter 5

The Gaussian Chain: Exact Score, Belief Propagation, and Locality

This chapter answers RQ1, RQ2, and the Gaussian half of RQ4. For the stationary AR(1) chain under the OU channel we compute the joint score exactly by two independent routes (linear algebra and Bayesian smoothing), characterise how its coupling structure evolves with diffusion time, derive the belief-propagation computation of the same score and identify it with the Kalman filter and smoother, and quantify how rapidly the influence of remote observations decays. Everything in this chapter is exact unless explicitly marked as a numerical check; each boxed formula is covered by a named check of the audit suite (Appendix C).

5.1 The joint law and the object of study

A data point is a trajectory $a = (a_0, \dots, a_{L-1})$ of the stationary AR(1) chain (2.2) with $\sigma_\eta^2 = 1 - \alpha^2$. Each frame passes independently through the OU channel (2.5) for a time t ,

$$x_k = e^{-t} a_k + \sqrt{\Delta_t} \xi_k, \quad \xi_k \sim \mathcal{N}(0, 1) \text{ i.i.d.}, \quad (5.1)$$

and the object of study is the joint score of the noisy sequence,

$$S(x, t) = \nabla_x \log P_t(x) \in \mathbb{R}^L, \quad P_t(x) = \int_{\mathbb{R}^L} P_0(a) \prod_{k=0}^{L-1} \mathcal{N}(x_k; e^{-t} a_k, \Delta_t) da. \quad (5.2)$$

The gradient is taken with respect to all L noisy frames simultaneously, under the law of the whole noisy trajectory (Section 2.4).

5.2 The clean law: dense covariance, sparse precision

The chain factorisation (2.3) gives the clean joint density

$$P_0(a) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{L-2} \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2) = \mathcal{N}(a; 0, \Sigma_0). \quad (5.3)$$

Proposition 5.1 (Stationary covariance). $(\Sigma_0)_{ij} = \alpha^{|i-j|}$: a dense, symmetric Toeplitz matrix with geometrically decaying off-diagonals, $\gamma(d) = \alpha^{|d|}$.

Proposition 5.2 (Tridiagonal precision). $Q_0 := \Sigma_0^{-1}$ is exactly tridiagonal:

$$(Q_0)_{kk} = \begin{cases} \frac{1}{\sigma_\eta^2}, & k \in \{0, L-1\}, \\ \frac{1+\alpha^2}{\sigma_\eta^2}, & 0 < k < L-1, \end{cases} \quad (Q_0)_{k,k\pm 1} = -\frac{\alpha}{\sigma_\eta^2}, \quad (5.4)$$

and every entry at distance ≥ 2 vanishes identically.

Toolbox 5.1 — Deriving the tridiagonal precision from the log-density

For a centred Gaussian, the precision is the Hessian of the negative log-density. From (5.3),

$$-2 \log P_0(a) = \frac{a_0^2}{\sigma_0^2} + \sum_{k=0}^{L-2} \frac{(a_{k+1} - \alpha a_k)^2}{\sigma_\eta^2} + \text{const}, \quad \sigma_0^2 = 1.$$

Read off the quadratic form term by term. The only term coupling a_k and a_{k+1} is the transition $(a_{k+1} - \alpha a_k)^2 / \sigma_\eta^2$, with cross coefficient $-2\alpha / \sigma_\eta^2$, hence precision entry $-\alpha / \sigma_\eta^2$; no term couples variables at distance ≥ 2 , so those entries are exactly zero. An

interior a_k appears in its left transition with coefficient $1/\sigma_\eta^2$ and in its right transition with α^2/σ_η^2 , summing to $(1 + \alpha^2)/\sigma_\eta^2$. At the left boundary the prior term $1/\sigma_0^2 = (1 - \alpha^2)/\sigma_\eta^2$ plus the right transition α^2/σ_η^2 give $1/\sigma_\eta^2$; the right boundary is symmetric. $\square$

Remark 5.3 (Tridiagonality is Markovianity, not Gaussianity). A zero in the precision matrix is a conditional independence given all other variables; by the Hammersley–Clifford theorem (Section 3.1), factorisation over the cliques of a graph and conditional independence with respect to that graph are the same property. Gaussianity entered the derivation only to identify the precision with the Hessian of $-\log P_0$; the support of that Hessian, nearest neighbours only, comes from the chain factorisation alone. Any Markov chain with smooth transitions has this sparsity pattern; the Gaussian case makes the entries constants.

The covariance is dense (every frame correlates with every other) while the precision is tridiagonal: correlation is global, conditional dependence is local. The score is governed by the second object.

5.3 The score, two ways

5.3.1 Linear algebra

Proposition 5.4 (Noised law). $P_t(x) = \mathcal{N}(x; 0, \Sigma_t)$ with

$$\boxed{\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I_L} \quad (5.5)$$

This is (2.6). The diagonal is $e^{-2t} + \Delta_t = 1$ at every t (variance preservation); only the off-diagonal entries, $\alpha^{|i-j|} e^{-2t}$, depend on the diffusion time, because the noise of different frames is independent. Since $\log P_t(x) = -\frac{1}{2} x^\top \Sigma_t^{-1} x + \text{const}$,

$$\boxed{S(x, t) = -Q_t x, \quad Q_t := \Sigma_t^{-1}.} \quad (5.6)$$

The joint score is linear in x , the hallmark of Gaussianity, and its entire structure is the noisy precision matrix Q_t : which frames couple, how strongly, and at which t , are questions about this one matrix, answered in Sections 5.4 and 5.7.

5.3.2 Bayesian inference

The score–posterior identity (2.10) expresses the same score through the posterior mean of the clean chain,

$$S_k(x, t) = \frac{e^{-t} \mathbb{E}[a_k | x] - x_k}{\Delta_t}, \quad (5.7)$$

and the posterior is Gaussian with parameters used throughout:

Proposition 5.5 (Posterior of the clean chain). $P(a | x) = \mathcal{N}(a; m, J^{-1})$ with

$$\boxed{J = \frac{e^{-2t}}{\Delta_t} I + Q_0, \quad J m = h := \frac{e^{-t}}{\Delta_t} x.} \quad (5.8)$$

Toolbox 5.2 — Posterior precision and field, by completing the square

Bayes' rule with the likelihood (5.1):

$$\begin{aligned} -2 \log P(a|x) &= a^\top Q_0 a + \sum_k \frac{(x_k - e^{-t} a_k)^2}{\Delta_t} + \text{const} \\ &= a^\top \left(Q_0 + \frac{e^{-2t}}{\Delta_t} I \right) a - 2 \frac{e^{-t}}{\Delta_t} x^\top a + \text{const}. \end{aligned}$$

Matching against the generic form $a^\top J a - 2h^\top a$ gives (5.8). Two structural observations. (i) The likelihood is diagonal in a : observing each frame through its own channel adds the channel precision e^{-2t}/Δ_t to every diagonal entry of the prior precision and touches nothing else. (ii) Consequently J is tridiagonal at every t : the *posterior* of the clean chain remains a nearest-neighbour Markov field; what fills up with t is the precision Q_t of the noisy marginal law. Keeping J (tridiagonal, posterior) and Q_t (dense, noisy marginal) distinct prevents much confusion. $\square$

Substituting $m = J^{-1}h$ into (5.7) must return $-Q_t x$; the operator identity behind this

is Woodbury’s (Appendix A), and the audit suite checks the two routes agree to 2.7×10^{-15} across sweeps of (L, α, t) . The agreement is an end-to-end check of two computations built from different formulas.

5.3.3 The smallest example, $L = 2$

For $L = 2$, with $r := \alpha e^{-2t}$,

$$\Sigma_t = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} \implies S_0(x, t) = -\frac{x_0 - r x_1}{1 - r^2}, \quad S_1(x, t) = -\frac{x_1 - r x_0}{1 - r^2}. \quad (5.9)$$

By contrast, each x_k is marginally $\mathcal{N}(0, 1)$ at every t (variance preservation), so the per-frame marginal score is $-x_k$: independent of t and of the other frame, it carries no information about the dynamics. The joint score contains the cross term $-rx_1/(1 - r^2)$: the denoiser of frame 0 uses evidence from frame 1 with a weight that equals α at $t = 0$ and decays as e^{-2t} . This is the coupling structure whose full lifecycle the next section derives, in miniature.

5.4 The spectral form and the lifecycle of the precision

How does Q_t interpolate between the tridiagonal Q_0 at $t = 0$ and the identity at $t = \infty$? The analysis rests on the spectral theorem and its affine-family corollary (Section D.2): affine images of a symmetric matrix share its eigenbasis, $(cA + dI)u_i = (c\omega_i + d)u_i$. Applying this to the forward map (5.5), which is affine in Σ_0 , gives the representation from which everything in this section follows.

Theorem 5.6 (Spectral form). *Let $\Sigma_0 = U \operatorname{diag}(\omega_i) U^\top$. Then for every $t \geq 0$,*

$$\Sigma_t = U \operatorname{diag}(\lambda_i(t)) U^\top, \quad Q_t = U \operatorname{diag}\left(\frac{1}{\lambda_i(t)}\right) U^\top, \quad \lambda_i(t) = e^{-2t} \omega_i + \Delta_t. \quad (5.10)$$

Diffusion never rotates the problem: the eigenvectors of Σ_t and Q_t are those of Σ_0 , frozen for all t , while each eigenvalue flows along the affine path $\omega_i \mapsto e^{-2t}\omega_i + \Delta_t$, monotonically towards 1. Because Σ_0 is symmetric Toeplitz, its eigenvectors are asymptotically the Fourier modes of the sequence (Section D.2): in frequency space the denoising problem is L independent scalar problems at every diffusion time. Two computational corollaries are recorded for later use: factoring Δ_t out of (5.5) gives the SNR form

$$Q_t = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1}, \quad \gamma_t := e^{-2t}/\Delta_t, \quad (5.11)$$

and multiplying (5.5) by Q_0 and inverting gives the resolvent form

$$Q_t = e^{2t} Q_0 (I + c_t Q_0)^{-1} = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}, \quad c_t := e^{2t} - 1, \quad (5.12)$$

whose Neumann series converges for $\|c_t Q_0\| < 1$, i.e. at small t .

5.4.1 Losing tridiagonality: when, how, and how much

Sparsity is a property of a matrix in a basis. In the eigenbasis the problem is diagonal at all times; in the frame basis, the operationally meaningful one, the zeros of Q_0 are a cancellation that diffusion destroys. The destruction is quantifiable, and it answers three questions: when tridiagonality is lost, how, and how much structure survives at each t .

When: immediately. By (5.10) every entry of Q_t is analytic in t . The distance- d entries, identically zero at $t = 0$ for $d \geq 2$, are non-zero for every $t > 0$: no exact zero survives any positive diffusion time.

How: one diagonal per order in t . The loss is nevertheless graded, and the resolvent form gives the exact rate at which each diagonal switches on.

Theorem 5.7 (Band-fill leading order). *For small t and frame distance $d \geq 1$,*

$$(Q_t)_{i,i+d} = (-1)^{d-1} (2t)^{d-1} (Q_0^d)_{i,i+d} + O(t^d). \quad (5.13)$$

Toolbox 5.3 — Band fill from the Neumann series

In the resolvent form (5.12), substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$:

$$Q_t = (1 + O(t)) \left[Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \dots \right].$$

Q_0 is tridiagonal, so Q_0^m has bandwidth m : the entry at distance d receives its first contribution from the single term $m = d$, with prefactor $(-c_t)^{d-1} = (-1)^{d-1} (2t)^{d-1} (1 + O(t))$; every later term is $O(t^d)$. Each additional frame of coupling range costs one power of t : the band fills outward, one diagonal per order. $\square$

Remark 5.8 (The prefactor carries no factorial, and why that is checked). Equation (5.13) has no $1/(d-1)!$: the leading distance- d term is a single Neumann term, with no combinatorial sum behind it. The distinction is easy to miss and worth verifying numerically, because slope fits in $\log t$ are blind to constant prefactors — a wrong constant survives every check on the exponent. The audit therefore tests the coefficient itself, and it separates the two candidates cleanly: the no-factorial form matches the measured $(Q_t)_{i,i+d}/t^{d-1}$ to better than 0.9% for every $d \in \{1, \dots, 5\}$ (at $\alpha = 0.9$, $L = 25$), while a factorial version would be wrong by 86% already at $d = 3$. Every prefactor in this thesis is audited on the same grounds, not only every exponent.

How much: the off-tridiagonal mass. Define the off-tridiagonal Frobenius mass of Q_t : the norm of everything outside the band that Markov structure would allow. At small t the band-fill law makes it rise from zero, dominated by the $d = 2$ diagonal, linearly in t . At large t , expanding the inverse of $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ gives

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}), \quad (5.14)$$

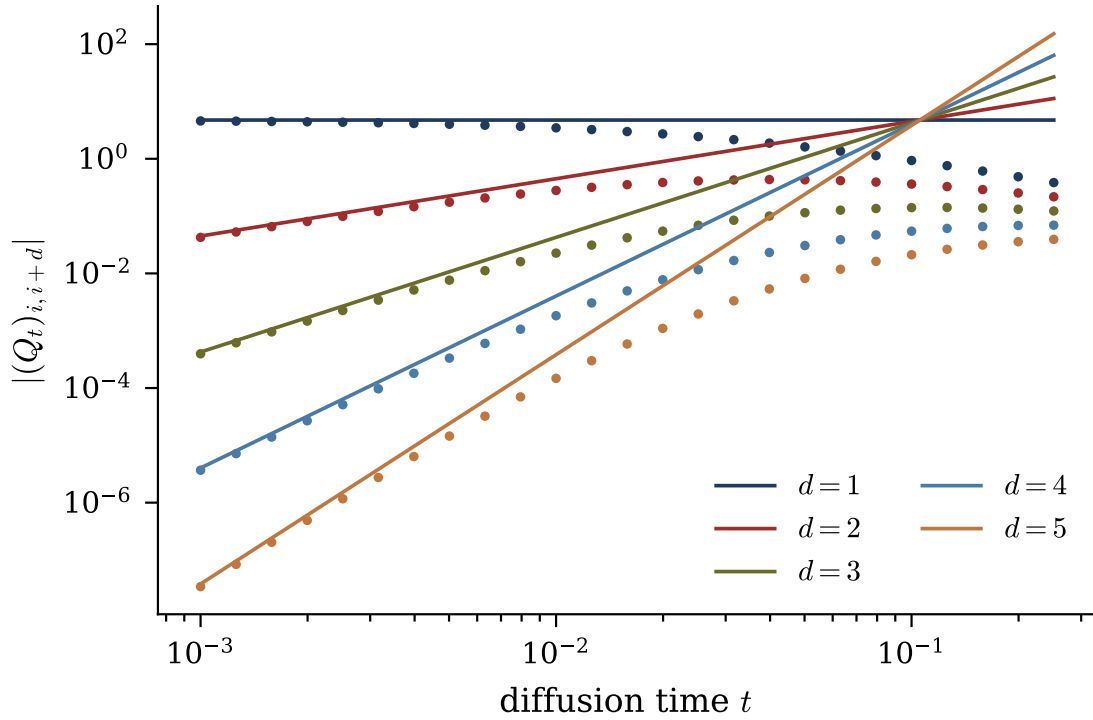


Figure 5.1: Band-fill law. Measured $|(Q_t)_{i,i+d}|$ against t (dots) and the leading order (5.13) (solid lines, no fitted parameters), for $\alpha = 0.9$, $L = 25$, $d = 1, \dots, 5$. The slopes are $d - 1$; agreement at small t is at the percent level for every d .

so $\|Q_t - I\|_F \simeq e^{-2t} \|\Sigma_0 - I\|_F$: all structure, off-tridiagonal and tridiagonal alike, decays at the rate e^{-2t} , and the score approaches the isotropic $-x$. In between, the off-tridiagonal mass peaks (Figure 5.2): the noisy sequence is most non-Markov at intermediate diffusion times. The lifecycle in one line: tridiagonal at $t = 0$; densifying at rate t^{d-1} per diagonal d ; maximally coupled at intermediate t ; isotropic as $t \rightarrow \infty$ at rate e^{-2t} .

5.5 Belief propagation on the chain

The matrix route inverts an $L \times L$ matrix, a global computation over the whole sequence. The score-posterior identity says the same object is a posterior expectation, and the posterior has a structure that supports local computation. This section develops that computation once, in full.

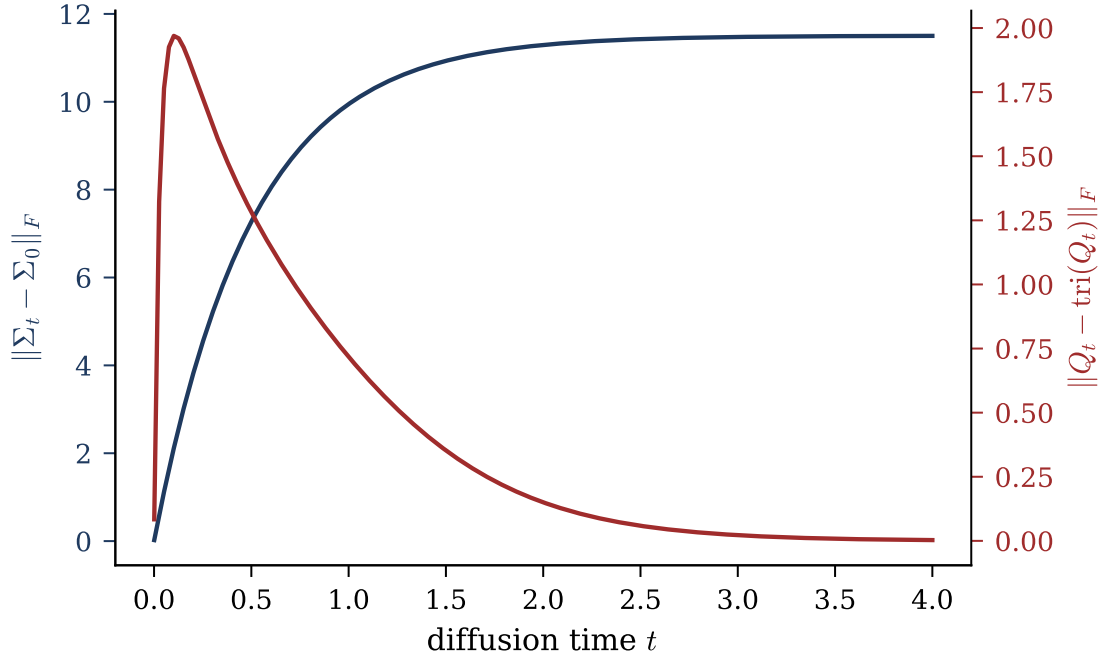


Figure 5.2: Structure across diffusion time ($\alpha = 0.8$, $L = 40$). $\|\Sigma_t - \Sigma_0\|_F$ (navy) grows monotonically as the covariance is driven to the identity; the off-tridiagonal Frobenius mass of Q_t (red) rises from zero, peaks at intermediate t , and decays to zero: the band fills, then empties.

5.5.1 The factor graph of the posterior

For any Markov-chain prior and per-frame OU evidence, Bayes' rule gives

$$p(a|x) \propto \psi_{\text{pr}}(a_0) \prod_{k=0}^{L-2} \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \prod_{k=0}^{L-1} \psi_{\text{ob}}^{(k)}(a_k), \quad \psi_{\text{ob}}^{(k)}(a_k) = \mathcal{N}(x_k; e^{-t}a_k, \Delta_t), \quad (5.15)$$

a hidden Markov model (Rabiner, 1989): a latent chain observed through independent per-frame noise. By (2.10), the k -th score component requires exactly the smoothing marginal $\mathbb{E}[a_k|x]$ of (5.15).

The factor graph of this posterior (Section 3.2) has L variable nodes, $2L$ factor nodes (one prior, $L-1$ transitions, L observations; the data x_k are constants clamped inside their factors) and $3L - 1$ edges. A connected graph with one fewer edge than nodes is a tree: here, a spine of variables and transition factors with one evidence leaf per variable.

Proposition 5.9. *For any Markov-chain prior and any per-frame observation model, the posterior $p(a | x)$ factorises over a tree-shaped factor graph, and all L smoothing marginals are computed exactly by one forward and one backward sweep of sum-product messages.*

Proposition 5.9 counts sweeps, not work: it says nothing about how the messages are represented, and for continuous variables each message is a function (Section 3.3). Keeping tree-exactness of the algorithm (a graph property) separate from representability of the messages (a property of the model’s function class) is the conceptual hinge of this thesis; the Gaussian case closes the gap, and Chapter 6 measures what happens when it no longer does.

5.5.2 The sum-product recursion, with the bookkeeping explicit

Specialising the sum-product rules (3.3) to this tree, the schedule collapses into one forward and one backward sweep (Theorem 3.3), and the bookkeeping deserves care.

Proposition 5.10 (Chain sweeps, Convention A). *Define messages*

$$\mu_{\rightarrow 0}(a_0) := \psi_{\text{pr}}(a_0), \quad \mu_{\rightarrow k+1}(a_{k+1}) := \int \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\rightarrow k}(a_k) da_k, \quad (5.16)$$

$$\mu_{\leftarrow L-1}(a_{L-1}) := 1, \quad \mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\text{tr}}^{(k-1)}(a_{k-1}, a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k. \quad (5.17)$$

Then for every k ,

$$p(a_k | x) \propto \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) : \quad (5.18)$$

the forward message into a_k carries the evidence $x_0, \dots, x_{k-1}$ only, the backward message carries $x_{k+1}, \dots, x_{L-1}$ only, and the local evidence enters once, at combination time.

The proof is direct marginalisation: integrating (5.15) over all a_j , $j \neq k$, splits at a_k by the Markov property into independent left and right halves, and peeling the innermost variable of each half yields the recursions.

Remark 5.11 (The double-counting trap). An alternative convention absorbs the local evidence into the forward message, $\mu_{\rightarrow k}^{\text{B}} = \mu_{\rightarrow k} \cdot \psi_{\text{ob}}^{(k)}$; the product (5.18) then counts x_k

twice and must be corrected by dividing one factor back out. The cost of getting this wrong is not subtle, which is why it is worth stating: under Convention A the BP score matches the matrix score to 10^{-14} , and under the naive alternative the disagreement is of order one. Every message in this thesis is therefore stated with its evidence content explicit.

5.6 The Gaussian closure and the Kalman smoother

For the Gaussian chain, the two closure lemmas of Toolbox D.1 are the only facts needed: the recursions (5.16)–(5.17) perform exactly one product and one affine marginalisation per step. The local evidence, read as a function of a_k , is itself Gaussian, $\psi_{\text{ob}}^{(k)}(a_k) \propto \mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$: a pseudo-observation at the deconvolved location $e^t x_k$ with precision e^{-2t}/Δ_t , the pinning strength that appears throughout this chapter.

Theorem 5.12 (Gaussian closure of BP on the chain). *For the Gaussian AR(1) chain under the OU channel, every BP message and every belief is exactly Gaussian, at every node, for every (L, α, t) . Each message is carried by two numbers, each update is a few arithmetic operations, and the two sweeps compute the exact posterior marginals, hence the exact joint score. Here, and only here, the sweep count of Proposition 5.9 translates into an actual computation of fixed, small cost per frame.*

Derivation. Induction along each sweep. The initial messages are Gaussian (the prior; the flat message is the zero-precision limit of the family). Each step of (5.16)–(5.17) is one application of Lemma 1 of Toolbox D.1 (absorb the local evidence) followed by one of Lemma 2 (push through the transition); the combination (5.18) is Lemma 1 twice. Exactness follows from tree exactness of sum-product (Theorem 3.3). $\square$

Remark 5.13 (Closure is algebra, not a central limit theorem). The closure just proved holds at node degree two exactly as it would at any degree: it is algebra, not an aggregation effect, as discussed in Section D.3. The central-limit argument matters only for the per-node closures of statistical physics (Section D.4), and on the chain those fail in a way

that can be made exact, with a breakdown time $t_c(\alpha)$ and a critical coupling $\alpha_c = \sqrt{2} - 1$ (Appendix B).

Executing the Gaussian sweeps in mean–variance arithmetic reproduces, update by update, the classical algorithms of linear estimation.

Toolbox 5.4 — The Kalman filter, derived from the BP sweep

Write the forward message in Convention A as $\mu_{\rightarrow k} = \mathcal{N}(a_k; m_{k|k-1}, P_{k|k-1})$, the predicted state given past observations. One BP step does two things.

Update (absorb the evidence; Lemma 1). Multiplying by the pseudo-observation $\mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$ gives the filtered Gaussian $\mathcal{N}(a_k; m_{k|k}, P_{k|k})$ with

$$\begin{aligned} \frac{1}{P_{k|k}} &= \frac{1}{P_{k|k-1}} + \frac{e^{-2t}}{\Delta_t}, \\ m_{k|k} &= P_{k|k} \left(\frac{m_{k|k-1}}{P_{k|k-1}} + \frac{e^{-t} x_k}{\Delta_t} \right) = m_{k|k-1} + G_k (e^t x_k - m_{k|k-1}), \end{aligned}$$

where $G_k = P_{k|k} e^{-2t} / \Delta_t \in (0, 1)$ is the Kalman gain: the filtered mean is the prediction corrected by a gain-weighted innovation (Kalman, 1960).

Predict (push through the dynamics; Lemma 2).

$$m_{k+1|k} = \alpha m_{k|k}, \quad P_{k+1|k} = \alpha^2 P_{k|k} + \sigma_\eta^2.$$

The forward BP sweep is the Kalman filter, message by message.

Smoothing (combine both directions). The belief (5.18) multiplies the filtered forward Gaussian by the backward message (Lemma 1 once more), producing the posterior marginal given all observations, $\mathcal{N}(a_k; m_{k|L-1}, P_{k|L-1})$; carrying out the same algebra for the backward recursion yields the fixed-interval smoother of Rauch et al. (1965). The two-sweep BP schedule on the chain is the Kalman filter followed by the RTS smoother, exactly as the generic correspondence of Section D.3 predicts. $\square$

Combining with the score–posterior identity answers RQ2: the k -th component of the

joint diffusion score is an affine function of the Kalman-smoothed estimate of frame k ,

$$S_k(x, t) = \frac{e^{-t} m_{k|L-1} - x_k}{\Delta_t}. \quad (5.19)$$

For Markov sequences, score computation is a smoothing problem, and the classical theory of linear smoothing applies directly.

Numerical agreement of the two routes. The audit suite runs BP against the matrix route over 80 configurations ($L \in \{2, 3, 5, 8, 16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$, random x): maximum disagreement 1.3×10^{-15} on posterior means, 3.9×10^{-15} on variances, 1.2×10^{-14} on scores, i.e. double-precision rounding accumulated along the recursion. An independent implementation, written from scratch against the recursion specification alone, reproduces the same agreement (Appendix C). One qualitative detail from a worked $L = 3$ instance: at $x = (1.2, -0.4, 0.8)$ the score component S_1 is positive although $x_1 < 0$, because the neighbours pull the posterior mean of the middle frame positive. No per-frame marginal score can produce this; it is inter-frame evidence sharing in numbers.

5.7 The bulk in closed form, and the price of locality

For a long chain the interior (bulk) is translation invariant and can be solved exactly. By Proposition 5.5 the posterior precision in the bulk is tridiagonal Toeplitz with

$$\boxed{J_d = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2}{\sigma_\eta^2}, \quad \beta = -\frac{\alpha}{\sigma_\eta^2}, \quad \sigma_\eta^2 = 1 - \alpha^2.} \quad (5.20)$$

The diagonal J_d is the sum of the evidence precision e^{-2t}/Δ_t (divergent as $1/2t$ at small t , vanishing at large t) and the constant prior precision; the coupling β is pure prior. The dimensionless ratio $|\beta|/J_d \in (0, \frac{1}{2})$ controls everything below.

Theorem 5.14 (Exact bulk variance and geometric covariance). *In the bulk of a long*

chain, the posterior variance and covariance are

$$V = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \quad (J^{-1})_{i,i+d} = q^d V, \quad q = \frac{J_d - \sqrt{J_d^2 - 4\beta^2}}{2|\beta|} \in (0, 1). \quad (5.21)$$

Toolbox 5.5 — Transfer-matrix inversion of a tridiagonal Toeplitz matrix

Seek the inverse row in geometric form, $(J^{-1})_{i,i+d} = Cq^d$ for $d \geq 0$, symmetric in $\pm d$. The defining relation $\sum_j J_{ij}(J^{-1})_{j,i+d} = \delta_{0d}$ reads, for $d \geq 1$,

$$\beta Cq^{d-1} + J_d Cq^d + \beta Cq^{d+1} = 0 \iff \beta q^2 + J_d q + \beta = 0,$$

a quadratic whose root inside the unit disc is $q = (J_d - \sqrt{J_d^2 - 4\beta^2})/(2|\beta|)$ (for $\alpha > 0$ all inverse entries are positive; for $\alpha < 0$ they alternate in sign with the same modulus). The $d = 0$ relation fixes the prefactor: $J_d C + 2\beta Cq = 1$, and since $2\beta q = -(J_d - \sqrt{J_d^2 - 4\beta^2})$, $C = 1/\sqrt{J_d^2 - 4\beta^2} = V$. Existence of the real square root requires $J_d > 2|\beta|$, which always holds here:

$$J_d - 2|\beta| = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2 - 2|\alpha|}{\sigma_\eta^2} = \frac{e^{-2t}}{\Delta_t} + \frac{(1 - |\alpha|)^2}{\sigma_\eta^2} > 0$$

for every $|\alpha| < 1$ and $t > 0$. This positivity margin is also what guarantees the belief-propagation fixed point exists for every coupling and diffusion time (Appendix B).

□

The number $q(\alpha, t)$ measures how far evidence travels along the chain: the posterior correlation length is $\xi = 1/\log(1/q)$. Its limits are

$$\begin{aligned} t \rightarrow 0 : \quad q &\approx \frac{|\beta|}{J_d} \rightarrow 0 \quad (\text{each frame pinned by its own observation}); \\ t \rightarrow \infty : \quad q &\rightarrow |\alpha| \quad (\text{posterior} \rightarrow \text{prior}), \end{aligned} \quad (5.22)$$

the second because $q \rightarrow ((1+\alpha^2) - (1-\alpha^2))/(2|\alpha|) = |\alpha|$ when the evidence term vanishes. As diffusion time runs, the posterior correlation length climbs monotonically from 0 to

the prior's own $1/\log(1/|\alpha|)$.

5.7.1 The locality law

RQ4 asks how rapidly the influence of remote observations decays. Define the local estimator of radius r : the posterior mean of a_k given only the observations within distance r , $m_k^{(r)} = \mathbb{E}[a_k | x_{k-r}, \dots, x_{k+r}]$. Two facts make the comparison clean. Any contiguous window of a stationary AR(1) chain is itself a stationary AR(1) chain, so $m_k^{(r)}$ is exact on its window; the only approximation is the truncation of range. And both estimators are linear in x , so the error $m_k - m_k^{(r)} = e^\top x$ has a closed-form RMS over the data law, $\text{RMS}_r = \sqrt{e^\top \Sigma_t e}$, a deterministic number with no sampling involved.

Proposition 5.15 (Geometric locality: influence coefficients). *In the bulk, the influence of the observation at distance d on the exact posterior mean is $(J^{-1})_{k,k\pm d} \propto q(\alpha, t)^d$, exactly, by Theorem 5.14. Truncating the evidence at radius r therefore discards a leading term of order q^{r+1} .*

What Proposition 5.15 controls is the size of the *discarded* evidence, and that part is exact. It does not by itself prove that the radius- r estimator's error inherits the rate: $m_k^{(r)}$ is the posterior mean of a *re-normalised* model on the window, so truncation changes the boundary conditions as well as removing the leading coefficient, and bounding the difference between those two effects is not done here.

The rate is instead confirmed numerically, and closely. Fitted slopes of $\log \text{RMS}_r$ over $r = 2, \dots, 13$ at the centre of an $L = 121$ chain match $\log q$ within 0.2% at $t \in \{0.1, 0.5, 2.0\}$ ($\alpha = 0.8$); see Figure 5.3, where the dashed lines carry the predicted slope with no fitting. The claim this thesis makes is therefore: an exact geometric law for the influence coefficients, and a measured geometric law with the same rate for the windowed estimator's error, over the stated range of r . A two-sided bound establishing the second from the first is left open.

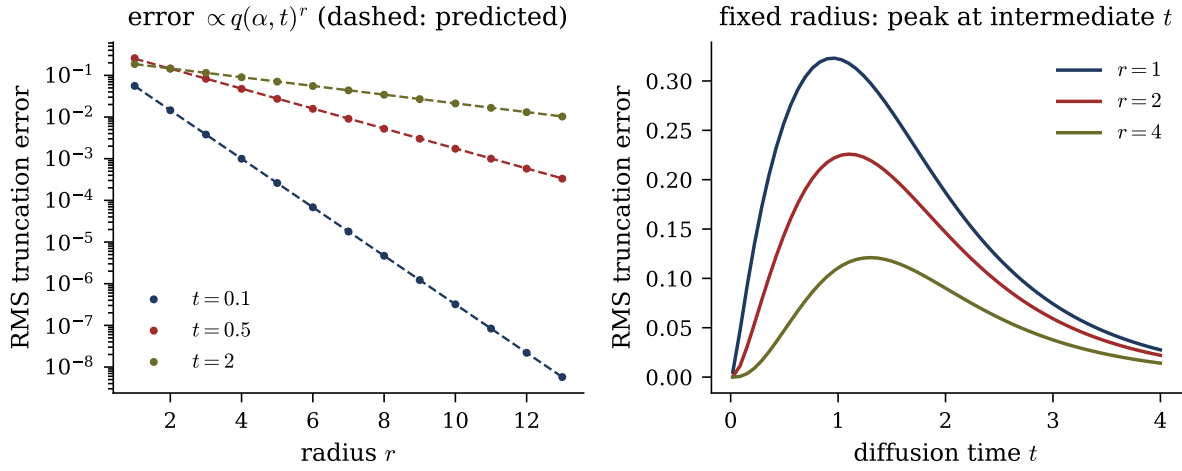


Figure 5.3: Locality of the joint score ($\alpha = 0.8$, $L = 121$, deterministic RMS over the data law). Left: exact truncation error of the radius- r local estimator (dots) at three diffusion times; the dashed lines have the predicted slopes $q(\alpha, t)^r$ from (5.21), with no fitting. Right: the same error at fixed radius as a function of t : it vanishes at both ends and peaks at intermediate t , where the band of Q_t is fullest.

5.7.2 Truncating the matrix versus truncating the inference

A truncated score can be built in two inequivalent ways. *Truncate the answer*: band the matrix, $\hat{S} = -Q_t^{(b)}x$ with entries beyond $|i - j| \leq b$ zeroed. *Truncate the algorithm*: let messages travel at most r hops, i.e. run exact inference on the window $x_{k-r}, \dots, x_{k+r}$ (the local estimator above). Both are linear in x , so their RMS errors are again deterministic. Table 5.1 compares them at equal locality budget ($b = r = 1$, $L = 40$, $\alpha = 0.8$):

Table 5.1: Truncating the answer against truncating the algorithm, at equal locality budget ($b = r = 1$, $L = 40$, $\alpha = 0.8$). Both estimators are linear in x , so both errors are deterministic rather than sampled. The cost of locality peaks at intermediate diffusion times and the ordering reverses nowhere.

relative RMS score error	$t = 0.05$	$t = 0.3$	$t = 1$	$t = 3$
banded matrix ($b = 1$)	0.210	0.302	0.135	0.0035
truncated inference ($r = 1$)	0.123	0.235	0.130	0.0035

Two observations. The cost of locality is non-monotone in t and peaks at intermediate times, consistent with Section 5.4.1. And at small t truncating the algorithm beats truncating the answer (12.3% against 21.0% at $t = 0.05$): the windowed inference solves a

correctly normalised problem on its window, while zeroing entries of an inverse produces a matrix that is the inverse of nothing and misstates the diagonal. The two coincide at large t , where the off-diagonal content vanishes. Either error is measured to decay geometrically in the allowed range at the rate $q(\alpha, t)$ of Proposition 5.15.

5.8 Status of the results

Classified in the vocabulary of Section 1.5. Exact, by derivation: the score formulas (5.6) and (5.7), the spectral form (5.10), the band-fill law (5.13) as a small- t asymptotic, the large- t law (5.14), tree exactness and Gaussian closure of BP, the Kalman identification, and the bulk formulas (5.20)–(5.21), including the q^d decay of the influence coefficients. Numerical, within stated grids and tolerances: the agreement of the two score routes, the audited band-fill coefficients, the locality slopes of the *windowed estimator* (whose rate is measured rather than derived, see Proposition 5.15), and the truncation table. Interpretation, limited to this model: within the Gaussian benchmark, a local score computation with receptive field $r \gtrsim \xi \log(1/\varepsilon)$, $\xi = 1/\log(1/q)$, achieves relative error ε , and the requirement is most demanding at intermediate diffusion times; whether trained score networks on real sequence data behave accordingly is not tested here.

Chapter 6

Beyond Gaussianity: Laplace Innovations and the Gaussian-Message Approximation

Every exact result of Chapter 5 rests on the closure of the Gaussian family under the two operations the model performs, multiplying densities and marginalising. This chapter removes that property in the most controlled way available and answers RQ3. We keep the AR(1) recursion, the OU channel, and the factor-graph structure, and change only the innovation law from Gaussian to Laplace: heavy-tailed and kinked at the origin, yet log-concave and analytically tractable, it is a minimal step outside the Gaussian family, so every deviation we find is attributable to non-Gaussianity alone. The chapter first establishes what remains exact (the score-posterior identity, the tree structure, one boundary integral, and a differential characterisation of the $L = 1$ marginal), then locates precisely where closed forms stop, and finally measures, against a validated numerical reference, what a Gaussian message family costs.

6.1 What changes, and what is preserved

The Laplace AR(1) chain replaces the Gaussian innovations of (2.2) with Laplace ones, keeping the initial law and the second-order structure fixed:

$$a_0 \sim \mathcal{N}(0, 1), \quad a_k = \alpha a_{k-1} + \varepsilon_k, \quad \varepsilon_k \sim \text{Laplace}(0, b) \text{ i.i.d.}, \quad k = 1, \dots, L-1, \quad (6.1)$$

observed through the same per-frame OU channel. The innovation scale is *not* free: $b = \sqrt{(1 - \alpha^2)/2}$, so that $\text{Var}(\varepsilon_k) = 1 - \alpha^2$ and the chain has unit marginal variance and $\text{Cov}(a_j, a_k) = \alpha^{|j-k|}$, exactly as in the Gaussian case. This is what makes the comparison in this chapter a comparison: the Gaussian and Laplace chains agree on every second moment and differ *only* beyond them, the innovation's excess kurtosis (0 against +3) being the single non-Gaussianity knob. The same convention — Gaussian initial state, variance-matched innovations — is used for every non-Gaussian family in this thesis, and carries forward unchanged to the estimated chain of (7.1). Three structural facts survive unchanged, because none of them used Gaussianity of the prior. The score-posterior identity (2.10) holds for any p_0 . The posterior factorises over the same tree-shaped factor graph (Proposition 5.9), so the sum-product recursion (5.16)–(5.17) still computes exact smoothing marginals in one forward and one backward sweep. And the OU likelihood factors are still Gaussian in a_k . What is lost is representability: the messages are no longer Gaussian, no finite-dimensional family we know of is closed under the updates, and the cost of an update is no longer bounded. The score also ceases to be linear in x , which is the first thing to make precise.

6.2 One frame: the score becomes nonlinear

For $L = 1$ there is no chain; the model isolates the ingredient “non-Gaussian prior meets Gaussian channel”. Let $a \sim \text{Laplace}(0, b)$, $p_0(a) = e^{-|a|/b}/2b$, and write $\mu := e^{-t}$. The

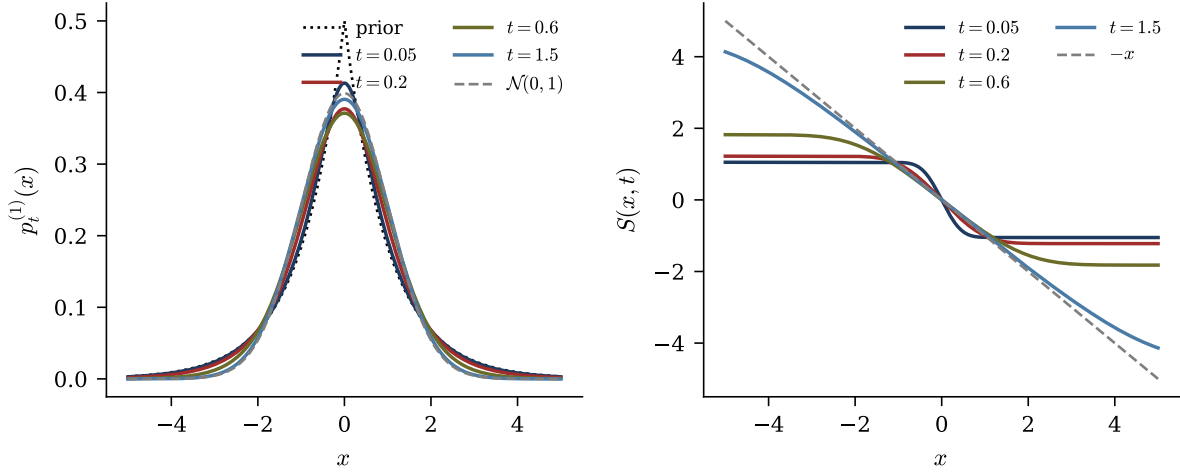


Figure 6.1: The Laplace $L = 1$ problem ($b = 1$). Left: noised marginal $p_t^{(1)}$ at increasing diffusion times, from the kinked Laplace prior (dotted) to the Gaussian attractor (dashed). Right: the score at the same times; the dashed line is the attractor $-x$. The nonlinearity lives near $x = 0$ and flattens as t grows; the limits are the exact statements (6.3). Curves computed by deterministic quadrature.

noised marginal is the convolution

$$p_t^{(1)}(x) = \int p_0(a) \mathcal{N}(x; \mu a, \Delta_t) da. \quad (6.2)$$

As $t \rightarrow 0$ the Gaussian kernel concentrates and $p_t^{(1)} \rightarrow p_0$; as $t \rightarrow \infty$, $p_t^{(1)} \rightarrow \mathcal{N}(0, 1)$, the OU attractor. By the score-posterior identity, $S(x, t) = (\mu \mathbb{E}[a|x] - x)/\Delta_t$; for a Laplace prior the conditional mean is a nonlinear function of x , and the score inherits the nonlinearity. Its limits are exact: for every fixed $x \neq 0$,

$$t \rightarrow 0 : \quad S(x, t) \longrightarrow \partial_x \log p_0(x) = -\frac{\text{sign}(x)}{b}, \quad t \rightarrow \infty : \quad S(x, t) \longrightarrow -x. \quad (6.3)$$

At small t the score converges pointwise to the sign-shaped score of the prior, with the nonlinearity concentrated in the kink near $x = 0$; at large t it flattens into the linear attractor score $-x$ (Figure 6.1). The audit verifies the posterior-expectation route against central finite differences of $\log p_t^{(1)}$ computed by quadrature: maximum disagreement 1.19×10^{-7} over a grid of (b, t, x) , at the truncation floor of the finite difference.

6.2.1 A differential characterisation of the noised marginal

The channel writes x as a sum of independent variables, so (6.2) is a convolution, which factorises in Fourier space:

$$\hat{p}_t^{(1)}(k) = \frac{1}{1 + b^2 \mu^2 k^2} e^{-\Delta_t k^2/2}, \quad (6.4)$$

the product of the (rescaled) Laplace and Gaussian characteristic functions.

Theorem 6.1 (Screened Poisson equation). *The Laplace noised marginal solves the linear, constant-coefficient ODE*

$$\boxed{(1 - b^2 \mu^2 \partial_x^2) p_t^{(1)}(x) = G_{\Delta_t}(x)}, \quad G_{\Delta_t}(x) = \frac{e^{-x^2/2\Delta_t}}{\sqrt{2\pi\Delta_t}}. \quad (6.5)$$

Toolbox 6.1 — From the characteristic function to the screened Poisson equation

Multiply (6.4) by $(1 + b^2 \mu^2 k^2)$:

$$(1 + b^2 \mu^2 k^2) \hat{p}_t^{(1)}(k) = e^{-\Delta_t k^2/2},$$

whose right-hand side is the characteristic function of G_{Δ_t} . Multiplication by k^2 in Fourier space is $-\partial_x^2$ in real space, so the polynomial $(1 + b^2 \mu^2 k^2)$ becomes the operator $(1 - b^2 \mu^2 \partial_x^2)$, and (6.5) follows by inverse transformation. The audit checks (6.4) against direct quadrature of the characteristic function to 2.9×10^{-6} , the floor of oscillatory quadrature. $\square$

Equation (6.5) states that the noised marginal is the solution of a screened Poisson equation with a Gaussian source, with screening length $b\mu = be^{-t}$: the memory of the prior's tails shrinks as e^{-t} . The deviation from Gaussianity is governed by a single local differential operator whose strength dies with diffusion time.

One further exact consequence identifies what a Gaussian approximation must get

wrong. Differentiating the score-posterior identity once more,

$$H_t(x) := -\partial_x^2 \log p_t^{(1)}(x) = \frac{1}{\Delta_t} \left(1 - \frac{\mu^2}{\Delta_t} \text{Var}[a|x] \right). \quad (6.6)$$

For a Gaussian prior the posterior variance is independent of x and H_t is constant. For the Laplace prior $\text{Var}[a|x]$ depends on x and H_t is a nontrivial function: the breakdown of a constant curvature is the second-order form of the breakdown of score linearity. A Gaussian message is exactly the object that cannot represent an x -dependent curvature, so (6.6) states what the Gaussian closure of Section 6.5 will miss, before it is measured.

6.3 The chain: where closed forms stop

Theorem 6.2 (Joint score from chain messages). *For any chain prior, in particular (6.1), and the OU likelihood, the joint score is, componentwise,*

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k|x] - x_k}{\Delta_t}, \quad \mathbb{E}[a_k|x] = \frac{\int a_k \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k}{\int \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k}, \quad (6.7)$$

with the messages of Proposition 5.10.

This is a complexity statement: the L -dimensional posterior integral is replaced by two one-dimensional message recursions plus one one-dimensional integral per frame. Computational locality survives non-Gaussianity; what does not survive is the collapse of those one-dimensional integrals into closed-form arithmetic. Exactly one such integral has a closed form.

Proposition 6.3 (One-step boundary integral). *For the Laplace transition $M(a'|a) = \frac{1}{2b} e^{-|a' - \alpha a|/b}$ and OU evidence $\mathcal{N}(x'; \mu a', \Delta_t)$,*

$$\int_{\mathbb{R}} M(a'|a) \mathcal{N}(x'; \mu a', \Delta_t) da' = p_t^{(1)}(x' - \mu \alpha a) : \quad (6.8)$$

the first backward message of the chain is the $L = 1$ noised marginal evaluated at a shifted

argument.

The proof is the innovation substitution $r := a' - \alpha a$, which turns the transition into the bare Laplace density at r and the evidence into $\mathcal{N}(x' - \mu\alpha a; \mu r, \Delta_t)$; the integral is then the definition (6.2). The shift $\mu\alpha a$ is what frame a predicts for the next noisy observation, and the whole $L = 1$ analysis, screened Poisson equation included, recycles as the chain's boundary building block.

Remark 6.4 (Why iteration destroys the closed form). Equation (6.8) works because the integrand contains exactly one Laplace kink times exactly one Gaussian. Every subsequent BP step violates this: the incoming message already carries the propagated structure of several frames, so the next integrand contains two or more absolute-value kinks. On each sign-quadrant the integral reduces to half-line Gaussians, but globally it is a sum of error functions whose arguments depend on the conditioning variable non-affinely, and the clean form does not survive one iteration. In the Gaussian chain every step is Gaussian-times-Gaussian and closure propagates forever; this is the precise structural location of the difference between the two models, and the reason the Gaussian projection below is an approximation with an error to be measured, not an identity.

A change of coordinates shows the obstruction is intrinsic, not an artefact of the frame basis. In innovation coordinates $r_0 = a_0$, $r_k = a_k - \alpha a_{k-1}$ (unit Jacobian), the prior factorises into independent variables — r_0 Gaussian and $r_{k \geq 1}$ i.i.d. Laplace — but the likelihood becomes $p(x|r) \propto \exp(-\frac{\mu^2}{2\Delta_t} r^\top B^\top B r + \frac{\mu}{\Delta_t} x^\top B r)$ with B the unit lower-triangular map $a = Br$, and $B^\top B$ is dense. The coupling does not disappear; it migrates from the prior to the likelihood. There is no basis in which both are diagonal at once.

6.4 A validated numerical reference: grid BP

Since functional messages admit no closed form, the exact recursion is approximated numerically: messages are represented by their values on N_g equispaced points in $[-A, A]$ and the update integrals are carried out by trapezoidal quadrature, each update touching

every pair of grid points. This *grid BP* is a numerical approximation of the functional-message recursion, and it can serve as a reference only inside a validated regime, which we establish on the exactly solvable Gaussian chain. There the trapezoid rule on analytic, rapidly decaying integrands converges spectrally, so failure is abrupt rather than gradual: at $N_g = 101$, $A = 8$ the score error sits at the 10^{-15} floor down to $t = 0.02$ and fails only when the likelihood width $\sqrt{2t}$ approaches the grid step. The working configuration for all non-Gaussian experiments is $N_g = 401$, $A = 8$, with a reference solution at $N_g = 801$ and the resolution constraint $\sqrt{2t} \gtrsim 3h$ on the grid spacing $h = 2A/(N_g - 1)$ checked automatically; grid self-convergence error remains 10^4 to 10^7 times smaller than every effect measured below. Within this regime we treat grid BP as the reference score; outside it no claim is made.

Remark 6.5 (Symbols that are easy to swap). Three pairs are worth fixing before the numerical sections use them. The grid resolution is N_g and the number of training *sequences* is N ; they are independent, and every configuration states both. The chain length is L and its internal index is u , never t , which is reserved for diffusion time throughout. And q always means the geometric decay rate $q(\alpha, t)$ of Section 5.7; the innovation variance is σ_η^2 , never q . The companion paper writes the sequence count M where this thesis writes N ; the grid symbol is the same in both.

All score errors are amplified posterior-mean errors: for any two solvers with means $\widehat{m}$ and m_{ref} , the score-posterior identity gives, exactly,

$$\widehat{s} - s_{\text{ref}} = \frac{e^{-t}}{\Delta_t} (\widehat{m} - m_{\text{ref}}), \quad \frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t} \quad (t \downarrow 0). \quad (6.9)$$

The numerical residual of this identity is logged in every experiment and stays at 10^{-15} to 10^{-12} . Equation (6.9) explains in advance why small diffusion times are the difficult regime: whatever an approximation does to the posterior mean is amplified by $1/2t$ in the score.

6.5 The Gaussian-message approximation, measured

The object under test is Gaussian-projected BP: the same sweeps, with every message forced into the Gaussian family by moment matching. This is the algorithm one would deploy when quadrature is unaffordable, and the chain analogue of the per-node closures of statistical physics (Appendix B).

Before measuring its error it is worth identifying exactly what it computes, because the answer turns this from an implementation-specific approximation study into a precise question. Write $\Pi_G f$ for the Gaussian density matching the mean and variance of $f/\int f$.

Theorem 6.6 (Gaussian closure is the LMMSE estimator). *Let moment-projected sum-product be the forward-backward recursion of (5.16)–(5.17) that applies Π_G to each message right after every transition integral, multiplying by the site's unary likelihood before the next. Take the chain (6.1) with $\alpha \neq 0$, Gaussian unary likelihoods, $\mathbb{E}[a_0] = 0$, an innovation law with mean zero and finite variance σ_η^2 , and an initial message that is Gaussian or projected by Π_G . Then its posterior means are exactly those of the covariance-matched Gaussian model:*

$$\hat{a}_{\text{LMMSE}} = \text{Cov}(a, x) \text{Cov}(x)^{-1} x = e^{-t\Sigma_0} \left(e^{-2t\Sigma_0} + \Delta_t I \right)^{-1} x, \quad (6.10)$$

and it returns the same estimator for every innovation law sharing the moments $(0, \sigma_\eta^2)$.

Toolbox 6.2 — Why the projection cannot see past two moments

The recursion is linear in the message except at the projection, and Π_G retains only the first two moments of what it is given. A Gaussian message multiplied by a Gaussian likelihood and pushed through a linear-AR transition integral is Gaussian with mean and variance depending on the incoming mean and variance alone, so by induction along the chain every message the recursion carries is determined by second-order information — and the second-order information of the Laplace chain is by construction (6.1) identical to that of the Gaussian chain. The full induction, and the four hypotheses the argument requires (finite second moments, zero mean, $\alpha \neq 0$, and

where the projection is applied, since Π_G does not commute with multiplication by the likelihood), are given in the companion paper.

Theorem 6.6 fixes the meaning of every number below. The comparison is not “grid BP against one particular approximation”: it is *full non-Gaussian posterior inference against the best second-order linear estimator*, and the measured gap is precisely the price of discarding everything beyond second moments. It also explains why the errors reported here are attributable to non-Gaussianity alone — the projected arm is literally blind to the only thing that distinguishes the two chains.

Table 6.1 reports the median relative score error of the Gaussian closure on the Laplace chain:

Table 6.1: The Gaussian-message approximation against validated grid BP on the Laplace chain ($\alpha = 0.85$, $L = 32$, 60 trials per t , reference $N_g = 801$). By Theorem 6.6 the first row is the price of discarding everything beyond second moments, not the error of one particular implementation. The second row is the share of trials exceeding 10% relative error, which reaches zero an order of magnitude in t before the median does.

t	0.05	0.32	0.98	2.07	3.0
median relative score error	0.24	4.6×10^{-2}	6.8×10^{-3}	2.5×10^{-4}	7.3×10^{-6}
$\mathbb{P}(\text{err} > 0.1)$	1.00	0.08	0.00	0.00	0.00

(Figure 6.2). Two thresholds are worth separating, because they are different statistics and sit an order of magnitude apart in t . The probability of a trial exceeding 10% relative error first reaches zero at $t \approx 0.98$; the *median* error first falls below 10^{-3} only at $t \approx 2.1$. In between — around $t \approx 1$, where the failure probability has already vanished — the typical error is still $\sim 7 \times 10^{-3}$.

The closure error is of order tens of percent at small diffusion time and decays as diffusion noise grows. The natural explanation is that the tilted posteriors approach Gaussianity, and it is consistent with the measurement without being established by it: the Tweedie prefactor e^{-t}/Δ_t also attenuates with t , so a falling score error does not by itself separate a posterior becoming more Gaussian from a prefactor damping the effect of any given posterior-mean discrepancy. Posterior non-Gaussianity is not measured directly

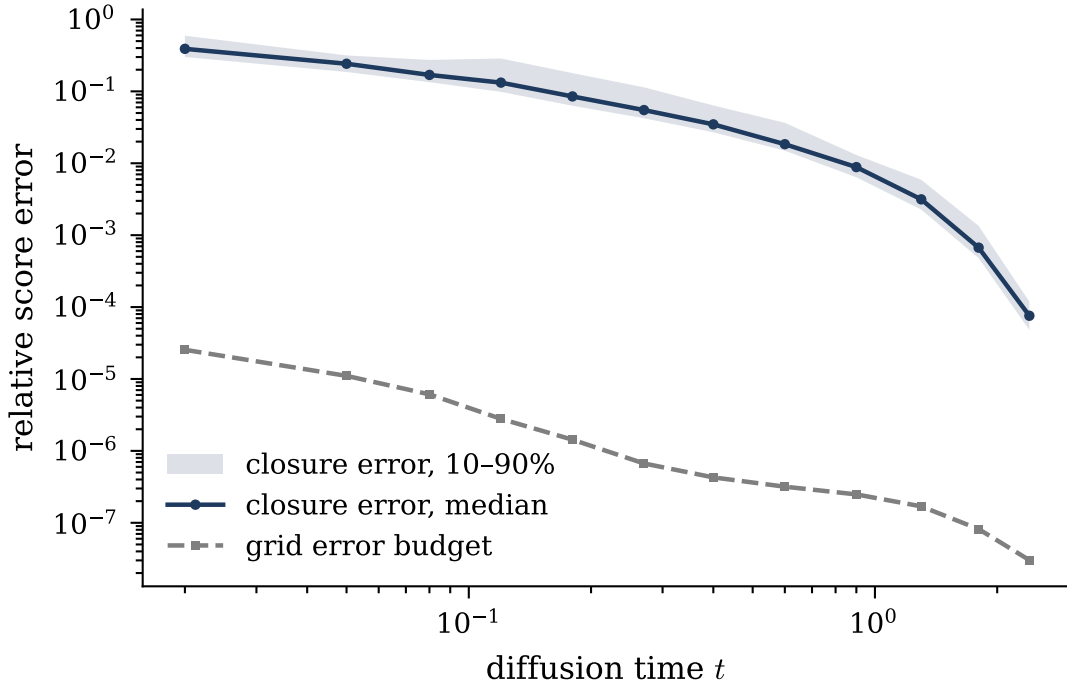


Figure 6.2: Gaussian-closure score error on the Laplace chain ($\alpha = 0.85$, $L = 32$, 60 trials per t ; median and 10–90% band) against the grid error budget measured on identical data. The closure error exceeds the reference error by 10^4 to 10^7 at every t , so the measurement is trustworthy where it is used.

here. Inspection of the worst trials shows the mechanism: the reference belief is sharply non-Gaussian, occasionally bimodal; the single Gaussian commits to a compromise mean; and (6.9) amplifies the shift. The score *direction* is much more robust than its magnitude: the median cosine similarity between projected and reference scores is 0.976 at $t = 0.05$ (where the magnitude error is 24%), exceeds 0.99 for all $t \geq 0.11$, and the worst single trial of the sweep stays above 0.879. These are numerical observations for the stated parameter ranges, not theorems.

6.5.1 Varying the innovation law

Sweeping six innovation families with identical covariance $\alpha^{|i-j|}$ (Laplace, Student- t with $\nu = 5, 8$, and symmetric Gaussian mixtures of increasing separation) isolates the shape of the innovation law as the only moving part. Three observations, all within the tested

grids. First, the noise level dominates: every family's error decays to about 10^{-3} or below by $t = 2.4$; all substantial closure error lives at $t \lesssim 0.4$. Second, bimodality is the worst case: at $t = 0.05$, $\alpha = 0.5$, the median error is 0.94 for a strongly bimodal mixture, 0.52 for Laplace, 0.32 for Student- $t_{\nu=5}$, and 0.07 for a mildly bimodal mixture; a single Gaussian cannot represent two posterior modes and commits to the gap between them. Third, stronger chain correlation helps rather than hurts: for Laplace at $t = 0.05$ the error is 0.43, 0.27, 0.13 at $\alpha = 0.5, 0.85, 0.95$; informative neighbours act like additional Gaussian observations, which is consistent with the local tilted posteriors becoming more Gaussian, again as an interpretation rather than a measurement.

6.6 What can and cannot be concluded

Classified in the vocabulary of Section 1.5. Exact, by derivation: the surviving structure of Section 6.1, the limits (6.3), the factorisation (6.4), the screened Poisson equation (6.5), the curvature identity (6.6), the message form (6.7), the boundary integral (6.8), the breakdown argument of Remark 6.4, the identification of Gaussian closure with the covariance-matched LMMSE estimator (Theorem 6.6), and the error identity (6.9). Exact for the discretised model: grid BP solves the finite-state chain that discretisation defines to floating-point accuracy — a different statement from reproducing the continuum recursion, and the gap between them is what the calibration of Section 6.4 bounds. Numerical, within validated regimes: the quadrature audits of Section 6.2, the grid-BP calibration, and the closure-error measurements, which are specific to the stated innovation laws, correlation levels, chain lengths, and diffusion times. Interpretation rather than measurement: that the decline of the closure error with t reflects the tilted posteriors becoming Gaussian — consistent with the data, but not separated here from the attenuation of the Tweedie prefactor. Not concluded: closed forms for general (non-boundary) messages at $L \geq 2$; any locality law for the non-Gaussian score beyond the structural delimitation above; and any statement about learned scores. No low-rank structure is asserted for the Laplace $L = 2$ posterior Hessian: it holds only in a narrow regime.

Chapter 7

Learning the Innovation Law: EM for Non-Gaussian Chains

Chapters 5 and 6 treated the transition law of the clean chain as given: Gaussian in the first case, a fixed non-Gaussian family in the second. Belief propagation then computed the diffusion score exactly, in the Gaussian case, or against a validated numerical reference, in the Laplace case. Neither chapter asked where the transition law itself comes from. This chapter does. The clean data generating process is the same first-order Markov chain of (6.1), but the innovation law is now a member of an estimated family, and the question is how much the Markov structure this thesis has exploited throughout still helps once part of the model must be learned rather than assumed, and what breaks first when the assumptions themselves are wrong.

How much does known Markov structure reduce the statistical and computational burden of learning a diffusion score, and what is lost when the transition law or the continuous messages must be approximated?

The chapter answers the *how* — the estimator, what is exact in it and what is numerical, and what the implementation was checked against. Chapter 8 answers the *how much*.

7.1 Model: an estimated mixture innovation

The clean chain is the one of (6.1), in the index convention of this chapter,

$$a_1 \sim \mathcal{N}(0, 1), \quad a_i = \alpha a_{i-1} + \varepsilon_i, \quad i = 2, \dots, n, \quad (7.1)$$

with innovations ε_i i.i.d., mean zero, variance $\sigma_\eta^2 = 1 - \alpha^2$ — the same Gaussian initial state and the same variance-matching that Chapter 6 used — but now drawn from a member of a *parametric family* rather than a fixed law. Forward noising is the coordinatewise Ornstein–Uhlenbeck channel of Section 2.2, $x = e^{-t}a + \sqrt{\Delta_t}\xi$, unchanged from every earlier chapter.

The estimated family is a C -component Gaussian mixture on the innovation. It is the transition kernel written $M(a' | a)$ in Chapters 2 and 6, carrying its parameters explicitly because they are now what is being estimated:

$$K_\theta(a' | a) = \sum_{c=1}^C \pi_c \mathcal{N}(a' - \alpha a; \nu_c, s_c^2), \quad \theta = \{\alpha, (\pi_c, \nu_c, s_c)_{c=1}^C\}, \quad (7.2)$$

so both the autoregressive coefficient α and the shape of the innovation density are free parameters, estimated jointly by maximum marginal likelihood from noised data alone. At $C = 1$ this collapses to a single Gaussian, recovering the model of Chapter 5; larger C can represent skew, heavy tails, and multimodality to the resolution of the grid. The initial law $a_1 \sim \mathcal{N}(0, 1)$ is *held fixed* at estimation time — it is the value used to generate every family compared in Chapter 8, so there is no misspecification in it, but neither is there direct evidence it could itself be recovered from noised data; that question is not addressed here.

Two assumptions carry the argument and, consistent with the practice of this thesis, are stated rather than tested by the estimator itself: the clean process is exactly first-order Markov, and the forward noise is exactly coordinatewise. Section 8.5 in the next chapter measures what happens when the first of these is false by a controlled amount; the second is never relaxed anywhere in this thesis.

Remark 7.1 (Covariance-stationary, not strictly stationary). Choosing $\sigma_\eta^2 = 1 - \alpha^2$ makes the chain covariance-stationary — $\text{Var}(a_i) = 1$ and $\text{Cov}(a_i, a_j) = \alpha^{|i-j|}$ hold exactly for *every* innovation law, by the same second-moment recursion as the Gaussian case. It does not make the chain strictly stationary: for a non-Gaussian innovation the invariant marginal law of the chain is not $\mathcal{N}(0, 1)$, so fixing $a_1 \sim \mathcal{N}(0, 1)$ starts every family off its own invariant distribution. Nothing compared in this or the next chapter depends on the distinction, because every family compared shares the same a_1 and the same covariance structure. The claim available here is covariance-stationarity, and it is the one used.

7.2 What remains exact

Two facts from Section 6.4 carry over unchanged, and everything the estimator does rests on them, so they are worth restating here.

Proposition 7.2 (The posterior factor graph is still the prior’s chain). *Coordinatewise forward noising contributes exactly one unary factor per site, $x_i \mapsto \phi_i(a_i) \propto \exp\left(- (x_i - e^{-t}a_i)^2 / 2\Delta_t\right)$. It reweights node potentials and creates no new edges among the latent a_i , for any innovation law, mixture or otherwise. The posterior over the clean chain given the noised observation therefore factorises on exactly the same chain graph as the prior.*

Proposition 7.3 (The E-step is exact). *Since the posterior graph is a tree, one forward and one backward sum-product sweep return the exact single-site and pairwise marginals of the posterior, at the level of functional messages, for any innovation law. Consequently the E-step of an EM procedure for θ requires no variational approximation and no sampling: the sufficient statistics EM needs are exact conditional expectations under the true posterior, mixture label included.*

Both are consequences of the chain being a tree (Theorem 3.3) and hold independently of whether the innovation law admits a closed-form message update. What is *not* exact is the representation of a functional message on a computer, which is where Section 7.4 and the grid-BP machinery of Section 6.4 enter.

7.3 The EM algorithm

Write $\Xi(\theta)$ for the vector of exact posterior sufficient statistics that BP supplies given the current parameter θ — the conditional expectations of the pairwise sums, mixture-label indicators, and second moments that the mixture kernel’s complete-data log-likelihood needs. The gradient of the *marginal* log-likelihood $\mathcal{L}(\theta)$ is given by Fisher’s identity,

$$\nabla_{\theta} \mathcal{L}(\theta) = \langle \Xi(\theta), \nabla_{\theta} \log K_{\theta} \rangle. \quad (7.3)$$

The practical content of (7.3) is that belief propagation is never differentiated through: BP supplies the conditional expectation $\Xi(\theta)$ at the current parameter, and the gradient with respect to θ is then an ordinary derivative of the kernel’s log-density. This is what makes the E-step of Proposition 7.3 directly usable inside an optimiser.

The M-step for the mixture kernel is *not* an exact maximisation. Given $\Xi(\theta^{(k)})$, updating $(\pi_c, \nu_c, s_c)_{c=1}^C$ exactly would require solving a mixture-likelihood problem with its own latent label, so the implementation alternates closed-form *conditional* maximisers — the mixture block and α — until the per-edge gain in Q falls below a tolerance or a cap of sixteen sweeps is reached, recording on the returned kernel which of the two occurred. The algorithm is therefore generalised EM, or ECM (Meng and Rubin, 1993), rather than exact EM: a sweep ascends Q without maximising it.

The cap of sixteen is set by measurement, not convention. Varying only this budget at $C = 8$, $n = 256$, and a fixed outer count of 40, the fitted innovation excess kurtosis moves by 28% between four sweeps and sixteen ($1.68 \rightarrow 2.14$, against a true value of 3.0) while α moves in the fourth decimal — the same rate asymmetry between shape and correlation that Section 7.5 finds in the outer loop, appearing one level down. The effect is confined to short outer runs: by 60 outer iterations the two loops substitute for one another and the inner budget stops mattering. Sixteen removes the interaction rather than requiring a reader to establish which regime a given cell is in.

Monotone ascent of $\mathcal{L}$ is guaranteed by the standard EM argument applied to the surrogate $Q(\theta \mid \theta^{(k)}) = \langle \Xi(\theta^{(k)}), \log K_{\theta} \rangle$ — ascending Q is all monotonicity requires —

and is checked directly: across an executable audit of the implementation, the minimum increase in Q over twelve consecutive steps was +2.61 for a Gaussian kernel, +4.34 for Laplace, and +4.18 for an eight-component mixture, with zero decreases recorded in any logged run. The returned kernel's evidence equals `trace.log_evidence[-1]` by construction, with a regression test asserting it, so the reported convergence trace always describes the model actually returned.

7.4 The numerical representation: grid BP, again

Messages remain functions on $\mathbb{R}$, so they are represented exactly as in Section 6.4: values on N_g equispaced points in $[-A, A]$, updates by trapezoidal quadrature. Two error sources are tracked separately here as well. Truncation error — the probability mass the finite domain $[-A, A]$ misses — is diagnosed by the maximum boundary mass over the sites of a chain. That is a statistic of a sampled trajectory rather than a bound, so it is recorded over 256 chains per configuration: the worst chain falls from 2.3×10^{-3} at $A = 4$ to 1.2×10^{-6} at $A = 8$, the 90th percentile from 2.7×10^{-4} to 1.5×10^{-8} , and neither depends on the grid resolution N_g . The gap between the two rows matters, because a figure read off a single trajectory sits near the median and so understates the tail by two orders of magnitude. Quadrature error, from the finite spacing $h = 2A/(N_g - 1)$ — the grid holds N_g points including both endpoints — on an otherwise exact interior update, falls as $O(h^2)$: a factor of four per halving of h , matching the trapezoidal rate on a smooth integrand. At the working configuration used for every experiment in this and the next chapter, $N_g = 401$, $A = 8$, both sources sit far below any effect subsequently measured: the discretised recursion reproduces the closed-form Gaussian posterior mean to 9.2×10^{-15} , and agrees with an independently coded brute-force enumeration — sharing no code with the BP recursion — to 1.6×10^{-14} on posterior means and 1.8×10^{-15} on the log-evidence.

Which finite-grid model is being optimised. “Exact for the discretised chain” needs one more specification than it has been given so far, because there are two conventions available and they lead to different objectives. The transition array used here is the continuum density $K_\theta(a' | a)$ evaluated at the grid points and combined with the trapezoidal weights, with *no* per-column renormalisation. BP is therefore exact — up to floating point — for the chain-shaped *factor model* that array defines, and that model is not literally a normalised finite-state transition kernel on the truncated domain. The distance between the two is precisely the column-mass residual already reported as the quadrature diagnostic. At the working configuration, the worst column with $|a| \leq A/2$ departs from unit mass by 9.6×10^{-4} ; such a column sits far enough inside the domain that its transition density has no tail outside the grid, so its residual measures quadrature alone. The worst column anywhere departs by 1.9×10^{-2} , at the domain edge where the density is truncated.

The alternative convention — renormalising each parent-conditioned column under the quadrature weights — would make the array a genuine finite-state kernel. Its normaliser would then depend on θ , most strongly near the boundary, and that dependence would have to be carried: the derivative of $\log Z_\theta$ would enter both Fisher’s identity and the M-step, and the surrogate could no longer be written as $Q(\theta | \theta^{(k)}) = \langle \Xi(\theta^{(k)}), \log K_\theta \rangle$ with K_θ the raw density. Because no renormalisation is applied, that term is absent by construction rather than by oversight, and the surrogate above is the correct one for the model actually fitted. The price is that the fitted object is the continuum kernel read through a quadrature rule, which is what the column-mass residual measures and what the resolution gate of Section 7.4 bounds.

Figure 7.1 separates the two error sources the paragraphs above keep apart: resolution and truncation are refined one at a time, so that neither diagnostic can be mistaken for a certificate of the other.

Remark 7.4 (The boundary-mass diagnostic is empirical, not a bound). A boundary-mass figure computed from one sampled chain does not bound the truncation error for every run, and the gap is large enough to matter. Running the same recursion on one chain and on

256 of them, at the working configuration and over all 12 noise levels, the batch maximum exceeds the single-chain reading by a median factor of 86 and by as much as 3,281; at the mildest level it is only 1. A single draw therefore lands somewhere in the body of a distribution with a very long tail, and where it lands is not knowable from the draw. This is why the figures above are reported as a maximum and a 90th percentile over chains rather than as one number. The diagnostic is an empirical estimate of typical truncation loss, not a certified bound. What it does *not* affect is the separation of truncation from quadrature error reported above, which is a pure function of (A, N_g) and the prior and involves no sampled chain at all — that comparison reproduces to 0.000 relative difference under reseeding and is exactly the result Section 6.4 already relies on.

One further identity is worth restating from Section 6.4 because it governs every result in the next chapter exactly as it governed Chapter 6: whatever an approximation — Gaussian closure, grid truncation, or an under-converged fit — does to the posterior mean is amplified by $e^{-t}/\Delta_t \sim 1/2t$ in the score, so small diffusion times remain the regime where any deficiency is most visible.

Figure 7.2 collects the optimisation-side checks: monotone ascent of the marginal likelihood, recovery of α from initialisations far from the truth, and the $n^{-1/2}$ decay of the recovery error. The first two are what make the procedure an EM rather than a heuristic; the third is what makes it an estimator.

7.5 What convergence actually costs

The last piece of machinery this chapter reports is not a theorem but a measurement, and it is the one that most directly qualifies the results of Chapter 8: how many EM iterations the algorithm of Section 7.3 actually needs.

Table 7.1 shows what the channel costs, in two parts: the through-channel fit at four updates, and where a clean and a channel arm each finish.

Table 7.1: What the observation channel costs in rate rather than in where the fit lands. Above: the through-channel fit at $N = 512$, $C = 8$, as a median over 16 seeds at four updates — the correlation is essentially in place by 30 while the innovation shape is still moving at 120. Below: where the two arms finish, fit on clean transition pairs and on the same chains once through the channel. Generating values are $\alpha = 0.85$ and excess kurtosis 3.0; a finite mixture does not contain the generating law exactly, so what is approached is the best fit within the family.

update	α	innovation excess kurtosis
1	0.4402	0.078
30	0.8412	1.275
120	0.8499	3.019
600	0.8504	2.980
<i>where each arm finishes</i>		
clean transition pairs	0.8509	2.939
through the channel	0.8526	2.729

The two halves say different things. The trace shows the coordinates separating: the correlation is essentially in place by thirty updates while the innovation shape is still climbing at a hundred and twenty, because the mixture label is a latent variable of its own and the channel compounds the missing information — the mechanism of Dempster et al. (1977) appearing in the convergence rate, where the theory places it.

The channel has two distinct effects, and they should not be collapsed into one. First, it changes the local rate of EM through the fraction of missing information, which is the effect the trace above displays. Second, at finite sample size it reduces the observed Fisher information and can therefore increase estimation uncertainty — an effect measured directly, by coordinate, in Section 8.6. Under identifiability and correct specification it need not move the population optimum, and that is the limited sense in which the clean and channel arms may approach the same endpoint while doing so at different rates. The endpoints here are consistent with that reading: the two arms finish within 0.002 of each other on the correlation and within a fifth of a unit on excess kurtosis, both close to the generating values. This is evidence of similar fitted endpoints in this configuration, not

of an absent information or accuracy penalty.

That is one configuration. A dedicated sweep measured the cost across sixteen seeds, recording for each fit the first update after which a coordinate stays within a fixed relative tolerance of its value at the end of the run. At $N = 512$, $C = 8$ and the uniform noise design, over 16 seeds run to 800 updates, the correlation coefficient settles at a median of 80 updates and the innovation shape at 229, the largest observed value being 638. Forming the ratio within each fit and then taking the median over fits gives 2.8; the quotient of the two medians is 2.9, and the paired figure is the one reported, because both coordinates are measured on the same fit.

Two things about what that measures. “Settled” is trace stability — the first update after which a coordinate stays within 10^{-3} of its end-of-run value for α , or 2×10^{-2} for the excess kurtosis — and not distance to an optimiser; a coordinate can sit still while the density it belongs to moves along a shallow direction. And the run has to be long enough for the statistic to mean anything: the sweep was run at two lengths, and these medians come from the 800-update shard, the only one in which no configuration settles near its own cap. Against them, the fixed budget of 40 iterations that several experiments in Chapter 8 used as a default falls short of the shape coordinate’s median by about a factor of six and leaves 105 of the sweep’s 112 configurations — both shards, since censoring can only shorten a settling time and so cannot inflate this count — unsettled in it.

Remark 7.5 (Fixed budgets compare convergence rates, not estimators). This is the single methodological lesson of the chapter, and it recurs throughout the next one. Whatever coordinate of a fit converges slower looks worse under a fixed iteration count, and a slower-converging configuration under-iterated by the same budget produces what looks like a genuine deficit rather than symmetric noise — which is precisely the signature that makes it easy to mistake for a real effect. Compared correctly, at eight seeds and a fixed budget of 200 updates, no channel penalty is resolved in the reported shape summary: the paired clean – noised difference at that operating point is -0.052 ± 0.217 , consistent with zero. The operating point is not a converged one — the chapter’s own median settling time for the shape coordinate is 229 updates — so this is a statement about the paired difference

at 200 updates, not about the converged estimators. A single non-converged draw would not have supported that comparison in either direction. Where Chapter 8 reports a result obtained at a fixed, possibly insufficient budget, it says so explicitly and states which quantities are and are not affected.

7.6 Summary

The estimator combines two facts that hold for any innovation law — Proposition 7.2 and Proposition 7.3 — with a generalised M-step and Fisher’s identity to turn exact BP conditional expectations into a marginal-likelihood gradient. Everything about the *posterior inference* step is exact up to the grid discretisation of Section 7.4, which is controlled to far below any effect measured. What is not free is optimisation time. Observing the chain only through the channel costs the autoregressive coefficient about an order of magnitude in iterations, and the innovation shape then settles some three times slower again, with a tail several times longer than its own median. Chapter 8 quantifies pointwise accuracy, capacity and robustness to controlled misspecification, and leaves the reverse-generation comparison open, because its fitted shape coordinates were not sufficiently settled under the available budget.

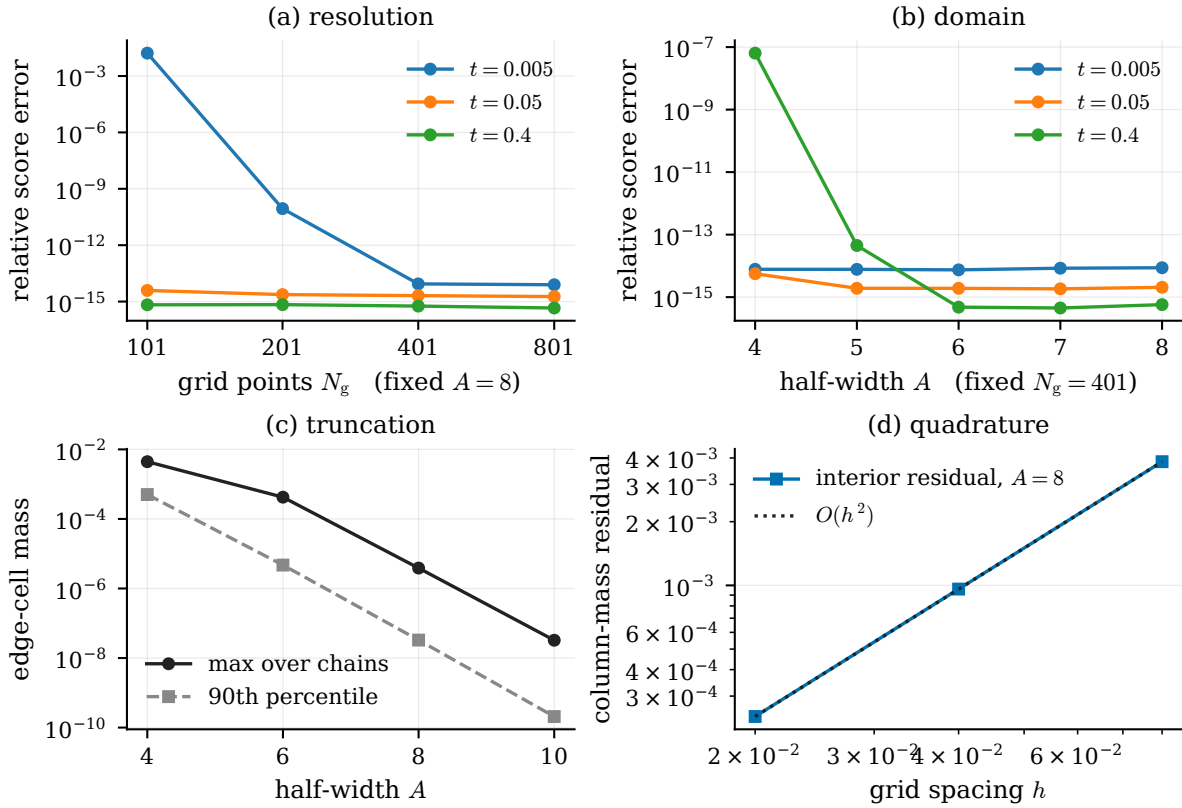


Figure 7.1: Grid and domain control on a Laplace chain ($\alpha = 0.85$, $L = 32$ sites), four panels. (a) Relative score error against a high-resolution reference as grid resolution N_g grows at fixed half-width $A = 8$; (b) the same as A grows at fixed $N_g = 401$ — both saturate at the floating-point floor well inside the working configuration. (c) Forward-message boundary mass against A , the truncation diagnostic, drawn as *both* the maximum and the 90th percentile over 256 chains, which differ by roughly two orders of magnitude for the reason given in Remark 7.4. (d) Interior column-mass residual against spacing at $A = 8$, the quadrature diagnostic, with an $O(h^2)$ reference. Neither diagnostic alone certifies convergence. Source: `outputs/frozen/exp_01/` and `outputs/frozen/exp_18/`.

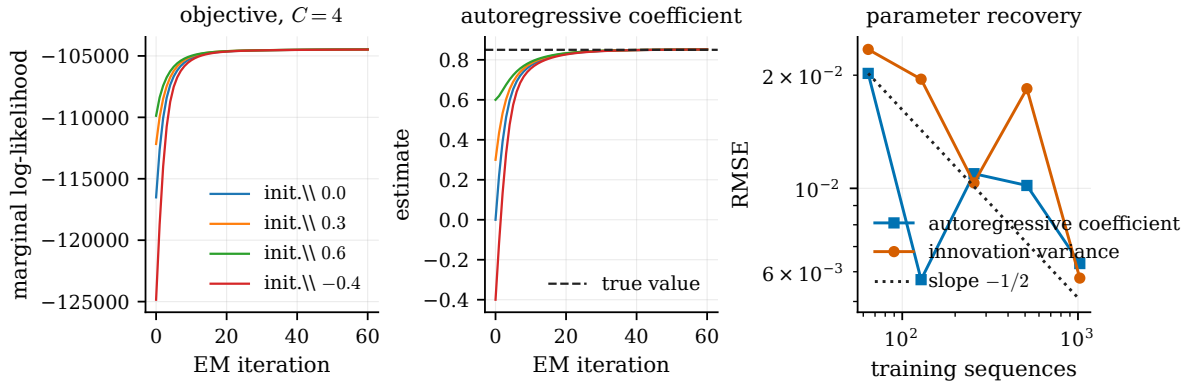


Figure 7.2: Optimisation diagnostics for the EM procedure of Section 7.3. Left: marginal log-likelihood by iteration from four initialisations of α ; the largest decrease across iterations is zero in every run. Centre: the estimated α over the same runs, converging to $[0.8517, 0.8520]$ against a truth of 0.85 from initialisations spanning $[-0.4, 0.6]$ — direct evidence that α is estimated, not supplied. Right: RMSE of the recovered α and innovation variance against the number of training sequences, with an $n^{-1/2}$ reference line. Source: `outputs/exp_18/` and `outputs/exp_06_em_parameter_recovery/`.

Chapter 8

What Structure Buys, and Where It Stops Paying

Chapter 7 built an estimator that is exact in its inference step and approximate only in its numerical representation and its optimisation budget. This chapter asks what that estimator is worth. Five questions organise it, in the order the evidence answers them most cleanly.

How much does the structured estimator reduce denoising error at a given amount of data; how much of that reduction survives against a baseline that is given the locality but not the model; how much mixture capacity does the estimator need; does the pointwise advantage survive when the estimated score is used to *generate*, by integrating the reverse process; and how far can the Markov assumption itself be wrong before the structure stops helping?

Three are answered outright. The capacity question is answered for the estimator as implemented — and in the opposite direction to the one a smaller design suggested — but not for mixture capacity in the abstract, and Section 8.3 says exactly where that line falls. The generative-fidelity question is not answered at all, and the section that addresses it says why the experiment as run cannot settle it. The summary at the end of the chapter states each of the five at the strength its evidence supports.

Every comparison below fits the mixture kernel of (7.2) by the EM procedure of Chapter 7 and compares it against neural denoisers trained by ordinary denoising score matching on the same noised data, with no access to the Markov factorisation, the linear kernel form, or homogeneity across sites — three structural facts the estimator is given and the networks are not. Two network architectures recur throughout: a fully connected MLP with no locality bias, and a weight-shared local network with a fixed receptive field. The second is a *locality-respecting* baseline, not one matched to the estimator: EM-BP smooths over the whole chain and has no radius of its own to match. The comparison measures the value of the structure supplied, not a general claim about neural denoisers.

8.1 Pointwise data efficiency

Under the structural assumptions of Section 7.1, at the data budgets tested, the structured estimator attains lower relative denoising error than the global MLP by a factor of 7.0–17.6 across the 8 training-set sizes (Table 8.1, Figure 8.1). The factor is *not* monotone in the budget, and it is the largest of several defensible aggregations of the same cells — Appendix F.2 gives the other three, with a paired interval on the one quoted here. These figures are generated by `tools/make_tab_efficiency.py` from the frozen outputs rather than transcribed, so the table, the figure and this sentence cannot drift apart.

That first comparison is against a network with neither a locality prior nor weight sharing. When those two biases are supplied through the window-head baseline, the observed ratio narrows to 2.3–6.1 $\times$, measured at the same headline protocol and reported with its own scoping in Section 8.2. This change does not identify a causal decomposition: architecture, parameter sharing, capacity and optimisation change together across the two baselines. It shows that the generic-MLP comparison overstates what can be attributed specifically to the supplied Markov model. The relevant locality result is the influence decay of Chapter 5.

Every tunable quantity either arm is allowed to choose is chosen on a validation bundle disjoint from both the training set and the test set, and the two arms are given the same

Table 8.1: Relative denoising error against the deterministic grid-BP reference under the true kernel, averaged over the noise schedule; 16 seeds $\pm$ one standard error, aggregated per seed. Both arms select their optimisation budget on a disjoint validation bundle. EM-BP is more accurate in 1,518 of 1,520 cells. The ratio column is the mean of the per-cell ratios and not the quotient of the two columns beside it, which is a smaller number; Appendix F.2 gives both, and three further aggregations. Protocol, cell exclusions and the budget calibration are in Appendix C.

sequences N	network	EM-BP (24 free)	ratio
32	0.4031 ± 0.0059	0.0589 ± 0.0026	7.0
64	0.3285 ± 0.0026	0.0501 ± 0.0041	7.8
128	0.2759 ± 0.0024	0.0343 ± 0.0017	8.8
256	0.2268 ± 0.0010	0.0258 ± 0.0014	9.7
512	0.1878 ± 0.0006	0.0219 ± 0.0011	9.0
1024	0.1569 ± 0.0006	0.0162 ± 0.0012	12.5
2048	0.1326 ± 0.0005	0.0132 ± 0.0008	15.5
4096	0.1171 ± 0.0004	0.0094 ± 0.0007	17.6

privilege: the network selects its output parameterisation — predicting the noise or the clean signal — *and* its training length, EM-BP its iteration count. That choice coincides with the test-set optimum in 81.1% of cells for the network and 77.6% for EM-BP. A dedicated audit of the selection rule, over 1,344 cells at 16 seeds, puts a size on what the remainder costs: the mean advantage is $13.07\times$ under validation selection and $13.06\times$ when each arm is instead given its test-set optimum, a difference far smaller than the spread across seeds. The protocol is the defensible one independently of the size of the correction it produces.

8.2 How much of the margin is the architecture?

A ratio against a fully connected network cannot separate knowing the Markov structure from sharing weights across sites, and the second is the cheaper explanation. This section measures against a structure-aware baseline at the headline protocol: the same twelve noise levels and the same sixteen seeds as Table 8.1, every site scored rather than the centre alone, the baseline’s architecture and radius fixed by a screen on separate development seeds, and each arm’s checkpoint chosen on a validation bundle disjoint from test. The

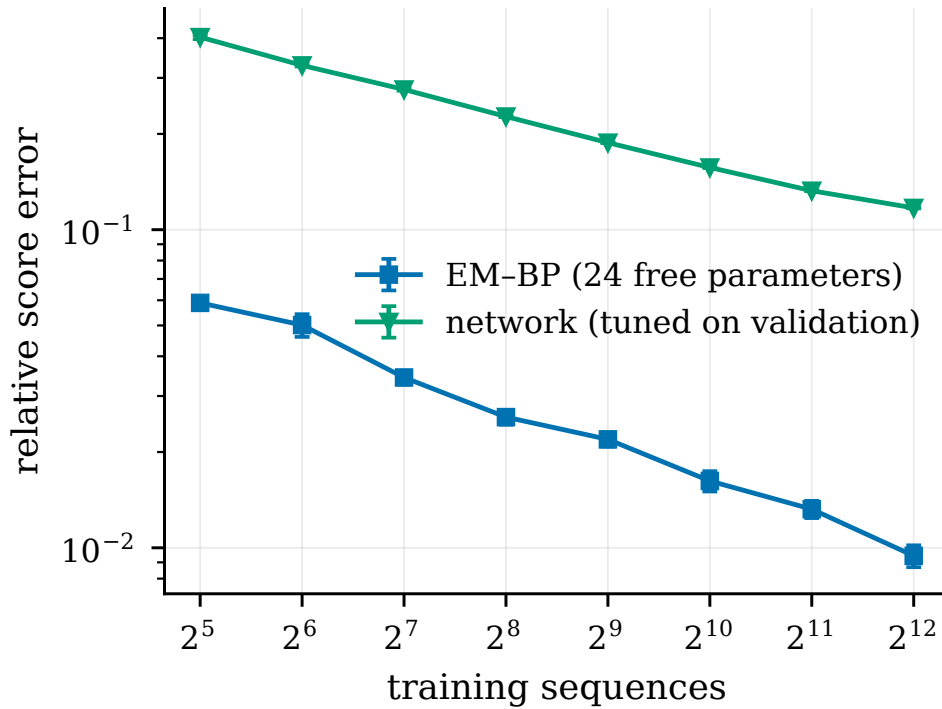


Figure 8.1: Relative score error against training-set size, mean over 16 replicates with standard errors, for the EM-BP estimator and the validation-tuned MLP. The window-head baseline of Section 8.2 scores every site rather than the aggregate plotted here and is reported as a ratio table (Table 8.3) rather than superimposed on this figure. Drawn from the same outputs, over the same size grid and with the same seed-first aggregation, as Table 8.1.

deployment and provenance conditions are in Appendix C.

Which architecture, and under what search. Three candidates were screened on 4 development seeds, disjoint from the 16 used below: a weight-shared window head (which cannot propagate past its radius however much data it sees), a dilated convolutional stack, and a bidirectional message-passing network (both able, in principle, to propagate across the whole chain). Each was run over its own hyperparameter grid — a shared learning-rate grid, both output parameterisations, and a common checkpoint ladder from 500 to 64000 steps — 32 configurations in all, scored on the bulk region. **The window head won**, at radius 4 and width 64, with the bidirectional message passing network next (0.137 against 0.172).

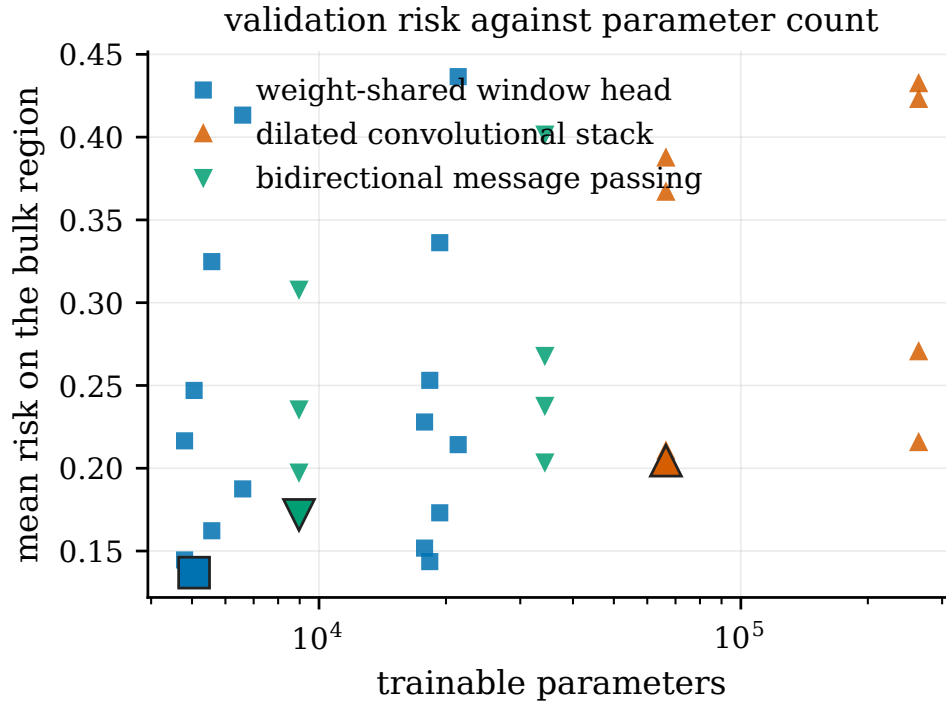


Figure 8.2: Every configuration of the screen, scored on the bulk region and averaged over the 4 development seeds. One marker per architecture; the outlined marker is each architecture’s best configuration. The horizontal axis is what makes the comparison readable: the dilated stack sits an order of magnitude to the right and does not reach the window head’s risk anywhere in the screen. Drawn from the same artifact as Table 8.2.

Table 8.2: The architecture screen that selected Table 8.3’s baseline. Each candidate architecture was run over its own hyperparameter grid on 4 development seeds (disjoint from the confirmatory seeds), at $N \in \{32, 2048\}$, with a shared learning-rate grid, both output parameterisations, and a shared checkpoint ladder; the reported risk is the mean over development seeds on the interior slice used for selection. $\star$ marks the winner carried into the confirmatory run. The dilated stack carries an order of magnitude more parameters than the winner and finishes last, so the outcome is not an artefact of one arm being given more capacity than another.

architecture	configurations	parameters	selection risk
weight-shared window head $\star$	16	4.8k–21.4k	0.137
dilated convolutional stack	8	66.4k–264k	0.205
bidirectional message passing	8	9.0k–34.3k	0.172

Figure 8.2 shows the whole search rather than its winner, and it is the more useful object: the three architectures separate by parameter count far more cleanly than by risk,

and the two arms built to propagate further occupy the expensive half of the axis without buying anything there.

What that does and does not license is worth stating, because the two are easily confused. It is a *model-selection* fact: under this search space, this training protocol and these development seeds, the simplest candidate scored best. It is not evidence that long-range propagation is intrinsically useless here, that radius 4 is the population-optimal receptive field, or that a dilated or message-passing architecture could not overtake it under different tuning. It does establish that the long-range arms were not offered a smaller budget: the dilated stack carries 66.4k–264k parameters against the window head’s 4.8k–21.4k, an order of magnitude more, and finished last of the three. A larger parameter count is not by itself proof of adequate optimisation or of a correct inductive bias, so this bounds one explanation rather than eliminating the class. Under the stated development screen the radius-4 window head achieved the lowest validation risk, and the confirmatory experiment below therefore compares EM–BP with the best-performing candidate in that screen. That scoping is carried in every claim that follows.

Table 8.3: EM–BP against the screened structure-aware baseline, at the headline protocol throughout. The baseline is a weight-shared window head whose architecture and hyperparameters were fixed by the screen of Table 8.2 on development seeds; only the checkpoint is chosen here, on a validation bundle disjoint from test, and EM–BP’s iteration count is chosen the same way. Risks are all-site and averaged over the 16 seeds; the ratio is $\text{risk}(\text{window})/\text{risk}(\text{EM–BP})$ formed within each seed and then averaged, $\pm$ one s.e. *All* scores every site, and is the headline; *centre* is a single-site diagnostic. A third region, the predeclared interior slice, is scored in the stored output and used for architecture selection but not reported here.

N	risk, all sites		ratio	
	EM–BP	window	all sites	centre site
32	0.1005	0.2006	2.34 ± 0.26	2.04 ± 0.24
128	0.0652	0.1594	2.82 ± 0.30	2.35 ± 0.26
512	0.0394	0.1353	3.58 ± 0.18	3.25 ± 0.18
2048	0.0224	0.1267	6.14 ± 0.45	5.43 ± 0.39

A ratio against a fully connected network is not comparable to this one: a large fraction of that headline margin disappears once locality and weight sharing are supplied. What

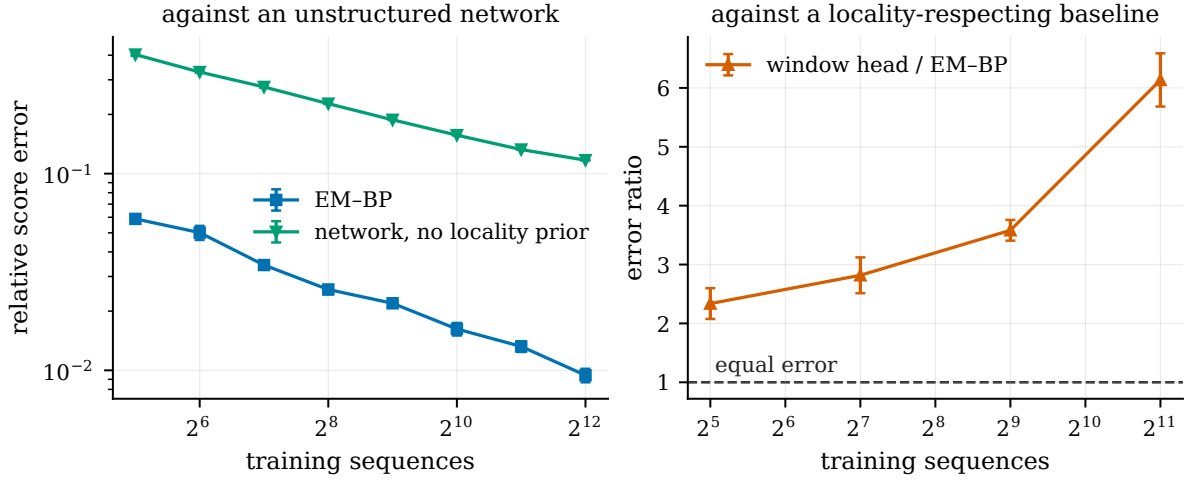


Figure 8.3: The two comparisons together. *Left:* relative score error against training-set size for EM-BP and a network given no locality prior, with one standard error over the 16 seeds. *Right:* the paired per-seed ratio of window-head risk to EM-BP risk, against a baseline that *does* have locality and weight sharing, with the break-even line drawn. The left gap is the headline 7.0–17.6 $\times$; the right one is 2.3–6.1 $\times$, and the difference between the two panels is how much of the headline margin does not survive supplying those two structural biases. Both ratios grow across the tested sizes. That is a description of four measured sizes, not a mechanism: different statistical rates and different optimisation limits produce the same signature, and no design here separates architecture, parameter sharing, capacity and optimisation, which change together when the baseline is swapped. Drawn from the same certified cells as Tables 8.1 and 8.3.

survives against a baseline that has them — the window head — is 2.3–6.1 $\times$, all-site scored, across the same four sizes as Table 8.1. EM-BP is more accurate in every one of the 192 scored cells; that count is 16 seeds $\times$ 4 sizes $\times$ 3 scoring regions (all, bulk, centre), of which the table displays two, and it is a descriptive diagnostic rather than a replication count — the regions and sizes share fitted models and training data, so the inferential unit behind each displayed row is the 16 paired seeds.

The ratio increases across the four sizes tested, from 2.3 $\times$ to 6.1 $\times$. That is a description of four measurements and nothing more is claimed from it. In particular it is not evidence that the residual is non-architectural. An architecture with an irreducible approximation floor b^2 would give $R_{\text{window}}/R_{\text{EM}} = (b^2 + c_w/N)/(c_e/N)$, which grows linearly in N for an entirely architectural reason. The raw risks in Table 8.3 are consistent with exactly that: EM-BP’s risk falls by a factor of about four and a half across the range,

while the window head's falls by less than two. Different statistical rates or different optimisation limits would produce the same signature. The direction is measured; the mechanism is not.

Both arms choose their budget symmetrically, on a validation bundle disjoint from test, *within the budget grids on offer*: 22% of EM-BP cells and 27% of window-head cells select the last rung of their respective ladder. Selecting the last rung is not proof of convergence — it is a sign that validation error was still falling where the grid ran out, and that a wider grid might have improved that arm further. So the reported values compare the two procedures at stated, validation-selected computational budgets, not at converged optima. Table F.2 reports the same ratio under 4 aggregations of the same cells; none of them, at any size, falls below $1.98\times$.

8.3 How much mixture capacity is worth having

Enlarging C in (7.2) should help until the mixture can represent the true innovation law and stop helping after. The design that tests this pairs each capacity against $C = 1$ — a single Gaussian innovation, the kernel the estimator would use if it did not model the innovation law at all — on identical training and evaluation data within each cell, at 16 seeds and two training sizes, with three initialisations per cell and the reported one selected on validation. The fits are run to their plateau rather than to a fixed iteration count, because Section 7.5 measures the fitted shape as settling far later than the correlation coefficient does, and shape is what capacity moves.

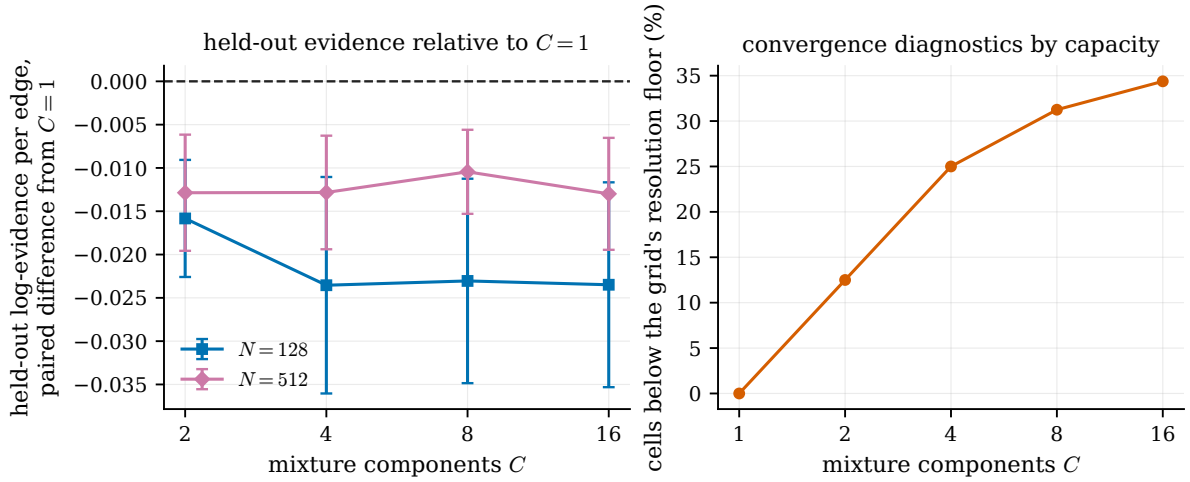


Figure 8.4: Left: the paired difference in held-out log-evidence per edge against $C = 1$, with one standard error over the 16 seeds. Every interval lies below zero at both training sizes, which is the result of this section. Right: the share of fits whose narrowest component falls under the grid’s resolution floor of 2 spacings, which climbs monotonically with C — the second caveat of Remark 8.1, and the reason the left panel is a statement about this estimator on this grid. Every cell at $C \geq 4$ reached the 1200-iteration cap rather than the convergence tolerance. Both panels are drawn from the cells of Table 8.4.

Table 8.4: Held-out log-evidence per edge at each mixture capacity, paired within seed against the single-Gaussian innovation $C = 1$ on identical training and evaluation data, 16 seeds, fits run to their plateau rather than to a fixed iteration count. Positive would mean the mixture buys something. No capacity does, at either size, and the sign count is the more honest summary of the two columns: the mean is dominated by a few large-magnitude seeds while the direction is consistent across most of them. The last column counts cells, over both sizes, whose narrowest fitted component falls below the grid’s resolution floor of 2 spacings — a share that climbs with C , and a reason not to read the table as a pure statement about capacity. Means and standard errors are in units of 10^{-2} .

C	$N = 128$			$N = 512$			under-resolved
	mean	s.e.	seeds better	mean	s.e.	seeds better	
2	-1.58	0.68	4/16	-1.29	0.67	3/16	4/32
4	-2.35	1.25	3/16	-1.28	0.66	2/16	8/32
8	-2.30	1.18	3/16	-1.04	0.49	2/16	10/32
16	-2.35	1.18	3/16	-1.30	0.65	2/16	11/32

The result is that **under this procedure — this grid, this generalised M-step, this validation rule and this iteration cap — no mixture capacity improves on a**

single Gaussian innovation at either training size. The qualification is quantitative. Every cell at $C \geq 4$ stops at the cap rather than at the tolerance, and the share of fits the grid cannot resolve grows with C , so the design cannot separate a statistical disadvantage of larger mixtures from an optimisation or a discretisation one. What it can say, it says at every seed. The means are dominated by a few large-magnitude seeds, so the sign count carries the claim: at $C = 8$ the mixture is ahead on only 2 to 3 of 16 paired seeds, with median paired differences of -5.73×10^{-3} and -9.89×10^{-3} per edge. The same holds on schedule-level score risk.

The predeclared equivalence test sharpens this rather than softening it. The region — $|\Delta(\text{evidence/edge})| < 10^{-4}$ together with $|\Delta R|/R < 1\%$ — was fixed before the contrast was computed, precisely so that “ $C = 8$ and $C = 16$ are equivalent” could be asserted or refused rather than inferred from a failure to reject. It is refused. At $N = 128$ the $C = 8 \rightarrow C = 16$ step is resolved against the region and in the direction of $C = 16$ being *worse*: evidence -4.4×10^{-4} with interval $[-9.2 \times 10^{-4}, -1.1 \times 10^{-4}]$, risk $+5.0 \times 10^{-3}$ with interval $[+9.7 \times 10^{-4}, +1.1 \times 10^{-2}]$, both lying wholly outside it. At $N = 512$ it is unresolved. So neither “capacity saturates by eight” nor “capacity keeps paying past eight” is supported. What is measured is narrower: at $N = 128$, the observed $C = 8 \rightarrow C = 16$ contrast is resolved outside the predeclared equivalence region, in the direction of worse held-out evidence and worse score risk for $C = 16$.

Remark 8.1 (Two things this does not isolate). Both point the same way, and both are reasons to read the table as a statement about this estimator on this grid rather than about mixture capacity as such. First, every cell at $C \geq 4$ stopped at the 1200-iteration cap rather than at the convergence tolerance (96 of 96). The cap was placed at the measured practical plateau. A representative $C = 16$ fit taken to 2000 iterations has per-edge gain 6.9×10^{-8} at 400, about 6×10^{-9} by 800, and then does not move: 6.08×10^{-9} at 1200 against 6.09×10^{-9} at 2000, never reaching the strict tolerance at any practical count. The residual asymmetry is therefore small, but it is not zero, and it disfavours the higher capacities. Second, the share of fits whose narrowest component falls below the grid’s resolution floor of 2 spacings climbs with capacity, from 0/32 at $C = 1$ to 11/32

at $C = 16$. Part of what the larger mixtures are fitting is therefore narrower than the grid can represent, which is a property of the discretisation and not of the innovation law. Separating the two would need a finer grid at fixed capacity, which is not run here.

A smaller design — six seeds, one training size, one initialisation and a fixed 40-iteration budget — reported the opposite sign, and the reversal is worth recording rather than quietly dropping: it does not reproduce at sixteen seeds and a plateau-stopped budget. Which of the two readings the budget produced, and by what mechanism, is not separately tested here; the companion compendium carries the comparison.

Remark 8.2 (Why the iteration budget, and not the capacity, has to be controlled). The reason the design above runs to a plateau is that a fixed budget cannot distinguish the two. Section 7.5 measures the innovation shape as settling at a median of 229 updates against 80 for the correlation coefficient — a factor of 2.8 formed within each fit and then aggregated, with a largest observed value of 638. Against a fixed 40-iteration budget, 105 of the 112 configurations in that sweep had not settled in the shape coordinate. Because a larger C has more shape parameters to move within the same budget, under-iteration alone produces a curve that looks exactly like a capacity effect whether or not capacity is doing anything. This is why any quantity resting on fitted or generated *shape* at a fixed budget — in particular the reverse-dynamics comparison of Section 8.4, which reuses forty-iteration fits — is read here as evidence about an under-converged operating point rather than about the converged estimator.

Per-iteration runtime changes little over the tested capacity range, $13.5 \rightarrow 14.3$ seconds from $C = 2$ to $C = 16$ while the number of free parameters grows $6 \rightarrow 48$, because the E-step's $O(L N_g^2)$ cost dominates the component-dependent M-step. This shows that increasing C adds little cost to one iteration. It does not remove the confounds above: the higher-capacity fits are precisely the ones for which the iteration-cap and grid-resolution diagnostics become less favourable.

8.4 Pointwise accuracy and generative fidelity

Whether the pointwise advantage transfers to generation is not settled by the experiments available here, and the reason is the budget rather than the measurement. Every fit in this comparison used a fixed budget of 40 EM iterations, while Section 7.5 measures the fitted innovation shape — which is precisely what the generative metric scores — as settling at a median of 229 iterations, and as still moving at 40 in 94% of the configurations swept. The comparison therefore evaluates procedures whose fitted shape coordinates are unequally settled under the available budgets, on the one coordinate that carries the generative metric, and a dissociation between pointwise and generative quality cannot be separated from that asymmetry.

The dissociation may well be real, and the design cannot distinguish it from its own iteration budget. No claim in either direction is made here. The measured values, the within-arm form of the comparison and the convergence diagnostic that settles it are recorded in the companion compendium’s claim audit. What would settle the question is the same correction that Section 8.3 applies: select both arms on a convergence criterion rather than a fixed count, then compare.

8.5 Where the Markov assumption stops paying

Every result so far assumes the clean chain is exactly first-order Markov. Two controlled violations were measured against an exact reference (linear Gaussian algebra for the contaminated model, so the estimator being wrong is not confused with the reference being wrong): a rank-one global latent shared across all sites at strength β , and a long-range coupling at strength γ added on top of the chain. This experiment has its own noise schedule and its own training size, so its ratios measure how the advantage responds to misspecification and are not on the same scale as Table 8.1’s; only the direction and the crossing point are read from them. Both arms train at the headline protocol’s budget.

The measurement is reported for two innovation laws. The Gaussian arm carries both mechanisms, since its contaminated model has an exact reference in closed form;

the Laplace arm carries the rank-one mechanism, where the same contamination can be applied to a non-Gaussian chain and the reference computed on the grid. Reading them together separates what is a property of the chain structure from what is a property of Gaussianity. On the rank-one mechanism the two agree in shape — the same ordering, the same smooth decay, no crossing — and differ at the margin. At $\beta = 1$ the Gaussian arm is still ahead at every noise level against both baselines, worst ratio 1.06; the Laplace arm is still ahead against the convolutional baseline, worst ratio 1.32, but reaches the break-even line against the fully connected one at the largest diffusion time, at 0.998. The ordering carries across both innovation laws tested; the size of the margin it leaves does not quite. Whether it carries beyond these two families is not established here.

Table 8.5: Two controlled violations of the Markov assumption, each evaluated against the appropriate reference: the Gaussian rows against an exact linear-algebra reference for the contaminated model, the Laplace rank-one rows against the validated grid-BP reference. The first two rows of each block are the baseline’s relative score error divided by EM–BP’s, averaged over the noise schedule, so a value above 1 means the structured estimator wins and a bold value means it loses. The third row is the fitted correlation minus the Chow–Liu correlation of the true contaminated law: the distance between the chain that was fitted and the best chain there is. The fourth is EM–BP’s own relative score error at its worst noise level, which says how much of a change in the ratio is the estimator degrading rather than the baseline improving.

	global latent, rank-one, strength β				
	0.00	0.10	0.25	0.50	1.00
Gaussian, vs CNN	16.41	14.84	6.68	3.21	2.16
Gaussian, vs MLP	29.34	22.48	11.28	4.00	1.67
Laplace, vs CNN	11.14	10.92	6.21	2.96	2.09
Laplace, vs MLP	16.51	17.64	10.13	3.67	1.62
Gaussian: $\hat{\alpha} - \alpha_{\text{CL}}$	−0.0025	−0.0008	+0.0008	+0.0051	+0.0144
Gaussian: EM–BP error, worst t	0.019	0.025	0.055	0.134	0.167
Laplace: EM–BP error, worst t	0.022	0.022	0.047	0.132	0.183
	long-range precision coupling, strength γ				
	0.00	0.05	0.10	0.20	0.40
Gaussian, vs CNN	16.07	1.24	0.97	0.88	0.75
Gaussian, vs MLP	27.84	1.66	1.19	0.84	0.70
Gaussian: $\hat{\alpha} - \alpha_{\text{CL}}$	+0.0018	+0.0376	+0.0557	+0.0806	+0.0976
Gaussian: EM–BP error, worst t	0.016	0.298	0.402	0.490	0.579

The two violations behave qualitatively differently, and the difference is principled rather than incidental. A global latent shared across every site makes the score residual *exactly* rank one, by a Woodbury identity, and over the range tested a chain-structured estimator accommodates that correction within its own fitted coefficients: the fitted correlation climbs from 0.848 on the clean chain to 0.939 at $\beta = 1$: the fitted chain partially accommodates the rank-one perturbation by changing its effective lag-one correlation. This

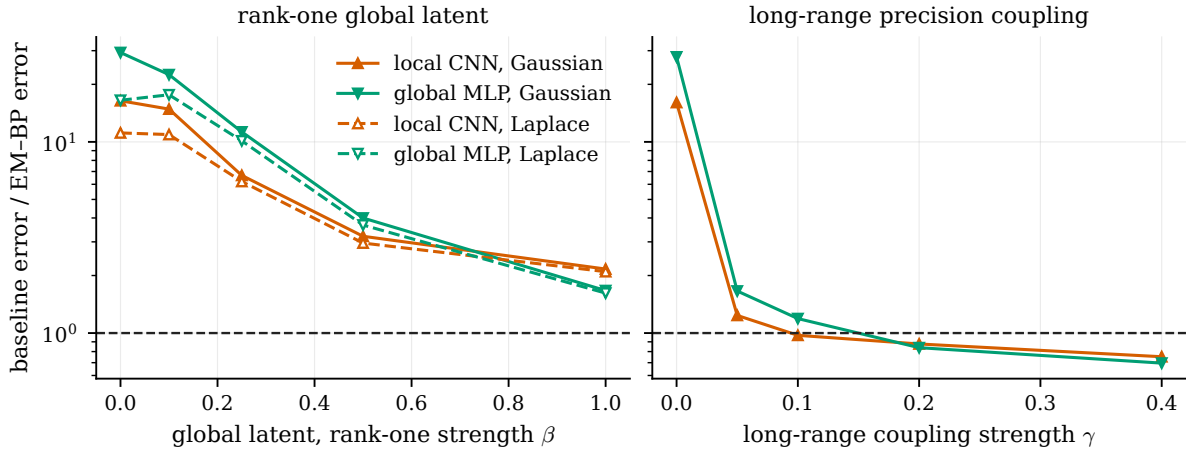


Figure 8.5: The same cells as Table 8.5, on a logarithmic vertical axis with the break-even line drawn; solid is the Gaussian innovation, dashed with open markers the Laplace one. The two mechanisms are not two amounts of the same thing: under a rank-one global latent the advantage decays smoothly and does not reach the line, even where half the marginal variance is a shared constant; under long-range coupling it crosses within the first tested strength and stays below. The Laplace curves track the Gaussian ones across the whole rank-one sweep, which shows that the rank-one result is not confined to the Gaussian experiment; its generality beyond these two innovation families remains open. The long-range panel is Gaussian only, since the contaminated Laplace model has no closed-form reference. Every curve is a mean over the noise schedule, so it shows the direction rather than the worst case — the tie reported in the text at $\beta = 1$ is at one noise level and is not visible here. All fall because the estimator degrades, not because the baselines improve: the last row of each block in Table 8.5 carries EM-BP’s own error.

is partial accommodation over a measured range, not an exact representation — a first-order Markov family does not contain the contaminated law — but the residual it cannot represent stays small enough that the advantage never inverts: at $\beta = 1$, where half the marginal variance is a shared global constant, the ratio at its worst noise level is still 1.06 against the weaker of the two baselines.

Long-range coupling is only *approximately* low rank, and a chain family cannot represent it at all. The advantage against the convolutional baseline is gone by $\gamma = 0.1$, having survived $\gamma = 0.05$, and by $\gamma = 0.2$ the estimator loses to both baselines rather than to one; at $\gamma = 0.4$ its worst ratio is 0.45 and its own relative score error reaches 58%.

Two rows of the table separate the mechanism from the outcome. The estimator’s own error is what moves in both blocks — the baselines barely change — so neither column is

a story about the networks improving. What distinguishes the two violations is the last row of each block: under rank-one contamination the fitted chain stays within 0.014 of the Chow–Liu correlation, the best first-order approximation of the contaminated truth, so its fitted lag-one correlation remains close to the corresponding first-order projection — one scalar agreeing, not the whole fitted law certified; under long-range coupling that gap is 7 times larger, which is what it looks like when the family can no longer represent what generated the data. The uncontaminated controls of the two blocks are the same configuration reached from two directions, and they agree to within 2% (16.41 and 16.07), which is what makes the rest of the table readable across blocks.

Remark 8.3. This extends the lesson of Chapter 6: there, what broke under a harder innovation law was the *representability of messages* in a finite-dimensional family, while the tree, the two-sweep schedule, and the score–posterior identity survived unconditionally. Here, the further question is what breaks when the graph is not a tree to begin with. The answer is not "misspecification degrades the advantage" uniformly — it is that a chain-structured estimator tolerates the one violation it can absorb into its own parameters and fails on the one it structurally cannot represent, and the two are not interchangeable amounts of "how wrong the model is": they are different in kind.

8.6 The channel removes shape information faster than correlation information

Section 7.5 found that the innovation shape converges far slower than the autoregressive coefficient under EM. A direct information-theoretic measurement runs alongside that finding.

This calculation complements the convergence experiment but does not derive its iteration count. The table below measures how much statistical information about each coordinate survives the observation channel. Local EM convergence is governed instead by the fraction of missing information, equivalently by the Jacobian of the EM update

near its fixed point. The two measurements point in the same direction here, but they answer different questions, and a flatter observed likelihood does not by itself entail a given number of updates.

What is computed. The innovation law is a generalised Gaussian $f_p(\varepsilon) \propto \exp(-|\varepsilon/s|^p)$, with s fixed by $\text{Var}(\varepsilon) = \sigma_\eta^2 = 1 - \alpha^2$ at every p , so p moves the fourth moment at *exactly fixed* variance and is a genuine shape coordinate rather than a reparameterised scale. For a three-parameter vector $\theta = (\alpha, \sigma_\eta^2, p)$, Fisher’s identity (7.3) supplies the exact per-chain score of the marginal log-likelihood from one forward–backward pass, and the information matrix is estimated in *outer-product* form, $J = D^{-1} \sum_d g_d g_d^\top$ over $D = 512$ chains of $L = 32$ sites, at $N_g = 401$, $A = 8$. This is the outer-product estimator, not the observed negative Hessian, and it is an estimate at the true parameter rather than at a fitted one. The reported quantity is the *efficient* information $I_{p,(\alpha,\sigma_\eta^2)} = 1/(J^{-1})_{pp}$, which is what remains about p once the nuisance coordinates must be estimated too — equal to the diagonal entry only when the coordinates are orthogonal and strictly smaller otherwise. The gap between J_{pp} and this quantity is recorded per cell as the nuisance cost, and the condition number of J is recorded alongside it, since the inversion is where a near-singular J would show up.

What it shows. The efficient information for p decays two to three orders of magnitude faster with diffusion noise than the information for α itself. Table 8.6 reads the loss between the smallest and largest diffusion time tested:

Table 8.6: How much Fisher information about each coordinate the channel destroys, read between the smallest and largest diffusion time tested ($t = 0.05$ against $t = 1.6$). The shape coordinate loses two to three orders of magnitude more than the correlation does, which is a property of the noised likelihood rather than of any optimiser. The reported quantity is the efficient information, so the nuisance coordinates are estimated rather than known.

true p	info loss, α	info loss, σ_η^2	info loss, p (shape)
1.0	$78\times$	$235\times$	8.7×10^3
1.5	$83\times$	$459\times$	6.4×10^4
3.0	$56\times$	$522\times$	6.8×10^4

This is a property of the noised likelihood itself, independent of any optimiser: at any fixed sample size, and at every diffusion time past the smallest ones, there is less information about the shape coordinate in noised data than about the correlation. That is consistent with the shape coordinate being the slow one, though as noted above it does not derive the rate. It is also the reason the capacity sweep of Section 8.3 is exactly the kind of measurement where an insufficient iteration budget is dangerous: the coordinate that is hardest to move is also the one with the least signal per observation.

Two limits on how far these figures should be pushed. They are ratios of outer-product estimates at $D = 512$ chains, so only the order is meaningful and the entries are printed to two significant digits for that reason. And the comparison is across the tested noise grid rather than a limit: the decay is measured between $t = 0.05$ and $t = 1.6$, and how it continues past the schedule’s end is not established here. What the table supports is the direction and the order — shape information collapses far faster than correlation information — which is the claim the rest of the chapter rests on.

8.7 Scope not covered here

The estimator of Chapter 7 was also extended, within the same research package, to video sequences and to natural-image patches, and to a wavelet-domain hidden-Markov-tree generalisation of the chain structure used throughout this thesis. Those extensions are reported in the package’s own notebooks rather than reproduced here. The image

comparison in particular fits the two kernel families with different and not directly comparable factories, so it does not support a ratio between them, and none is quoted in this thesis.

8.8 Summary

Three of the five questions this chapter opened with are answered outright, one is answered for the estimator as implemented rather than in general, and one is left open.

Pointwise, against an unstructured network. At equal training-set size the structured estimator attains relative denoising error lower than the MLP by a factor of 7.0–17.6, across 8 sizes and 16 replicates. This is a ratio of errors at fixed data, not a ratio of the data required to reach a fixed error: the two curves do not cross in the range measured, so no data-equivalence factor is quoted.

Pointwise, against a structure-aware network. A large fraction of that margin is attributable to the MLP’s lack of any locality prior rather than to the estimator. Against a weight-shared window head, at the same headline protocol and every site scored, the advantage is $2.3\text{--}6.1\times$, and it increases across the four sizes tested rather than falling. The increase is a description of those four measurements; an architecture with an approximation floor would produce the same signature, and this design does not separate the two (Section 8.2). The window head was itself the winner of a screen against two architectures built to propagate further across the chain, which bounds what the baseline is — the strongest this screen found, not the strongest conceivable.

Capacity, for this estimator. Under this grid, this M-step and this iteration cap, no mixture capacity improves on a single Gaussian innovation at either training size, once fits are run to their plateau rather than to a fixed iteration count. Nothing here shows that a mixture cannot help a better-resolved or longer-run estimator. The predeclared equivalence test refuses “saturates by eight” as well: at $N = 128$ the $C = 8 \rightarrow C = 16$

step is resolved against the region in the direction of $C = 16$ being worse. Two things are not isolated — the iteration cap and the grid’s resolution floor, both of which disfavour the larger mixtures (Remark 8.1) — so the claim is about this estimator on this grid, not about mixture capacity in general.

Misspecification. The Markov structure the estimator relies on is not a single quantity of “how much structure”. It survives a violation it can absorb into its own parameters — a rank-one global latent, where the advantage never inverts even when half the marginal variance is a shared constant — and fails, well before generic intuition about misspecification would suggest, on a long-range coupling it structurally cannot represent.

Generation: open. Whether the pointwise advantage transfers to generation is not resolved here. The comparison available evaluates procedures whose fitted shape coordinates are unequally settled on the single coordinate that carries the metric, so Section 8.4 withdraws the reading rather than reversing it. The correction is the one applied to capacity above: select on a convergence criterion, then compare.

Chapter 9

The Rotating Ring: A Two-Dimensional Dynamic Object

The main line of the thesis closes with Chapter 8: a chain of scalar frames, its joint score computed exactly, its transition learned from noised data, and the value of that structure measured against trained networks. This chapter is a separate extension, placed after the main results because nothing in them depends on it.

Chapters 5 and 6 studied the scalar chain, where the joint score is either exactly Gaussian or a controlled departure from it. Chapter 4 set aside a second family of dynamic objects — planar trajectories near a circle — because they are nonlinear in the coordinates in which the diffusion acts. This chapter takes up the first of them. It shows that a Cartesian surrogate of the model is exactly solvable, that the model as drawn is solvable numerically to a measured accuracy, and that both admit the same structural description of the joint score.

It is included because it settles, exactly and in a model where the calculation can be done, a question the chain model raises but cannot answer — whether per-frame marginals can carry the information that distinguishes one dynamics from another. Chapter 4 settles this for the stationary Gaussian chain, where the marginals are α -free outright (Proposition 4.1), and shows that the same argument does not extend to non-Gaussian innovations. Here it becomes a theorem of a different kind: for one specific dynamical parameter, in a

model whose symmetry makes the calculation exact, the one-frame marginals carry *zero* Fisher information while the trajectory carries a positive amount.

The question the chapter answers is narrower than the models it uses.

How can the Markov structure of the clean rotating trajectory be exploited to compute and interpret the joint diffusion score?

Four measurable subquestions make this operational: how far the joint score differs from a collection of one-frame marginal scores; how strongly the observation at frame j influences the score block at frame k ; how that influence decays with the temporal lag $|j - k|$; and how many neighbouring frames are needed to reproduce the score block to a stated accuracy. Every answer below is a statement about one model at one parameter set, in the sense fixed in Section 1.3.

9.1 Model and notation

9.1.1 The clean process

The clean state is a radius and a phase, evolving in the internal time u as two independent diffusions,

$$dR_u = -\kappa(R_u - 1) du + \sqrt{2D_r} dB_u^{(r)}, \quad (9.1)$$

$$d\Theta_u = \omega du + \sqrt{2D_\theta} dB_u^{(\theta)}, \quad (9.2)$$

observed through the polar embedding

$$a_u = R_u \begin{pmatrix} \cos \Theta_u, & \sin \Theta_u \end{pmatrix} \in \mathbb{R}^2. \quad (9.3)$$

We use the simplest constant-angular-drift reading of the model; Section 9.7 records the alternatives.

A point of $\mathbb{R}^2$ is described by its distance from the origin and the angle it makes with the first axis, and at each point the plane carries the orthonormal moving frame

$$e_r = (\cos \theta, \sin \theta), \quad e_\theta = (-\sin \theta, \cos \theta), \quad (9.4)$$

in which e_θ is e_r turned by a quarter revolution. Moving along e_r changes the radius at fixed angle, moving along e_θ changes the angle at fixed radius, and the area element is $dx dy = r dr d\theta$. The decomposition matters because the two directions are treated very differently by (9.1) and (9.2): the radius is pulled back towards one, the phase drifts and diffuses freely. Measuring the consequence of that asymmetry for the score is one purpose of the chapter.

9.1.2 The radial drift is a confining force

Equation (9.1) is an overdamped Langevin equation, $dx_u = -\nabla V(x_u) du + \sqrt{2D} dB_u$, for the quadratic potential

$$V(r) = \frac{\kappa}{2} (r - 1)^2. \quad (9.5)$$

A potential stores energy, and the force it generates is minus its slope,

$$F(r) = -V'(r) = -\kappa (r - 1). \quad (9.6)$$

The sign carries the content: a system driven by $-V'$ moves downhill in V , since $\frac{d}{du} V(r_u) = V'(r) \dot{r} = -V'(r)^2 \leq 0$. Reading the two cases of (9.6), if $r > 1$ then $F < 0$ and the force points inward, and if $r < 1$ then $F > 0$ and it points outward. The radius is driven towards one from either side at a rate proportional to the displacement, with κ setting the stiffness. Overdamped Langevin dynamics equilibrates to $p_\infty \propto e^{-V/D}$, so the stationary radial law is $\mathcal{N}(1, D_r/\kappa)$.

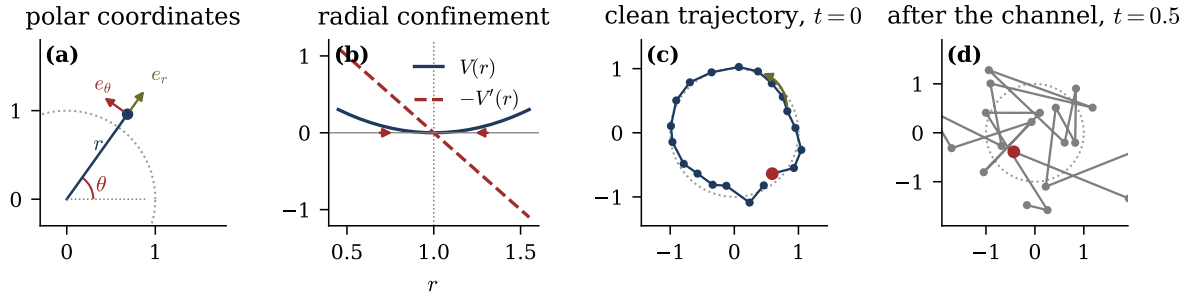


Figure 9.1: The rotating-ring model, at the parameters used throughout this chapter ($L = 20$, $\kappa = 2$, $D_r = 0.015$, $D_\theta = 0.005$, $\omega = 1$, $h = 2\pi/20$). (a) Polar coordinates and the moving frame (9.4). (b) The potential (9.5) and the force (9.6) it generates; the force changes sign at $r = 1$, which is therefore the stable radius. (c) One clean trajectory, a noisy coherently turning path near the unit circle; the arrow marks the sense of rotation. (d) The same trajectory after the channel (9.8) at $t = 0.5$, from a single standardised noise realisation.

9.1.3 One trajectory is one data point

The trajectory is sampled at internal times $u_k = kh$ for $k = 0, \dots, L-1$, and the complete datum is the stacked vector

$$a = (a_0, a_1, \dots, a_{L-1}) \in \mathbb{R}^{2L}. \quad (9.7)$$

The L frames are correlated coordinates of one data point, not L samples. Relative to Chapter 5 the only change of setting is that each frame is now two-dimensional, so the score has L blocks of size two rather than L scalar components; the forward channel is unchanged,

$$x = e^{-t}a + \sqrt{\Delta_t}\xi, \quad \xi \sim \mathcal{N}(0, I_{2L}), \quad (9.8)$$

which is the channel of Section 2.2 acting on all $2L$ coordinates at once. Figure 9.1 shows the model and the two clocks: internal time orders the frames along the path, diffusion time corrupts them simultaneously.

9.2 The gauge, and the exact score

9.2.1 The surrogate, and what it keeps

The surrogate retains a ring, a coherent rotation, and dependence along the trajectory, and drops the rest:

$$z_0 \sim p_\lambda(z) \propto \exp\left[-\frac{(\|z\| - 1)^2}{2\lambda}\right], \quad (9.9)$$

$$z_{k+1} = R_\psi z_k + \sigma \eta_{k+1}, \quad \eta_{k+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2), \quad (9.10)$$

with R_ψ the planar rotation through the one-frame angle $\psi = \omega h$.

Remark 9.1 (The surrogate is not a discretisation of the polar model). In (9.10) the ring acts only on z_0 . Later frames form a rotating Cartesian random walk and their radial variance grows without bound, whereas in (9.1) a confining force acts at every internal time. The surrogate is a solvable companion whose role is to exhibit the mechanism in closed form, not to approximate the model of Section 9.1.

9.2.2 The known rotation is an orthogonal change of frame

The rotation is the object to be removed, so we write it out:

$$R_\psi = \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix}. \quad (9.11)$$

Its columns are the standard basis vectors, each turned through ψ . They are unit vectors, since $\cos^2 \psi + \sin^2 \psi = 1$, and orthogonal, since $\cos \psi(-\sin \psi) + \sin \psi \cos \psi = 0$. In matrix form,

$$R_\psi^\top R_\psi = I_2, \quad \det R_\psi = 1, \quad R_\psi^\top = R_{-\psi}, \quad R_\psi R_\varphi = R_{\psi+\varphi}, \quad (9.12)$$

and orthogonality gives $\|R_\psi z\|^2 = z^\top R_\psi^\top R_\psi z = \|z\|^2$: lengths and inner products are preserved.

Lemma 9.2 (Rotations preserve the standard isotropic Gaussian). *If $\eta \sim \mathcal{N}(0, I_2)$ and R is orthogonal, then $R\eta \sim \mathcal{N}(0, I_2)$.*

Proof. Appendix E. □

The isotropy is essential. A rotation would not preserve $\mathcal{N}(0, \Sigma)$ for general Σ , and the reduction below is exact only because both the innovation in (9.10) and the channel (9.8) are isotropic.

Lemma 9.3 (Gauge reduction). *Define the de-rotated frame $y_k = R_{-k\psi} z_k$ and stack the per-frame rotations into $U_\psi = \text{diag}(I_2, R_{-\psi}, \dots, R_{-(L-1)\psi})$. Then*

$$y_{k+1} = y_k + \sigma \tilde{\eta}_{k+1}, \quad \tilde{\eta}_{k+1} \stackrel{iid}{\sim} \mathcal{N}(0, I_2), \quad (9.13)$$

U_ψ is orthogonal, and the noised density and score satisfy

$$P_t(z | \psi) = P_t^{(0)}(U_\psi z), \quad S_\psi(z, t) = U_\psi^\top S^{(0)}(U_\psi z, t), \quad (9.14)$$

where the superscript zero denotes the quantities of the non-rotating model.

Proof. Appendix E. □

The term *gauge* is used in the sense standard in physics, of a change of description that leaves the content untouched and may be chosen to simplify the equations. Equation (9.14) says that a known rotation adds no statistical difficulty: de-rotate the data, solve the non-rotating problem, rotate the score back. Since $U_\psi^\top = U_\psi^{-1}$, the operation is exactly invertible and destroys no information. It would not hold for an *unknown* ψ , which is a genuine latent variable (Section 9.7). Figure 9.2 shows the two frames.

9.2.3 The clean law after the gauge

Iterating (9.13) and cancelling the intermediate terms gives the anchored representation (E.1), whose derivation, together with the second-difference identity it satisfies, is in Appendix E. The identity is verified to machine zero (Table 9.1).

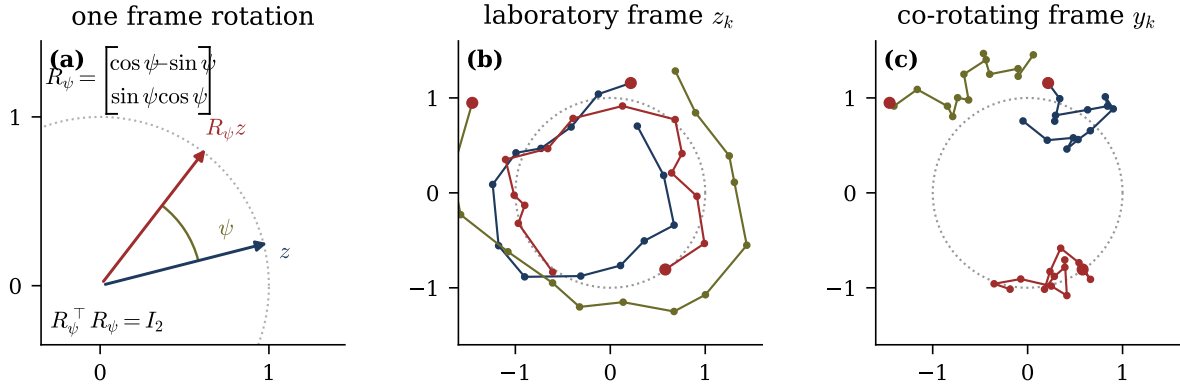


Figure 9.2: The rotation and the gauge. (a) R_ψ turns every vector through the same angle without changing its length, which is the statement $R_\psi^\top R_\psi = I_2$. (b) Three surrogate trajectories in the laboratory frame ($L = 14$, $\sigma = 0.16$, $\psi = 2\pi/14$; the marked point is $k = 0$), coherently circulating. (c) The same trajectories after U_ψ : a random walk anchored on the ring. Because U_ψ is orthogonal the reduction is exact at every diffusion time; the numerical round trip agrees to machine precision (Table 9.1).

9.2.4 The noised law and the exact score

Conditioning on the anchor, (E.1) gives $\mathbb{E}[y_i | A = \alpha] = \alpha$ and

$$\text{Cov}(y_i, y_j | A) = \sigma^2 \mathbb{E} \left[\left(\sum_{r \leq i} \tilde{\eta}_r \right) \left(\sum_{s \leq j} \tilde{\eta}_s \right)^\top \right] = \sigma^2 \min(i, j) I_2, \quad (9.15)$$

since $\mathbb{E}[\tilde{\eta}_r \tilde{\eta}_s^\top] = \delta_{rs} I_2$ leaves only the $\min(i, j)$ indices common to both sums. Compactly, $y | A = \alpha \sim \mathcal{N}(\mathbf{1} \otimes \alpha, G \otimes I_2)$ with $G_{ij} = \sigma^2 \min(i, j)$, where $\mathbf{1} \in \mathbb{R}^L$ is the all-ones vector. The two Kronecker products separate the temporal index from the spatial one: $\mathbf{1} \otimes \alpha$ repeats the same two-dimensional anchor in every frame, and $(G \otimes I_2)_{(i, \alpha'), (j, \beta')} = G_{ij} \delta_{\alpha' \beta'}$, so G carries all the temporal correlation while I_2 states that the two spatial coordinates are uncorrelated and share the same temporal covariance. Applying the channel and using $I_{2L} = I_L \otimes I_2$ to combine the covariances,

$$x | A = \alpha \sim \mathcal{N}(e^{-t} \mathbf{1} \otimes \alpha, C_t \otimes I_2), \quad C_t = e^{-2t} G + \Delta_t I_L, \quad (9.16)$$

in which C_t does not depend on α : the noised law is a two-parameter location mixture of one fixed Gaussian.

Proposition 9.4 (Exact surrogate score and response). *Let $q = C_t^{-1}\mathbf{1}$, $\beta = e^{-2t}\mathbf{1}^\top C_t^{-1}\mathbf{1}$ and $h = e^{-t}\sum_j q_j x_j \in \mathbb{R}^2$. Then the anchor posterior is*

$$p(\alpha | x) \propto \exp \left[-\frac{(\|\alpha\| - 1)^2}{2\lambda} - \frac{\beta}{2} \|\alpha\|^2 + h^\top \alpha \right], \quad (9.17)$$

and the score and response blocks are

$$S_k(x, t) = -\left(C_t^{-1}x\right)_k + e^{-t}q_k \mathbb{E}[A | x], \quad J_{kj}(x, t) = -\left(C_t^{-1}\right)_{kj}I_2 + e^{-2t}q_k q_j \text{Cov}(A | x). \quad (9.18)$$

Proof. Appendix E. □

The $2L$ -dimensional integral over clean trajectories has collapsed to a two-dimensional posterior over the shared anchor, which is the whole simplification and a direct consequence of (E.1). Its two moments are cheap: the prior and the $\|\alpha\|^2$ term are rotationally symmetric, so the posterior depends on the direction of α only through $h^\top \alpha$, and $\mathbb{E}[A | x] = \mu(\|h\|)\hat{h}$ with $\text{Cov}(A | x) = v_\perp I_2 + (v_\parallel - v_\perp)\hat{h}\hat{h}^\top$, three scalar functions of one scalar argument, tabulated once per diffusion time by one-dimensional quadrature.

The structural content of (9.18) is that the temporal coefficient of the second term is the outer product $qq^\top$, which is of rank one. The nonlocal part of the response is therefore not a chain of local couplings but a single shared latent variable seen by every frame. Figure 9.3 separates the two terms.

Proposition 9.5 (What the surrogate establishes). *Clean Markov locality does not survive marginalisation through the OU channel as locality of the noised score: by (9.18) the response has nonzero blocks at every lag for every $t > 0$. The nonlocal part is nonetheless structured, decomposing exactly into a near-diagonal chain term and a rank-one correction generated by the shared ring anchor.*

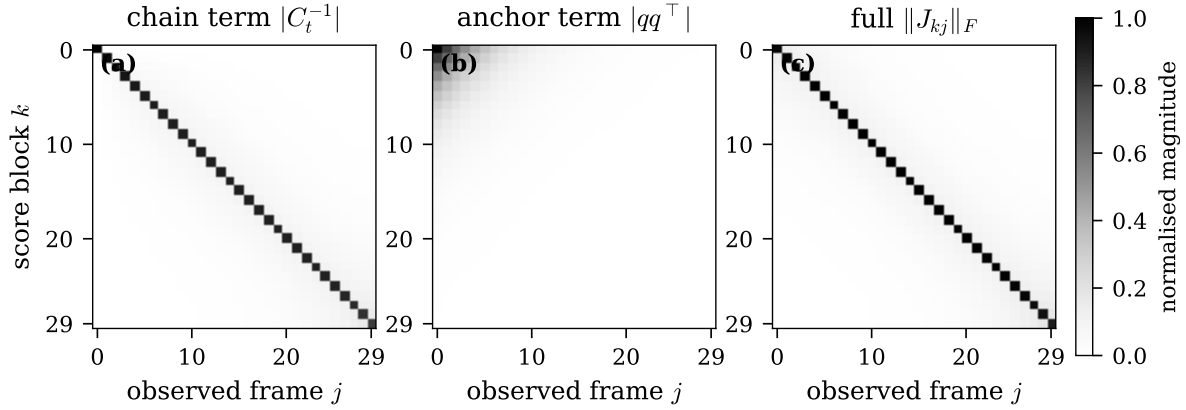


Figure 9.3: Exact decomposition of the surrogate response (9.18) at $t = 0.20$, $L = 30$, $\lambda = 0.05$, $\sigma = 0.15$, averaged over 250 noisy trajectories, each panel normalised by its own maximum. (a) The chain term $|C_t^{-1}|$ is concentrated near the diagonal. (b) The anchor coefficient $|qq^\top|$ is rank one and spread over all lags. (c) The mean norm of the complete 2×2 blocks. The analytic Jacobian agrees with central finite differences to 9.2×10^{-6} (Table 9.1).

Table 9.1: Validation of the rotating-ring computations. The response identity and the analytic surrogate Jacobian are compared with central finite differences of the score; the grid check compares two polar resolutions.

check	discrepancy	tolerance
posterior response identity vs finite differences	6.63×10^{-9}	10^{-7}
analytic surrogate Jacobian (9.18) vs finite differences	9.18×10^{-6}	10^{-4}
41×128 against 51×192 polar response grid	3.04×10^{-11}	10^{-7}
rotation gauge round trip (Lemma 9.3)	2.22×10^{-16}	10^{-12}
clean score on an exact co-rotating ring path	0 (exact)	10^{-12}

9.3 Numerical validation

Every identity of this chapter is checked against an independently computed reference, in the protocol of Appendix C.

Beyond Table 9.1 the checks include row normalisation of the transition kernels, global rotation equivariance of the score, stability under halving the finite-difference step, and the exact vanishing of the clean surrogate score on a perfect co-rotating ring path. One caveat on grid convergence: the difference between the 41×128 and 51×192 grids is small because the baseline likelihoods and kernels are already smooth at those resolutions,

and a deliberately coarse 11×24 grid does show visible deviations. The production grid is converged for these parameters rather than trivially grid-independent.

9.4 Marginal blindness

Everything above concerns how far influence *reaches* in the noised score. This section establishes something stronger and of a different kind: there is a parameter of this model about which the per-frame marginals carry not less information, but none.

Theorem 9.6 (Marginal blindness). *For the rotating ring with uniform initial phase, every single-frame marginal is independent of the rotation ψ , at every noise level. Consequently a model built from per-frame marginals carries exactly zero Fisher information about the rotation.*

Proof. Unrolling the recursion, $z_u = R_\psi^u z_0 + \sigma \sum_{k \leq u} R_\psi^{u-k} \eta_k$. Each η_k is isotropic Gaussian, and a rotation of an isotropic Gaussian has the same law, so $R_\psi^{u-k} \eta_k \stackrel{d}{=} \eta_k$ independently across k . The initial phase θ_0 is uniform on the circle, so z_0 has a rotationally invariant law and $R_\psi^u z_0 \stackrel{d}{=} z_0$. The two terms are independent, hence

$$z_u \stackrel{d}{=} z_0 + \sigma \sqrt{u} \xi, \quad \xi \sim \mathcal{N}(0, I_2), \quad (9.19)$$

in which ψ does not appear. The channel acts coordinatewise and independently of ψ , so the noised frame is likewise ψ -free. Therefore $\partial_\psi \log P_t(x_u) \equiv 0$ for every frame and every noise level, and a sum of per-frame log-densities cannot contain information no term contains. $\square$

Remark 9.7 (What the theorem does not say). The marginals are not uninformative in general. They determine the width and centre of the confining well and the walk noise perfectly well, and by (9.19) the per-frame law is a visible, non-Gaussian ring whose radius grows as $\sqrt{u}$. They are blind to the *rotation* specifically. Nor is the joint easy: the joint carries the information, and extracting it is what the estimator does.

The sharpness is what matters. A per-frame model is not less informative about the dynamics, nor informative only at low noise. It carries zero, exactly, at every noise level. Radii are not evidence.

9.5 Three estimation targets

The theorem furnishes an unusually clean experiment. This model carries a parameter the marginals can see — the confining potential $V(r) = (r - r_*)^2/2\lambda$ — and a parameter they provably cannot, so running both through the same estimator isolates the effect of the structure rather than of the method. Table 9.2 sets out three targets, each run in two arms differing only in which likelihood scores a trajectory:

Table 9.2: The three estimation targets of this chapter, and which of them a one-frame marginal can see. Target (a) is the control that makes the contrast readable and is not itself reported; (b) and (c) are what Theorem 9.6 makes invisible to marginals and the joint law nonetheless recovers.

	target	free parameters	visible to marginals?
(a)	the potential (λ, r_*)	two	yes
(b)	a single rotation ψ	one	no
(c)	a law over rotations $p(\psi)$	a distribution	no

Target (a) defines the contrast rather than supplying a result: it is the marginal-visible control against which (b) and (c) are read, and it is what makes the design isolate the structure rather than the estimator. **No measurement of (a) is reported in this thesis.** What follows is (b) and (c).

For (c) the estimator is expectation–maximisation over a grid of angles, with responsibilities $r_\mu(\psi) \propto p(\psi)P_t(x^\mu \mid \psi)$ and the update $p(\psi) \leftarrow M^{-1} \sum_\mu r_\mu(\psi)$. The same sufficiency that makes the expected-transition statistic a one-shot quantity on the chain (Chapter 7) appears here: the ψ -conditional log-likelihood does not depend on $p(\psi)$, so the matrix of log-densities is computed once and every iteration reweights it.

The blind arm is not a baseline chosen to lose. It is the theorem’s own control: its likelihood is constant in ψ , so its responsibilities equal $p(\psi)$, its update is the identity,

and $p(\psi)$ cannot leave its uniform initialisation. If it moves, the implementation is wrong rather than the theorem.

The blind arm returns exactly the null. Over sixteen seeds, seven budgets and twelve noise levels — 1344 cells — the per-frame arm’s estimate moves 0.0000 in total variation from its uniform initialisation, and over the 480 cells of target (b) its profile likelihood in ψ has range 0.00×10^0 nats. This is a machine-precision zero rather than a small number, which is what a theorem predicting exactly no information requires.

It is worth being precise about what that zero does and does not evidence. The per-frame likelihood is implemented by a function that takes no ψ argument, so its independence of ψ is a property of the code path, not a measurement: the zero confirms that the estimator is wired to the marginal it is supposed to use, and would catch a leak, but it cannot corroborate the theorem against an independently constructed alternative.

The independent check is a paired one. Trajectories are simulated separately at $\psi \in \{0, \pi/6, \pi/3, 1.1\}$, sharing no code with the estimator, and two statistics are computed on the *samples*. As a negative control, the energy distance between one-frame clouds at different ψ is compared against the same distance between two independent draws at *equal* ψ , which is the sampling floor: the floor is 1.5×10^{-2} and the cross- ψ distances are 2.6 to 4.7×10^{-3} — strictly *below* the noise of having drawn twice, both on clean trajectories and after OU noising at $t = 0.5$. As a positive control, the lag-one cross-covariance of those same samples must equal $s^2 R_\psi$, so its antisymmetric part is $\sin \psi$; measured, it is $-0.001, 0.503, 0.865, 0.893$ against $\sin \psi = 0, 0.500, 0.866, 0.891$.

The pairing is the point. A null result on its own cannot distinguish a true invariance from a statistic too blunt to detect anything; here the same samples, at the same size, move by 0.9 in the joint statistic and by less than their own resampling noise in the marginal one. Both controls are in `tests/test_ring_blindness_controls.py`.

The joint arm recovers the rotation, and loses it to the channel. Estimating a single angle, the error is $2.1 \pm 0.7^\circ$ pooled over the whole schedule. Recovering the two-

species law $p^*(\psi) = \frac{1}{2}\delta_{+\omega} + \frac{1}{2}\delta_{-\omega}$ against a truth of $\omega = 30^\circ$ and a null of 90° , Table 9.3 reports:

Table 9.3: Recovery of the two-species rotation law from the joint likelihood, against a truth of $\omega = 30^\circ$ and a null of 90° . The rotation survives the channel until $t \approx 1$ and has returned to the null by $t = 3$; the blind arm never leaves the null at any noise level.

t	0.05	0.32	0.68	0.98	1.43	3.00
median $ \hat{\omega} - \omega $	0.15°	0.24°	0.90°	2.86°	7.09°	58.8°

Information about the rotation survives until $t \approx 1$ and is gone by $t = 3$, where the estimate has returned to the null. That is the channel doing what a channel does, and it is legible here in a way it is not on the chain, because the null is known exactly rather than estimated.

Remark 9.8 (A normalisation that fails silently). The radial density of the confining well is naturally written unnormalised, and its normaliser depends on both λ and r_* . While only ψ is estimated the term is constant and cancels in every responsibility — which is why it can be absent for a long time without anything appearing wrong. The moment λ is estimated it stops being constant: a wider well contains more mass, the likelihood grows without bound in λ , and the estimate pins to the top of whatever grid it is given. The failure produces a smooth, confident, wrong answer rather than an error. It is recorded here because the same shape of mistake — a quantity that is constant in the case one has been testing, and is not constant in the case one moves to — is the most common way a correct derivation becomes an incorrect measurement.

9.6 What this chapter establishes

Three things, in increasing order of how much they constrain what a method can do.

First, the rotation is a *gauge*. Because U_ψ is orthogonal it commutes with the isotropic channel, so at known ψ the model reduces to $\psi = 0$ and the conditional density is closed form at any trajectory length — a quadratic form plus a single one-dimensional radial

Table 9.4: Status of the results of this chapter.

result	status
Gauge reduction and score transformation (Lemma 9.3)	exact
Exact surrogate score and rank-one response (Proposition 9.4)	exact
Marginal blindness (Theorem 9.6)	exact
Recovery of $p(\psi)$ from the joint, and its decay with t	numerical, sixteen seeds
The blind arm returning the null to machine precision	numerical, 1,824 cells; a consistency check on the estimator's wiring, not on the theorem
Paired sample-level controls: one-frame law invariant to ψ below the resampling floor, lag-one joint tracking $\sin \psi$	numerical, independent of the estimator
Recovery of the confining potential (λ, r_*)	not reported here (defines the control, see §9.5)
Behaviour in L , and inference of the phase rather than its use	open

integral. There is a ground truth here, computed rather than estimated, and it is what everything else is measured against.

Second, the clean law is local but the noised score is not. For $t > 0$, $S_k(x, t)$ is not a function of x_{k-1}, x_k, x_{k+1} alone: integrating out uncertain latent states creates posterior correlations at every lag, and the anchor term of (9.18) is rank one but touches every pair of frames. Clean Markovity buys a local *generative* law, not a local score.

Third, and this is the result that does not have an analogue on the chain, there is a parameter of the model about which per-frame marginals carry *exactly zero* information (Theorem 9.6), and it is nonetheless recoverable from the joint. The blind arm of §9.5 does not merely do worse; it cannot move off its initialisation, and over 1,824 cells it does not, to machine precision. That is a sharper statement than a comparison of error rates, because the null is known exactly rather than estimated. Table 9.4 places each result of the chapter in the vocabulary of Section 1.5, separating what is proved from what is measured and from what is left open.

9.7 Limitations

1. **One parameter set for the geometry.** The ring width, walk noise and trajectory length are held at a single baseline. Only the noise level and the data budget are swept, and nothing is claimed about scaling in L .
2. **Radius positivity.** The radial process is formally supported on all of $\mathbb{R}$. At the baseline the probability of a negative radius is negligible, but a strict model would reflect at the origin or use a positive diffusion.
3. **Known rotation in the gauge argument.** Lemma 9.3 assumes ψ known; §9.5 is what happens when it is not, and the two-species case is the smallest version of the speciation phenomenon discussed in Section 2.5.
4. **Grid approximation.** The radial integral is numerically converged at these parameters but is a quadrature, not a symbolic closed form. The validation of §9.3 bounds the error; it does not remove it.
5. **The marginal-visible control is not measured.** Target (a) of §9.5 specifies the contrast but no result for it is reported, so the claim that the two targets behave differently rests on the theorem rather than on a matched pair of measurements. Nothing else in the chapter depends on it.

The immediate extensions are to vary L systematically, to infer the angular phase rather than use an oracle basis, to add an unknown rotation class, and to test whether a time-dependent local architecture with receptive field $L_\epsilon(t)$ can match the full score.

Chapter 10

Discussion and Conclusions

10.1 Answers to the research questions

RQ1: the joint score and its couplings. For the stationary AR(1) chain under coordinatewise OU corruption the joint score is linear and known in closed form, $S(x, t) = -Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$, equal by the score-posterior identity to $(e^{-t}\mathbb{E}[a|x] - x)/\Delta_t$. Its coupling structure has a closed-form lifecycle: exactly tridiagonal at $t = 0$, filling one further diagonal per order in t by the band-fill law (5.13), maximally coupled at intermediate times, and returning to the identity at rate e^{-2t} . In the bulk a single quantity $q(\alpha, t) \in (0, |\alpha|]$ governs the posterior correlation length. This is exact for the Gaussian AR(1) model and for no other: it is the linearity of that model, not the chain structure alone, that makes the score a matrix.

RQ2: belief propagation and the Kalman smoother. The posterior of the noised chain lives on a tree-shaped factor graph (Figure 3.1), because the channel corrupts each coordinate independently and every observation therefore enters through a unary factor. One forward and one backward sweep compute all smoothing marginals. On the Gaussian chain the messages close algebraically in the Gaussian family, and the recursion is the Kalman filter followed by the RTS smoother, update by update. The two routes agree to 10^{-15} – 10^{-14} across the tested (L, α, t) grid, with an independently written implementation

reproducing the agreement. Tree exactness is a property of the graph and holds for any innovation law; closure in a finite-dimensional family is a property of the model, and holds here only because the model is Gaussian.

RQ3: what Gaussian messages cost. On Laplace chains, Gaussian-projected BP measured against validated grid quadrature has median relative score error 0.24 at $t = 0.05$, falling below 10^{-2} by $t \approx 1$ and below 10^{-3} by $t \approx 2.1$; no trial exceeds 10% relative error beyond $t \approx 0.98$, and the score direction is preserved far better than its magnitude (median cosine ≥ 0.976 throughout, worst single trial 0.879). What that number measures is identified exactly by Theorem 6.6: moment-projected sum-product returns the LMMSE estimator of the covariance-matched Gaussian model, so the comparison is full non-Gaussian posterior inference against the best second-order linear estimator, and nothing else. The error is amplified at small t by the exact factor e^{-t}/Δ_t , is worst for bimodal innovation laws, and decreases with stronger chain correlation. The reference is numerical, so these are measurements within its validated resolution regime rather than formulas.

RQ4: how fast remote evidence stops mattering. The influence of evidence at distance d on the posterior mean decays exactly as $q(\alpha, t)^d$. Separately, the error of the radius- r windowed score estimator is *measured* to decay at the same geometric rate over $r = 2, \dots, 13$. The two are distinct claims: windowing changes the boundary conditions of the recursion in a way the influence-coefficient argument does not cover, so the second is a numerical law over the tested range and not a corollary of the first. At equal locality budget, truncating the inference is more accurate than truncating the score matrix at small t (12.3% against 21.0% at $t = 0.05$). On the same model the per-node AMP/TAP closure returns the exact score — the mean fixed point is closure-independent — but a different posterior variance, with existence boundary, breakdown time $t_c(\alpha)$ and critical coupling $\alpha_c = \sqrt{2} - 1$ in closed form (Appendix B).

RQ5: recovering an unknown transition, and what must be checked first.

Fisher’s identity turns one complete forward–backward pass into the exact gradient of the marginal log-likelihood, with no differentiation through the recursion, so the E-step is structurally exact for any innovation law; the implementation adds controlled grid quadrature and a generalised M-step. The checks that recovery requires are the substance of the answer rather than a caveat on it. The channel acts twice — on the local EM rate through the fraction of missing information, and on the observed Fisher information at finite sample size — and it acts unevenly across coordinates: the fitted innovation shape settles at a median of 229 updates against 80 for the correlation, so a stopping rule read off the correlation trace certifies a kernel that has not converged in the coordinate every non-Gaussian claim depends on. A fixed iteration budget therefore compares convergence rates, not estimators, and any claim resting on fitted shape at a fixed budget must say so.

RQ6: what supplying the Markov assumption is worth. At equal training-set size the structured estimator attains $7.0\text{--}17.6\times$ lower pointwise denoising error than networks trained by denoising score matching on identical data, across the 8 sizes tested. Against a baseline that is given locality and weight sharing — the window head selected by a screen over three architectures on disjoint seeds — the advantage is $2.3\text{--}6.1\times$ (Figure 8.3). Three scope conditions travel with those numbers. The two arms are asymmetric in supplied information by construction, so this measures what a correct structural assumption is worth on a family where it holds exactly, and is not a ranking of learning algorithms. The difference between the two ratios is not a decomposition into an architecture term and an estimation term: architecture, capacity, parameter sharing, optimisation and implicit regularisation change together when the baseline is swapped. And both are ratios of errors at fixed data, not ratios of the data each method needs to reach a fixed error — the curves do not cross in the measured range, so no data-equivalence factor follows.

10.2 What is new

The exact structure of a chain diffusion score. The joint score of a noised Markov chain is characterised completely for the Gaussian case: its closed form, its eigenbasis at every diffusion time, the law by which its couplings fill and then decay, and the single parameter that controls posterior influence range. The identification of chain BP with the Kalman–RTS recursion makes the two classical literatures one object rather than two analogous ones.

An exact reading of what Gaussian closure computes. Where previous practice measured the error of moment-projected message passing as the error of an approximation, Theorem 6.6 identifies the estimator it returns. The measured Laplace gap is therefore the price of discarding everything beyond second moments — a property of the model class, shared by every innovation law with the same first two moments — and not a property of one implementation. The validated grid reference that makes the measurement possible is itself a contribution, since the functional recursion has no closed form.

Transition learning, and the measured value of structure. The transition law is estimated from noised sequences alone by exact chain inference inside EM, and the resulting estimator is compared against trained networks at equal data under a protocol that selects both arms on held-out validation. Two boundaries of the advantage are mapped rather than asserted: mixture capacity beyond a single Gaussian innovation buys nothing for this estimator under the grid and cap implemented here, and a chain-structured estimator partially accommodates a rank-one violation of the Markov assumption while failing on a long-range one at a coupling strength of roughly one tenth. The convergence measurement that separates rate from estimator is a methodological result in its own right, and it is what withdrew an earlier reading of the generative comparison.

10.3 Limitations

1. **Model class.** The exact solutions concern a one-dimensional state per frame and linear AR(1) dynamics, extended to a two-dimensional nonlinear state only in Chapter 9. The locality laws are proved for the Gaussian bulk; their non-Gaussian counterparts are only structurally delimited, and no low-rank structure is asserted for the Laplace $L = 2$ posterior Hessian beyond the narrow regime where it holds.
2. **The numerical reference.** The non-Gaussian results are numerical. Grid BP is a reference only within its validated resolution regime ($\sqrt{2t} \gtrsim 3h$ on the grid spacing h), and the Laplace closure error is a measured curve rather than a formula.
3. **Optimisation and finite seeds.** The learning results are statistical rather than exact: estimates over a finite number of seeds, reported with paired uncertainty, several resolved at only one of the two training sizes tested. Every capacity cell above $C = 4$ stops at an iteration cap rather than at tolerance, so a statistical disadvantage of larger mixtures is not separated from an optimisation or a discretisation one. The misspecification study probes exactly two violations of the Markov assumption; it establishes that the two are not interchangeable in effect, not a general theory of how much non-Markov structure a chain estimator tolerates.
4. **Generative fidelity, and transfer.** Whether pointwise accuracy and generative fidelity can move in opposite directions within one estimator is unresolved here: the comparison that addresses it evaluates procedures whose fitted shape coordinates are unequally settled, on the single coordinate the generative metric scores, so its reading is withdrawn rather than reversed. More broadly, these are properties of one controlled model. Whether trained score networks on real sequence data inherit them is untested here. The image and video extensions explored in the underlying research package are outside the scope of this thesis and are not reported.

10.4 Future work

Four directions, in the order in which they would most change what this thesis can claim.

First, the reverse-generation comparison, done with converged fits. Section 8.4 integrates the reverse SDE under the estimated EM–BP score and under two trained neural scores, but on fits the convergence analysis shows to be unequally settled. Selecting both arms on a convergence criterion rather than a fixed iteration count, then comparing, would answer the question the experiment was built for and would close the one question this thesis leaves open rather than answers.

Second, a grid-by-capacity design. The capacity result leaves two mechanisms entangled. A finer grid at fixed capacity would say how much of the larger mixtures’ disadvantage is their fits falling below the grid’s resolution rather than their overfitting the data, since only the second should survive a mesh refinement; a budget ladder at fixed capacity would do the same for the iteration cap.

Third, richer departures from the Markov assumption. The present study probes two violations at fixed strengths. A paired design across violation strengths, and further violation types beyond a global latent and long-range coupling, would test whether the finding that the two mechanisms are not interchangeable generalises. Discrete chains are untouched here: with a finite alphabet the BP messages are finite vectors and inference is exact with no closure at all, which would give a second solvable benchmark alongside the Gaussian one.

Fourth, extension to image and video models. The reverse-dynamics comparison used a small chain length and one innovation family. Whether the structural results survive longer sequences, other non-Gaussian innovation laws, and the move to data with spatial as well as temporal structure is the question that decides how far any of this carries beyond the solvable setting.

Appendix A

Gaussian Identities Used Throughout

This appendix collects, with proofs, the handful of Gaussian identities on which Chapter 5 relies. Notation: a Gaussian density in *moment* parameters is $\mathcal{N}(a; m, v)$; in *natural* parameters it is $\exp(-\frac{\lambda}{2}a^2 + \theta a + \text{const})$ with $\lambda = 1/v$, $\theta = m/v$.

A.1 Quadratic forms and Gaussian normalisation

For $J \succ 0$ (symmetric positive definite) and any h ,

$$\int_{\mathbb{R}^L} \exp\left(-\frac{1}{2}a^\top J a + h^\top a\right) da = \frac{(2\pi)^{L/2}}{\sqrt{\det J}} \exp\left(\frac{1}{2}h^\top J^{-1}h\right), \quad (\text{A.1})$$

by completing the square, $-\frac{1}{2}a^\top J a + h^\top a = -\frac{1}{2}(a - J^{-1}h)^\top J(a - J^{-1}h) + \frac{1}{2}h^\top J^{-1}h$, and translating. Consequently a density $p(a) \propto \exp(-\frac{1}{2}a^\top J a + h^\top a)$ is the Gaussian with mean $m = J^{-1}h$ and covariance J^{-1} : *reading off (J, h) from a log-density is the fastest route to posterior parameters*, used in Proposition 5.5.

A.2 Products of Gaussians in the same variable

$$\mathcal{N}(a; m_1, v_1) \mathcal{N}(a; m_2, v_2) \propto \mathcal{N}(a; m, v), \quad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \quad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}. \quad (\text{A.2})$$

Proof. Add the exponents; the sum of two quadratics in a is a quadratic in a whose a^2 -coefficient is $-\frac{1}{2}(1/v_1 + 1/v_2)$ and whose linear coefficient is $m_1/v_1 + m_2/v_2$; apply (A.1) in one dimension. In natural parameters the rule is addition: $(\lambda, \theta) = (\lambda_1 + \lambda_2, \theta_1 + \theta_2)$, which is why the BP updates of Chapter 5 are two additions per message. $\square$

A.3 Affine propagation (marginalisation)

If $a' = \alpha a + \eta$ with $\eta \sim \mathcal{N}(0, \sigma^2)$ independent of $a \sim \mathcal{N}(m, v)$, then

$$\int \mathcal{N}(a'; \alpha a, \sigma^2) \mathcal{N}(a; m, v) da = \mathcal{N}(a'; \alpha m, \alpha^2 v + \sigma^2). \quad (\text{A.3})$$

Proof. a' is a sum of independent Gaussians, hence Gaussian; its mean is αm and its variance $\alpha^2 v + \sigma^2$ by linearity of expectation and independence. Read in the reverse direction (solving for a given a Gaussian summary of a'), the same computation gives $\mathcal{N}(a; m'/\alpha, (v' + \sigma^2)/\alpha^2)$, the backward half-step of the BP sweep. $\square$

A.4 Gaussian conditioning

For a jointly Gaussian pair with precision blocks $\begin{pmatrix} J_{11} & J_{12} \\ J_{12}^T & J_{22} \end{pmatrix}$, the conditional of block 1 given block 2 is Gaussian with precision J_{11} (conditioning *reads off* the precision block) and mean shifted by $-J_{11}^{-1} J_{12} a_2$; marginalisation, dually, is trivial in covariance parameters. This precision/covariance duality (conditioning is local in J , marginalising is local in Σ) is the algebraic engine behind the cavity recursion (B.1): integrating a neighbour out of a tridiagonal precision produces the rank-one correction $-\beta^2/\lambda$ on the adjacent diagonal

entry, by the Schur complement

$$\begin{pmatrix} \lambda & \beta \\ \beta & J_d \end{pmatrix} \longrightarrow J_d - \frac{\beta^2}{\lambda}. \quad (\text{A.4})$$

A.5 Woodbury and the two routes to the score

The Woodbury identity $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$, specialised to $\Sigma_t = \Delta_t I + e^{-2t}\Sigma_0$, connects the noisy precision Q_t and the posterior precision $J = (e^{-2t}/\Delta_t)I + Q_0$:

$$Q_t = \frac{1}{\Delta_t} \left(I - \frac{e^{-2t}}{\Delta_t} J^{-1} \right), \quad (\text{A.5})$$

which is exactly the operator form of the score-posterior identity: substituting the posterior mean $m = J^{-1}h = J^{-1}(e^{-t}/\Delta_t)x$ into $S = (e^{-t}m - x)/\Delta_t$ gives $S = -Q_t x$ identically. The machine-precision agreement of the two score routes (Section 5.3) is the numerical shadow of (A.5).

Appendix B

AMP, TAP, and the Bulk Fixed Point

This appendix develops the per-node Gaussian closures of statistical physics, AMP and the TAP equations, on the Gaussian chain, and proves the exact breakdown announced in Remark 5.13. The results are complete, audited, and independently re-verified; they are collected here rather than in Chapter 5 because they branch off the thesis’s main line. Summary: on the chain, AMP still returns the exact score, but its variance closure fails past an exact breakdown time $t_c(\alpha)$, with critical coupling $\alpha_c = \sqrt{2} - 1$.

B.1 The bulk cavity fixed point of BP

Chapter 5 solved the bulk covariance by matrix algebra; the same formulas re-emerge from the message recursion, which is worth exhibiting because it localises where BP and AMP differ. In the bulk, parametrise the forward message (evidence absorbed) by its precision λ_k ; the closure lemmas of Section 5.6 give the scalar recursion

$$\lambda_{k+1} = J_d - \frac{\beta^2}{\lambda_k}, \tag{B.1}$$

with J_d, β the bulk parameters (5.20): absorbing one neighbour's summary costs the Gaussian-integration correction $-\beta^2/\lambda$. The fixed points are $\lambda^\pm = (J_d \pm \sqrt{J_d^2 - 4\beta^2})/2$, and since $J_d - 2|\beta| = e^{-2t}/\Delta_t + (1 - |\alpha|)^2/\sigma_\eta^2 > 0$ (Chapter 5), the attractive upper fixed point

$$\lambda^* = \frac{J_d + \sqrt{J_d^2 - 4\beta^2}}{2} \quad (\text{B.2})$$

exists for every $\alpha \in (-1, 1)$ and $t > 0$, with $|f'(\lambda^*)| = \beta^2/\lambda^{*2} < 1$: the sweep converges geometrically from any initialisation. Recombining the two directions and the local term, and using the fixed-point relation $\beta^2/\lambda^* = J_d - \lambda^*$,

$$\Lambda_{\text{bulk}} = J_d - \frac{2\beta^2}{\lambda^*} = 2\lambda^* - J_d = \sqrt{J_d^2 - 4\beta^2} \implies V_{\text{exact}} = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \quad (\text{B.3})$$

in exact agreement with Theorem 5.14. Belief propagation on the Gaussian chain has no phase transition: the cavity fixed point exists always, and the discriminant $J_d^2 - 4\beta^2$ is the quantity to watch as the closure is modified.

B.2 AMP/TAP: same score, different variance

AMP, the algorithmic form of the TAP equations (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016), is the message passing one uses when the factor graph is dense: each message is then a sum of many weak contributions, approximately Gaussian by a central-limit argument, and one tracks per-node means and variances instead of per-edge messages, with the Onsager term correcting the self-influence. Rigorous derivations of AMP from BP exist in that dense limit (Genovese and Piana, 2026). The chain is the opposite regime in every respect (diagonal measurement, coupling in the prior, degree two), so it isolates what the per-node closure costs. Viewing the posterior as a Gaussian Markov field with precision J and field h , the three schemes solve, at each node,

$$\text{mean field: } V_i = \frac{1}{J_{ii}}; \quad \text{AMP/TAP: } V_i = \frac{1}{J_{ii} - \sum_{k \neq i} J_{ik}^2 V_k}; \quad \text{BP: } V_i = (J^{-1})_{ii}. \quad (\text{B.4})$$

The AMP closure is the BP cavity with the exclusion of the receiving neighbour dropped ($V_{k \rightarrow i} \approx V_k$): an $O(1/N)$ change on a dense graph, but a change of fixed, order-one size on a chain with two neighbours.

Theorem B.1 (The score is closure-independent). *For a Gaussian posterior, the fixed-point condition for the means is the same linear system $Jm = h$ under mean field, BP, and AMP. All three schemes hence return the exact posterior mean and, via the score-posterior identity, the same exact joint score, for every (L, α, t) , regardless of whether their variance closures are accurate or even solvable.*

The proof is one line: in each scheme the mean update at node i reads $m_i \leftarrow (h_i - \sum_{k \neq i} J_{ik} m_k) / J_{ii}$ at the fixed point; the closures modify the uncertainty bookkeeping, not the linear relation among means, and the unique solution of $Jm = h$ is the exact posterior mean. Numerically, AMP means agree with exact means to 4.3×10^{-12} and AMP scores with BP scores to 1.7×10^{-12} across all tested configurations.

For the variances the story splits. In the bulk the AMP closure (B.4) reads $V = 1/(J_d - 2\beta^2 V)$; the BP cavity is $V_{\text{cav}} = 1/(J_d - \beta^2 V_{\text{cav}})$. The entire difference between BP and AMP on the chain is a factor of two: both neighbours included where BP excludes one.

Theorem B.2 (AMP bulk variance, breakdown time, critical coupling). *The AMP bulk closure is the quadratic $2\beta^2 V^2 - J_d V + 1 = 0$, with physical root*

$$V_{\text{AMP}} = \frac{J_d - \sqrt{J_d^2 - 8\beta^2}}{4\beta^2}, \quad (\text{B.5})$$

which exists iff $J_d \geq 2\sqrt{2}|\beta|$, strictly stronger than BP's always-true $J_d > 2|\beta|$. Since $J_d(t)$ decreases monotonically in t , the existence condition fails, if ever, past a single time; solving it as an equality gives

$$\boxed{t_c(\alpha) = -\frac{1}{2} \log \frac{g}{1+g}, \quad g(\alpha) = \frac{2\sqrt{2}|\alpha| - 1 - \alpha^2}{1 - \alpha^2},} \quad (\text{B.6})$$

finite exactly when $g > 0$, i.e.

$$\boxed{|\alpha| > \alpha_c = \sqrt{2} - 1 \approx 0.4142.} \quad (\text{B.7})$$

For $|\alpha| \leq \alpha_c$ the AMP variance has a fixed point at every diffusion time; for $|\alpha| > \alpha_c$ it ceases to exist for all $t > t_c(\alpha)$. In the weak-coupling regime the two closures agree to second order, with

$$V_{\text{AMP}} - V_{\text{exact}} = \frac{2\beta^4}{J_d^5} + O\left(\frac{\beta^6}{J_d^7}\right) > 0 : \quad (\text{B.8})$$

AMP always overestimates the bulk variance, at fourth order in the coupling.

Toolbox B.1 — Roots, branch, and the phase boundary

Branch selection. The quadratic has roots $V_{\pm} = (J_d \pm \sqrt{J_d^2 - 8\beta^2})/(4\beta^2)$; as $\beta \rightarrow 0$, $V_- \rightarrow 1/J_d$ (the decoupled answer) while V_+ diverges, so V_- is the physical branch, and the attractive fixed point of the damped iteration used in practice. *Critical coupling.* $g > 0 \iff \alpha^2 - 2\sqrt{2}|\alpha| + 1 < 0 \iff |\alpha| \in (\sqrt{2} - 1, \sqrt{2} + 1)$; intersecting with $|\alpha| < 1$ leaves $|\alpha| > \sqrt{2} - 1$. Consistency check at $t = \infty$: there the posterior is the bare prior, and the existence condition applied to Q_0 itself reads $(1 + \alpha^2) \geq 2\sqrt{2}|\alpha|$, i.e. AMP applied to a bare AR(1) chain has a variance fixed point iff $|\alpha| \leq \sqrt{2} - 1$, which is (B.7) read backwards. *Weak coupling.* With $\epsilon := \beta^2/J_d^2$: $(1 - 4\epsilon)^{-1/2} = 1 + 2\epsilon + 6\epsilon^2 + O(\epsilon^3)$ for the exact variance; $1 - \sqrt{1 - 8\epsilon} = 4\epsilon + 8\epsilon^2 + O(\epsilon^3)$, divided by $4\epsilon J_d$, gives $V_{\text{AMP}} = (1 + 2\epsilon + 8\epsilon^2 + O(\epsilon^3))/J_d$; subtract. $\square$

Numerically, on an $L = 12$ chain at $\alpha = 0.8$: the AMP score error stays at 10^{-13} – 10^{-12} for every t , while the variance error is 1.3×10^{-4} at $t = 0.05$, 3.5×10^{-2} at $t = 0.2$, and past $t \approx 0.23$ the variance update has no positive fixed point at all. The audit further verifies the closed form (B.5) against the converged iteration to 2.5×10^{-12} ; the existence condition at all 180 points of an (α, t) scan; t_c against the bisected flip point of the iteration to 6.6×10^{-6} relative; convergence below α_c even at $t = 50$; and the weak-coupling ratio $(V_{\text{AMP}} - V_{\text{exact}})/(2\beta^4/J_d^5) \rightarrow 1$ within 0.4%. The interpretation is not that AMP is wrong; on a dense benchmark (random $N = 600$ symmetric positive-definite

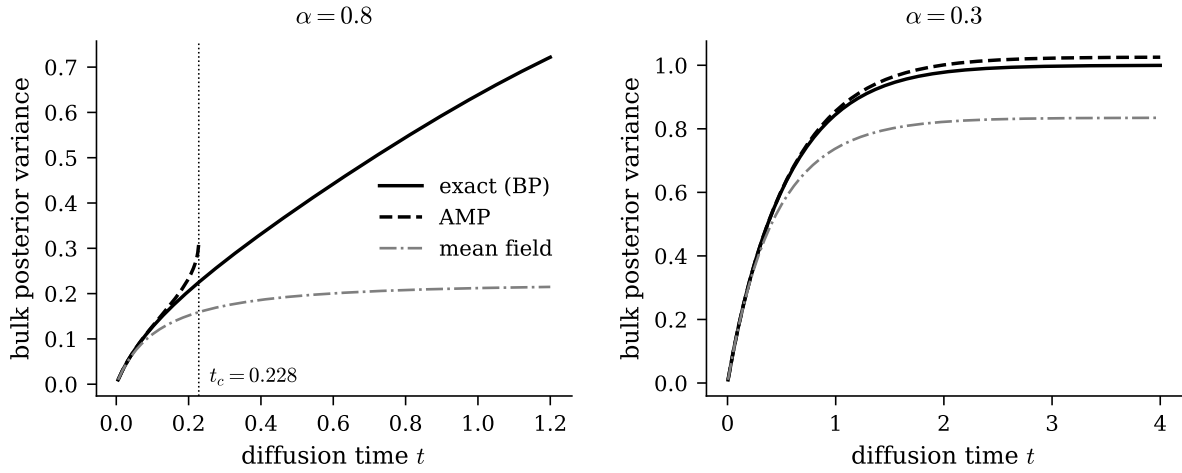


Figure B.1: The three closures in closed form. Left ($\alpha = 0.8 > \alpha_c$): exact = BP, AMP (B.5), and mean field; the AMP branch terminates with a square-root singularity at the breakdown time $t_c(0.8) = 0.228$ from (B.6) (dotted line); beyond it AMP has no variance fixed point. Right ($\alpha = 0.3 < \alpha_c$): AMP exists at every t and slightly overestimates the variance, as predicted by (B.8).

precision) the same closure gets variances to 1.7×10^{-3} where mean field errs at 4×10^{-2} . It is built for a different graph: the central-limit justification of the per-node closure needs many incoming messages, and a chain provides two.

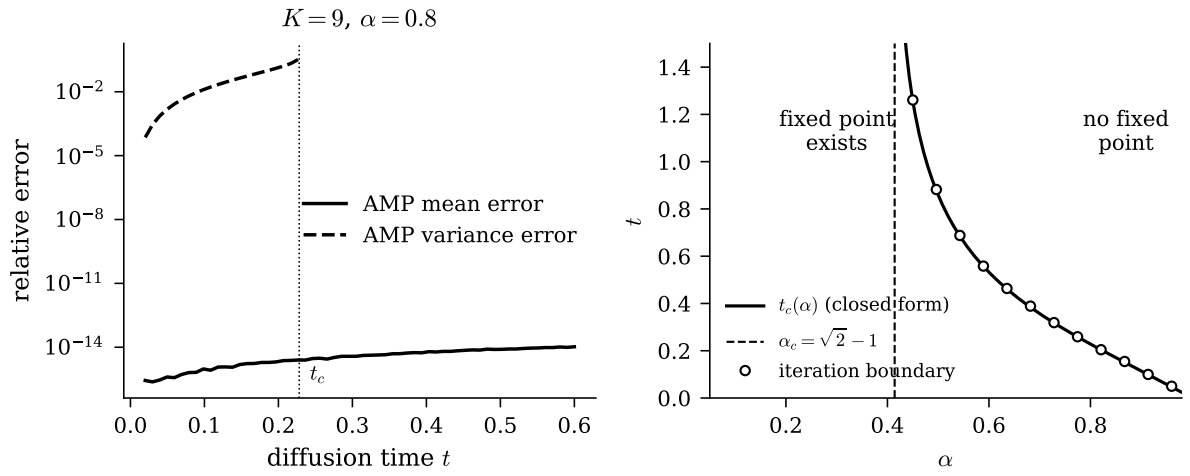


Figure B.2: BP against AMP on a finite chain ($L = 9$, $\alpha = 0.8$). Left: the AMP mean error (about 10^{-12} , flat in t : Theorem B.1) against the AMP variance error, which grows with t up to the breakdown time t_c (dotted line). Right: the phase boundary of the AMP variance fixed point in the (α, t) plane: the solid curve is the closed form $t_c(\alpha)$ of (B.6), the dashed line is $\alpha_c = \sqrt{2} - 1$, and the open circles mark the breakdown as located independently by damped fixed-point iteration; the audit suite performs the same comparison over 180 points of this plane.

Appendix C

Reproducibility

C.1 Repository layout

The background derivations this thesis cites but does not carry — statistical mechanics, stochastic calculus and the diffusion-model constructions — are in the companion compendium, at full length, together with the three background chapters as they were first written before being merged into Chapter 2. Nothing that was cut for length was discarded.

All source material lives in a single repository, laid out as in Table C.1.

Table C.1: Where each part of the work lives. Paths are relative to the repository root.

Path	Content
<code>overleaf/</code>	the four documents, their shared figures and the generated table and macro files
<code>research/gaussian-bp/</code>	Gaussian chain note, code, 72-check audit
<code>research/bp-from-scratch/</code>	independent BP derivation and implementation
<code>research/unified-note/</code>	consolidated working note, code, 55-check audit
<code>research/nongaussian-bp/</code> <code>src/, experiments/</code>	grid BP, EM, and the estimator comparisons implementation and one driver per experiment
<code>hpc/, tools/</code>	deployment/provenance, and the table and figure generators
<code>outputs/frozen/</code>	the certified run directories the generators read
<code>research/gaussian-ar1-bp/</code>	Gaussian AR(1) package (BP = matrix score)
<code>research/initial-experiments/</code>	early experiments and toy-model material
<code>research/experiment1-rotating-ring/</code>	rotating-ring package: code, data, validation

C.2 Audit protocol

Every closed-form claim is verified by a deterministic check against an independently computed reference (brute-force linear algebra, dense grid quadrature, or finite differences), collected in audit drivers that exit non-zero on any failure. The audit driver of `research/gaussian-bp` runs 72 checks: assembly and inversion of Σ_t ; the score-posterior route against the matrix score; the $L = 2$ closed forms; Convention-A BP against the matrix posterior (means, variances, scores); bulk variance, covariance decay

q^d , and the recombination identity; AMP mean and score closure-independence; the AMP variance formula, existence boundary, breakdown time, and weak-coupling law; the locality slopes; the band-fill coefficients; spectral preservation and the large- t return. The audit driver of `research/unified-note` runs 55 checks on the consolidated note, including the Laplace $L = 1$ quadrature checks and the one-step boundary-message identity. `research/nongaussian-bp/` contains experiment drivers with logged master-error-identity residuals (10^{-15} – 10^{-12} throughout), grid self-convergence gates, and per-trial CSV outputs; the grid calibration of Section 6.4 is experiment 01, the closure measurement of Section 6.5 experiment 02. `research/experiment1-rotating-ring/` carries the package behind Chapter 9: the exact surrogate score and Jacobian, the polar-grid recursion, the response diagnostics in a single module, and a validation driver that writes `validation.json` and exits non-zero on any failure. Its five checks are the ones reported in Table 9.1, and every number quoted in Chapter 9 is regenerated from the curated CSVs of that package rather than entered by hand. All checks were re-executed on the build machine at the time of writing: 72/72, 55/55 and 5/5 pass, and all experiment gates are green.

C.3 Figure-to-code map

Every figure is committed as a vector PDF under `overleaf/shared/figures/`, one directory for all four documents so that two of them cannot render different pictures under the same name. All are white background; randomness, where used, is seeded. Three packages compute the panels, and Table C.2 says which computes which.

The learning figures come from `tools/make_figures.py` in the `nongaussian-bp` package, which reads the frozen outputs directly and is the same script that writes the generated tables. The Gaussian-chain and AMP figures come from `code/experiments.py` in `gaussian-bp`, which recomputes each panel from the closed forms of that package’s audit; the same panels appear there under `figures/` as PNG, with the same file names. The ring figures come from `code/make_figures.py` in the `rotating-ring` package, whose

closed-form panels are recomputed at build time and whose measured panels read that package's curated CSVs; they are carried here under the `fig_ring_*` names rather than that script's numbered ones.

One caveat belongs with this. The restyling step that turned the older packages' outputs into the uniform vector PDFs used here was never committed and no longer exists, so those figures cannot be regenerated in their present style; this is why they, alone among the figures here, still carry Type 3 fonts. The computations behind every panel remain in the packages named above and are re-runnable, so what was lost is a styling pass rather than a result, and no quantitative claim depends on it: every number is generated, and none is read off a figure.

Table C.2: Every figure in this thesis, the file it is committed as, and the package whose code computes the numbers in it. The styling caveat above applies to the figures from the older packages; the two schematics are drawn in $\text{\TeX}$ and compute nothing.

Figure	File	Numbers computed by
1.1	drawn in $\text{\TeX}$ (<i>TikZ</i>)	schematic; no computed quantity
3.1	drawn in $\text{\TeX}$ (<i>TikZ</i>)	schematic; no computed quantity
4.1	fig_toymodel_score.pdf	nongaussian-bp, closed forms
5.1	fig_band_fill.pdf	gaussian-bp, closed forms
5.2	fig_tridiag_loss.pdf	gaussian-bp, closed forms
5.3	fig_local_vs_full.pdf	gaussian-bp, closed forms
6.1	fig_laplace_k1.pdf	unified-note, quadrature
6.2	fig_laplace_closure.pdf	nongaussian-bp exp. 02, per-trial CSV
7.1	fig_grid_domain.pdf	nongaussian-bp exp. 01 and 18
7.2	fig_em_diagnostics.pdf	nongaussian-bp exp. 06 and 18
8.1	fig_sample_efficiency.pdf	nongaussian-bp exp. 07, frozen
8.3	fig_value_of_structure.pdf	nongaussian-bp exp. 07 and 31, frozen
8.2	fig_screening.pdf	nongaussian-bp exp. 31 screen, frozen
8.4	fig_capacity.pdf	nongaussian-bp exp. 32, frozen
8.5	fig_nonmarkov.pdf	nongaussian-bp exp. 21, clean rerun
B.1	fig_bulk_variance.pdf	gaussian-bp, closed forms
B.2	fig_bp_vs_amp.pdf	gaussian-bp, closed forms and iteration
9.1	fig_ring_model.pdf	ring package, closed forms and simulation
9.2	fig_ring_gauge.pdf	ring package, closed forms and simulation
9.3	fig_ring_surrogate_response.pdf	ring package, exact surrogate score

C.4 The learning experiments of Chapters 7 and 8

The closed-form and quadrature studies above are deterministic and run on a laptop. The estimator comparisons are neither, and they are gated differently.

Provenance. Runs deployed by `hpc/deploy_clean.sh` are certified: the script refuses to build from a dirty tree, ships a `git archive` of the commit rather than the working tree, records the SHA-256 of that tarball, and extracts on the cluster under the commit id so that a run cannot overwrite or edit its own source. The motivating failure is preserved in that script’s header: an earlier two-step stamp-then-rsync path produced `outputs/exp_12_scaled/`, whose recorded commit cannot have run the recorded command, and whose numbers are therefore not recoverable and are not cited in this thesis. `tools/provenance_gate.py` intersects a run’s recorded dirty-file list with the import closure of its own experiment module, so that uncommitted work elsewhere in the tree is distinguished from uncommitted work on the code path that produced the numbers. That distinction does the work below, because the two are not equally serious.

How each run is certified. Not every experiment here was run after `deploy_clean.sh` existed, so three routes to certification appear. Nothing cited in this thesis is uncertified. The development record behind each route — which deployments failed, and why each check was added — is in the companion compendium.

Certified. `exp_31`, the architecture screen and the confirmatory comparison, and `exp_32`, capacity, were deployed by the script and pass `require_clean` unmodified. Every number in Sections 8.2 and 8.3 comes from them.

Legacy digest, empty intersection. `exp_07` — the headline efficiency comparison and its validation-selection audit — with `exp_27` behind the settling times of Section 7.5, and `exp_18`, predate the script. They record a commit, a host, a job id and a resolved configuration, but no archive digest, so the plain gate refuses them. What makes them citable is narrower and checkable: the intersection of each run’s recorded dirty list with its own import closure is *empty*, so no uncommitted edit could have reached the code that produced the numbers. `make_tab_efficiency.py` recomputes that intersection on every run and refuses if it is ever non-empty, rather than resting on a check performed once by hand. The one `exp_07` row where the intersection was *not* empty — $N = 8192$, whose dirty list contained the experiment, the frozen configuration and `src/em.py` — is

withdrawn and appears nowhere in this thesis.

Certified after rerun. `exp_21`, behind Table 8.5 and Figure 8.5, is reported from `outputs/frozen/exp_21_clean`: fifteen cells from two git-archive deployments, every one with an empty dirty list. It replaces an earlier run whose recorded commit did not reconstruct the program, and the Gaussian arm reproduces that run on every number quoted here, to the precision printed. Because the Laplace and Gaussian arms were deployed separately, the rerun spans two commits, so `make_tab_nonmarkov.py` refuses unless no single cell pools commits *and* every file in the experiment’s sixteen-file import closure is byte-identical between them. Both hold, and the check runs on every regeneration.

Generated numbers. No efficiency, structured-baseline or aggregation number in this thesis is typed. Each is defined once as a macro in `overleaf/shared/sections/`, written there by one of ten generators under `research/nongaussian-bp/tools/`, each reading the recorded outputs directly:

```
make_tab_efficiency  make_aggregation_robustness
make_tab_structured  make_aggregation_structured
make_tab_screening   make_convergence_numbers
make_tab_capacity     make_grid_diagnostics
make_tab_nonmarkov   make_figures
```

The prose then cites the macro rather than the value, so tables, figures and text share a size grid and an aggregation rule by construction rather than by inspection.

What Table 8.1 covers. Four qualifications, kept here rather than in the caption, which had grown into a methods subsection set in small type. Both arms select their optimisation budget on a disjoint validation bundle — the network its parameterisation *and* training length, EM–BP its iteration count — and that choice agrees with the test-set optimum in 81.1% and 77.6% of cells respectively, so the protocol is symmetric and the selection is close to an oracle without being one. The $N = 4096$ row uses a raised budget, the caps having been set for $N \leq 2048$; rerunning $N = 2048$ at that budget on the same

seeds moves its ratio by -0.16 ± 0.18 , so the caps are selected without binding. The resolution gate excludes 16 of 1,536 cells, because in those the narrowest fitted mixture component is under two grid cells wide; including them changes no ratio by more than 0.08. The network wins in 2 of the 1,520 reported cells, both at the smallest sizes — Appendix F.2 resolves where the advantage is largest.

What Table 8.3 covers. The confirmatory run scores three regions per cell — all, bulk, centre — of which the table displays two; the third is the predeclared interior slice, and it is the one the architecture screen selects on. That is why the 192 scored cells are $16 \times 4 \times 3$ and not the $16 \times 4 \times 2$ a two-column table suggests. The count is a descriptive diagnostic and not a replication count: regions are a scoring choice applied after fitting, and sizes share nothing but the seed, so the inferential unit behind each displayed row is the 16 paired seeds. Both arms are checkpointed and selected on a validation bundle disjoint from the test bundle, and neither bundle is shared with training.

The architecture screen. Table 8.2 reports it in full. Its source is `screening.csv` under `outputs/frozen/exp_31_screen/` at commit 569a67c, which is the artifact the confirmatory run names in its own recorded configuration (`winners=auto:` that path) rather than a reconstruction. 32 configurations over 4 development seeds, at two training sizes, with learning rates 0.001, 0.003, both output parameterisations, and a checkpoint ladder from 500 to 64000 steps shared across all three architectures. Selection is on mean risk over the development seeds, scored on the bulk region; the development seeds are numbered from 10000 and share no seed with the confirmatory set, so the reported interval is not a selection artefact. The winner is the window head at radius 4, width 64, learning rate 0.003, eps parameterisation. Parameter counts across the grids are 4.8k–21.4k (window), 9.0k–34.3k (message passing) and 66.4k–264k (dilated convolution): the arm with the most capacity placed last, which is the relevant check on whether the comparison was tuned evenly.

The convergence sweep, and the one exception to the gate above. The settling times of Section 7.5 come from `outputs/frozen/exp_27_*`, run on 16 August 2026 under SLURM job 628601 and its shards. That sweep predates `hpc/deploy_clean.sh`, so its records carry the job id, host and resolved configuration but no source-archive digest, and `provenance_gate` refuses it by default; `tools/make_convergence_numbers.py` passes `allow_legacy` and this paragraph is the disclosure that goes with it. The quoted numbers come from the 800-iteration shard, the only run length in the sweep at which no configuration settles at its own cap; the 400-iteration shard is censored by its cap and its settling times are lower bounds, so it is used only for the count of configurations that miss a 40-iteration budget, which censoring cannot inflate. The measurement replaces a figure of “on the order of 2000 iterations” that appeared in earlier drafts and was never measured: it had been carried over from a diagnostic that ran a fit *out to* 2000 iterations to show its per-edge gain had plateaued.

The transition array’s normalisation. Stated in Section 7.4 and repeated here because it determines which probabilistic object the M-step optimises: columns are *not* renormalised under the quadrature weights, so the fitted object is the continuum kernel read through a quadrature rule and the surrogate carries no θ -dependent log-normaliser. The column-mass residual is the measured distance to a normalised finite-state kernel.

Excluded inputs. $N = 8192$ is excluded from both Table 8.1 and Figure 8.1: the intersection of its dirty list with `exp_07`’s import closure contains the experiment, the frozen configuration and `src/em.py` themselves, so the run is not recoverable. The $N = 4096$ rows are retained: their intersection is empty, so no edit in that tree could have moved those numbers.

C.5 Environment

Python ≥ 3.10 with NumPy, SciPy, pandas and Matplotlib. The neural baselines of Chapter 8 are implemented in pure NumPy (`src/nnet.py` for the fully connected de-

noiser, `src/seq_nets.py` and `src/local_head.py` for the locality-respecting architectures) with an explicit `numpy.random.Generator` driving initialisation and minibatch order, so training is stochastic but reproducible from the recorded seed. Two optional dependencies are used and neither is on the exactness path: CuPy, selected by the `BP_DEVICE` environment variable, runs the same single BP implementation on a GPU — `tests/test_backend_parity.py` asserts the two devices agree, so a device change fails a test rather than moving a number quietly — and PyTorch, imported lazily in `src/fid.py` alone, for the generative metric. Arithmetic is float64 throughout.

The sweeps of Chapters 7 and 8 run under SLURM from the batch scripts in `hpc/`, one per experiment. The document compiles with a single run of `tectonic main.tex`.

Appendix D

Gaussian Vectors and Matrix Tools

The multivariate Gaussian in both parametrisations, the spectral theorem, and the matrix-inversion identities used throughout the thesis.

D.1 The multivariate Gaussian, in both parametrisations

Definition D.1 (Multivariate Gaussian). A random vector $a \in \mathbb{R}^L$ is Gaussian with mean m and (positive definite) covariance Σ , written $a \sim \mathcal{N}(m, \Sigma)$, if its density is

$$\mathcal{N}(a; m, \Sigma) = \frac{1}{(2\pi)^{L/2} \sqrt{\det \Sigma}} \exp \left[-\frac{1}{2} (a - m)^\top \Sigma^{-1} (a - m) \right]. \quad (\text{D.1})$$

Expanding the quadratic form shows that the same density can be written in *natural* (information) parameters,

$$\mathcal{N}(a; m, \Sigma) \propto \exp \left[-\frac{1}{2} a^\top J a + h^\top a \right], \quad J = \Sigma^{-1}, \quad h = \Sigma^{-1} m, \quad (\text{D.2})$$

with J the *precision matrix* and h the precision-weighted mean. The two parametrisations are dual, and the duality organises everything that follows (Bishop, 2006; Koller and Friedman, 2009):

- *Linear maps.* If $a \sim \mathcal{N}(m, \Sigma)$ then $Aa + b \sim \mathcal{N}(Am + b, A\Sigma A^\top)$: Gaussianity is preserved by affine transformations, with moments transforming linearly. In particular a sum of independent Gaussians is Gaussian with means and covariances adding.
- *Marginalisation is easy in Σ .* The marginal of a sub-block a_1 of $a = (a_1, a_2)$ is Gaussian with mean m_1 and covariance Σ_{11} : simply read off the block.
- *Conditioning is easy in J .* Partitioning $J = \begin{pmatrix} J_{11} & J_{12} \\ J_{12}^\top & J_{22} \end{pmatrix}$, the conditional law of a_1 given a_2 is Gaussian with precision J_{11} (read off the block) and mean shifted by $-J_{11}^{-1} J_{12} a_2$. Eliminating a variable instead produces the *Schur complement* correction on its neighbours. Proofs of both statements are in Appendix A.
- *Products are easy in J .* Multiplying two Gaussian densities in the same variable adds their natural parameters, $(J, h) \mapsto (J_1 + J_2, h_1 + h_2)$. Bayesian updating with Gaussian likelihoods is addition.

The dual reading of zeros is used constantly in this thesis. A zero in the *covariance* means marginal independence of two coordinates. A zero in the *precision* means *conditional* independence given all the others: from the conditioning rule, if $J_{ij} = 0$ then a_i and a_j do not interact once the rest is fixed. A Gaussian vector can be, and in this thesis always is, globally correlated (dense Σ) while conditionally local (sparse J). Which matrix one should look at depends on the question, and confusing the two is a common source of error in reasoning about Gaussian dependence.

D.2 Matrix tools: the spectral theorem and matrix inversion

Since Gaussian computation is matrix computation, we record the two linear-algebra facts the research chapters use constantly.

Theorem D.2 (Spectral theorem for symmetric matrices). *Every symmetric $A \in \mathbb{R}^{L \times L}$ has an orthonormal basis of eigenvectors: $A = U \operatorname{diag}(\omega_1, \dots, \omega_L) U^\top$ with $U^\top U = I$ and real eigenvalues ω_i .*

Three consequences matter here. First, analytic operations act on eigenvalues alone: if no eigenvalue vanishes,

$$A^{-1} = U \operatorname{diag}(1/\omega_i) U^\top, \quad \text{and more generally} \quad f(A) = U \operatorname{diag}(f(\omega_i)) U^\top.$$

Second, affine images share the eigenbasis: $(cA + dI)u_i = (c\omega_i + d)u_i$, so a whole family of matrices of the form $c(t)A + d(t)I$ is diagonalised once and for all by the U of A . Chapter 5 uses exactly this fact to diagonalise the diffusion problem at every noise level simultaneously. Third, for the symmetric Toeplitz covariances produced by stationary processes (Section 2.1), the eigenvectors are asymptotically the Fourier modes of the sequence (Brockwell and Davis, 1991): stationary Gaussian problems decouple, in the large- L limit, into independent scalar problems per frequency.

Inversion, when the inverse is not wanted eigenvalue by eigenvalue, has two workhorse expansions. The *Neumann series*

$$(I - B)^{-1} = \sum_{m \geq 0} B^m, \quad \text{convergent for } \|B\| < 1, \quad (\text{D.3})$$

expresses the inverse of a near-identity matrix as a sum of powers; it is the tool behind the small- t expansions of Chapter 5, where powers of a banded matrix have growing but controlled bandwidth. The *Woodbury identity* relates the inverse of a matrix to the inverse of a low-rank or structured perturbation of it; its statement, proof, and the specific consequence used by the research chapters are collected in Appendix A. Finally, for the tridiagonal Toeplitz matrices that Markov chains produce, the inverse is available in closed form through a transfer-matrix (geometric-ansatz) computation, carried out where it is needed (Chapter 5).

D.3 Gaussian belief propagation and the Kalman filter

The Gaussian family is the canonical closed message family, and its closure is pure algebra.

Toolbox D.1 — The two closure lemmas of Gaussian message passing

Lemma 1 (products). For Gaussians in the same variable,

$$\mathcal{N}(a; m_1, v_1) \times \mathcal{N}(a; m_2, v_2) \propto \mathcal{N}(a; m, v), \quad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \quad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2} :$$

the sum of two quadratics in a is a quadratic in a ; precisions and precision-weighted means add, as the natural-parameter rule of Section D.1 requires.

Lemma 2 (affine marginalisation). Pushing a Gaussian through a linear-Gaussian transition,

$$\int \mathcal{N}(a'; \alpha a, \sigma^2) \mathcal{N}(a; m, v) da = \mathcal{N}(a'; \alpha m, \alpha^2 v + \sigma^2),$$

because $a' = \alpha a + \eta$ is a sum of independent Gaussians. Read in the reverse direction the same computation gives $\mathcal{N}(a; m'/\alpha, (v' + \sigma^2)/\alpha^2)$.

A model whose factors are Gaussian in each variable performs, in (3.3), only these two operations. For such models every message is Gaussian, carried exactly by two numbers, and each update is a few arithmetic operations. $\square$

The oldest and most famous instance predates factor graphs by decades. For linear-Gaussian state-space models (a Gaussian Markov chain observed through linear Gaussian noise), the forward sum-product sweep is the *Kalman filter* (Kalman, 1960), alternating a *predict* step (Lemma 2, push the belief through the dynamics) and an *update* step (Lemma 1, absorb the new observation, with the correction weighted by the Kalman gain); the backward sweep combined with the forward one yields the fixed-interval smoother of Rauch et al. (1965), which conditions each state on *all* observations. That the classical filter and smoother are exactly the Gaussian specialisation of sum-product on the chain's factor graph was made explicit by Kschischang et al. (2001); textbook treatments are

Bishop (2006); Koller and Friedman (2009). Chapter 5 derives the filter from the BP sweep, line by line, for our diffusion posterior, where the observation factors take a specific form (a pseudo-observation whose strength depends on the diffusion time), and verifies the equivalence numerically to double-precision rounding.

One remark, recorded here because it resolves a doubt raised repeatedly in supervision: the Gaussian closure of this section is *algebra*, valid at any node degree, and invokes no limit theorem. It must not be confused with the central-limit justification of the *approximate* Gaussian closures of the next section, which does require many incoming messages and which a chain, with two neighbours per node, does not provide.

D.4 From BP to TAP and AMP

Statistical physics developed its own message-passing calculus for *dense* graphs, where every node interacts weakly with many others. On such graphs the incoming messages to a node are numerous and nearly independent, so their product is governed by a central-limit argument: the cavity field is approximately Gaussian, and each message can be summarised by a mean and a variance even when the underlying variables are not Gaussian. Historically the resulting self-consistency equations appeared first as the Thouless–Anderson–Palmer (TAP) equations of the Sherrington–Kirkpatrick spin glass (Thouless et al., 1977), whose distinctive ingredient is the Onsager reaction term, the correction that removes a node’s own influence from the field it feels. Three decades later the same structure re-emerged in compressed sensing as *approximate message passing* (AMP) (Donoho et al., 2009): a reduction of BP on a dense bipartite graph to iterations on means and variances, with the Onsager term reappearing as a memory correction. The derivation of AMP from BP by moment expansion is carried out for the ℓ_2 minimisation problem by Genovese and Piana (2026); the general technology, its state-evolution analysis, and its phase diagrams are reviewed by Zdeborová and Krzakala (2016) and, in lecture-note form, by Krzakala and Zdeborová (2024). Mézard (2017) derived the TAP equations of the Hopfield model, connecting this calculus back to the networks of Chapter 2.

The logic of the CLT justification should be kept explicit, because this thesis probes exactly its boundary. On a dense graph with N weak interactions per node, each message removes one interaction out of N , message and marginal differ at order $1/N$, and sums of many weak, nearly independent terms are Gaussian: the per-node Gaussian closure is controlled. On a chain every node has two neighbours: no aggregation takes place, each message carries the entire influence of half the system, and nothing justifies a Gaussian form for it beyond the special cases where it is exact. What happens when one imposes the closure anyway is a quantitative question. This thesis answers it on the chain in two complementary ways: exactly, for the per-node (AMP/TAP) closure on the Gaussian chain, where the breakdown admits a closed-form phase boundary (Appendix B); and numerically, for the Gaussian-projected messages on genuinely non-Gaussian chains (Chapter 6).

Appendix E

Derivations for the Rotating Ring

The mechanics behind Chapter 9. The chapter states the gauge reduction, the exact surrogate score and their consequences, which is what the argument needs; the algebra establishing them is here, where it can be checked without interrupting that argument.

E.1 Rotations preserve the standard isotropic Gaussian

Proof of Lemma 9.2.

Proof. A linear map of a Gaussian is Gaussian, so $R\eta$ is fixed by its first two moments: $\mathbb{E}[R\eta] = R\mathbb{E}[\eta] = 0$ and $\text{Cov}(R\eta) = RI_2R^\top = RR^\top = I_2$. Equivalently, in terms of densities, $(2\pi)^{-1}e^{-\|z\|^2/2}$ depends on z only through $\|z\|$, and the change of variables $z \mapsto Rz$ changes neither $\|R^{-1}z\| = \|z\|$ nor $|\det R^{-1}| = 1$: the level sets are circles centred at the origin and a rotation maps each circle to itself. $\square$

E.2 The gauge reduction

Proof of Lemma 9.3.

Proof. Substituting (9.10) and using the composition rule of (9.12),

$$y_{k+1} = R_{-(k+1)\psi} \left(R_\psi z_k + \sigma \eta_{k+1} \right) = \underbrace{R_{-(k+1)\psi} R_\psi}_{=R_{-k\psi}} z_k + \sigma R_{-(k+1)\psi} \eta_{k+1} = y_k + \sigma \tilde{\eta}_{k+1},$$

and by Lemma 9.2 each $\tilde{\eta}_{k+1} \sim \mathcal{N}(0, I_2)$; they stay independent because each is a fixed linear map of a different independent η . Next, U_ψ is block diagonal with orthogonal blocks, so $U_\psi^\top U_\psi = \text{diag}(I_2, \dots, I_2) = I_{2L}$ and $\det U_\psi = \prod_k \det R_{-k\psi} = 1$. Applying U_ψ to (9.8) gives $U_\psi x = e^{-t} U_\psi a + \sqrt{\Delta_t} U_\psi \xi$ with $U_\psi \xi \sim \mathcal{N}(0, I_{2L})$, so $U_\psi x$ has exactly the law of the noised non-rotating trajectory: the de-rotation commutes with the channel. For the densities, $z = U_\psi^\top y$ and $P_t(z | \psi) = P_t^{(0)}(U_\psi z) |\det U_\psi|$, the Jacobian factor being one. Differentiating that identity componentwise, with $\partial y_b / \partial z_c = (U_\psi)_{bc}$,

$$\frac{\partial}{\partial z_c} \log P_t^{(0)}(U_\psi z) = \sum_b (U_\psi)_{bc} \left(\nabla \log P_t^{(0)} \right)_b (U_\psi z) = \left[U_\psi^\top S^{(0)}(U_\psi z, t) \right]_c.$$

The transpose is not cosmetic: a score is a gradient, so it transforms with $U_\psi^\top$ while a point transforms with U_ψ . $\square$

E.3 The clean law after the gauge

Iterating (9.13) and letting the intermediate terms cancel,

$$y_k - y_0 = \sum_{r=1}^k (y_r - y_{r-1}) = \sigma \sum_{r=1}^k \tilde{\eta}_r, \quad \text{so} \quad y_k = A + \sigma \sum_{r=1}^k \tilde{\eta}_r, \quad (\text{E.1})$$

with $A := y_0 \sim p_\lambda$ and the empty sum equal to zero. Every frame contains the *same* random anchor, drawn once on the ring, plus a partial sum of independent innovations; the anchor is the only non-Gaussian ingredient of the model.

The clean law factorises over the ring prior and the one-step transitions, so $\log P_0(y) = -(\|y_0\| - 1)^2 / 2\lambda - (2\sigma^2)^{-1} \sum_{k=0}^{L-2} \|y_{k+1} - y_k\|^2 + \text{const.}$ Differentiating is a counting exercise: an interior y_k appears in exactly two summands, the edge arriving at it and the edge

leaving it, whose derivatives are $-(y_k - y_{k-1})/\sigma^2$ and $+(y_{k+1} - y_k)/\sigma^2$. Adding them, and using $\nabla_y \|y\| = y/\|y\|$ for the ring term,

$$\begin{aligned} S_0(y, 0) &= \frac{1 - \|y_0\|}{\lambda} \frac{y_0}{\|y_0\|} + \frac{y_1 - y_0}{\sigma^2}, \\ S_k(y, 0) &= \frac{y_{k-1} - 2y_k + y_{k+1}}{\sigma^2}, \quad 1 \leq k \leq L-2, \\ S_{L-1}(y, 0) &= \frac{y_{L-2} - y_{L-1}}{\sigma^2}. \end{aligned} \tag{E.2}$$

The interior clean score is a discrete second derivative along internal time, that is a discrete Laplacian: each block measures how far frame k departs from the average of its neighbours. It is strictly local, and it vanishes identically on the model's own noiseless solution, because a trajectory rotating at radius exactly one maps under the gauge to a constant y_k , for which every second difference vanishes and, with $\|y_0\| = 1$, so does the ring term. That vanishing is verified to machine zero (Table 9.1).

E.4 The exact surrogate score

Proof of Proposition 9.4.

Proof. Expanding the exponent of (9.16) in powers of α , with the correlated covariance $C_t \otimes I_2$, gives (9.17). Writing $\log P_t(x) = \text{const} - \frac{1}{2}x^\top (C_t^{-1} \otimes I_2)x + \log Z(h(x))$, the gradient of the first term is $-(C_t^{-1}x)_k$ in block k , while $\partial h/\partial x_k = e^{-t}q_k I_2$ and $\nabla_h \log Z = \mathbb{E}[A | x]$, the first cumulant identity for the exponential family in h . Differentiating once more, so that the second derivative of $\log Z$ returns $\text{Cov}(A | x)$, gives the response. $\square$

Appendix F

How the Reported Ratios Were Aggregated

A ratio is a nonlinear function of two means, so it depends on the order in which the averaging is done: $\mathbb{E}[X]/\mathbb{E}[Y]$ is not $\mathbb{E}[X/Y]$, and the gap between them grows with the spread. Both ratios this thesis reports therefore need the same question asked of them — whether the number is a property of the estimator or of the estimand chosen to summarise it. Neither section below refits anything; both recompute the same frozen cells under alternative aggregations.

F.1 The unstructured comparison

The ratio quoted in Section 8.1 and in the conclusions is one of several summaries the same cells support, and it is systematically the largest of them. That is a fact about the estimand rather than about the estimator, so it belongs with the number wherever the number appears. The section below is generated by `tools/make_aggregation_robustness.py` from the same frozen outputs that produce the table itself, and is shared verbatim with the paper.

F.2 Does the headline depend on how the ratio was averaged?

A ratio is a nonlinear function of two means, so its value depends on the order of averaging. Table 8.1 reports the mean of per-cell ratios, aggregated per seed. That is the right inferential unit — the twelve noise levels share a training set and a fitted model — but being the right unit does not make it the only defensible summary, so here is the same data under four, with a paired interval on the one the table reports.

Table F.1: The headline under four estimands. (1) is what Table 8.1 reports. The interval is a paired bootstrap over seeds, 20,000 resamples; seeds are resampled rather than cells, because cells within a seed share a fitted model.

N	(1) per-cell	(2) ratio of means	(3) geometric	(4) median	(5) paired 95% CI
32	7.0	6.8	6.1	6.8	[6.1, 8.1]
64	7.8	6.6	6.7	6.6	[6.3, 9.5]
128	8.8	8.0	7.8	8.6	[7.9, 9.6]
256	9.7	8.8	8.6	9.2	[8.5, 10.9]
512	9.0	8.6	7.9	8.2	[8.1, 9.9]
1024	12.5	9.7	10.8	11.1	[10.5, 14.7]
2048	15.5	10.0	13.2	12.9	[13.0, 18.9]
4096	17.6	12.4	15.6	16.0	[15.4, 19.9]

Two things follow, and the second is the one worth stating plainly.

The conclusion does not depend on the choice: no estimand at any size falls below 6.1, every bootstrap interval sits well clear of 1, and no aggregation produces a reversal anywhere.

But the estimand we report is the largest of the four at every size, and the margin widens with N over most of the range, reaching 5.5 at $N = 2048$. The ratio of per-seed averaged errors, which is the summary a reader is most likely to have in mind, runs 6.6 to 12.4 rather than 7.0 to 17.6.

That ordering is a measurement, not a theorem, and it is worth being precise about

why no inequality delivers it. Jensen gives $\mathbb{E}[1/Y] \geq 1/\mathbb{E}[Y]$, but $\mathbb{E}[X/Y] \geq \mathbb{E}[X]/\mathbb{E}[Y]$ does *not* follow for arbitrary positive correlated X and Y — the dependence between numerator and denominator can push the comparison either way. What holds here is empirical: denominator heterogeneity across noise levels interacts with that dependence, and in this dataset the mean of cellwise ratios comes out largest. It is a property of these cells, which is exactly why the estimand has to be named rather than assumed.

Resolving the average over noise levels shows where the aggregation hides structure: the ratio is not flat in t but peaks near $t \approx 0.5$ and falls at both ends. Two different “weakest” quantities have to be kept apart. The weakest seed-averaged (N, t) aggregate in the grid is 3.6; the weakest *individual* cell is 0.92, and EM–BP is beaten outright in 2 of the 1,520 cells summarised here. The estimator’s advantage is largest at moderate noise, where the posterior is dominated by neither the likelihood nor the prior — which is where knowing the transition structure should help most.

F.3 The structure-aware comparison

The same question applies to Table 8.3, and with more force, because that comparison’s endpoints are quoted as a range. Here the seed is the inferential unit by construction rather than by an averaging choice — the noise schedule is already reduced into each cell’s risk, with numerator and denominator summed before the square root — so what remains optional is only how the ratio is formed and pooled.

Table F.2: Table 8.3’s all-site ratio under four aggregations of the same cells, with the paired bootstrap interval over seeds. (1) is what the table reports. The conclusion does not depend on the choice: no estimand at any size falls below $1.98\times$, and the largest departure from the reported value is 0.48.

N	(1) per-seed	(2) ratio of means	(3) geometric	(4) median	(5) 95% CI on (1)
32	2.34	2.00	2.15	1.98	[1.88, 2.86]
128	2.82	2.45	2.61	2.62	[2.29, 3.44]
512	3.58	3.43	3.51	3.59	[3.25, 3.91]
2048	6.14	5.66	5.89	5.93	[5.30, 7.02]

The direction does not depend on the choice, and neither does the order of magnitude. The reported estimand is not uniformly the largest here, as it is for the unstructured comparison, but the spread is small enough that the qualitative reading — a low single-digit multiple at the smallest training set, growing to roughly six at the largest — survives every one of them.

Bibliography

- Abramson, J., Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., Ronneberger, O., Willmore, L., Ballard, A. J., Bambrick, J., Bodenstein, S. W., Evans, D. A., Hung, C.-C., O'Neill, M., Reiman, D., Tunyasuvunakool, K., Wu, Z., Žemgulytė, A., and Jumper, J. M. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500.
- Achilli, B., Benedetti, M., Biroli, G., and Mézard, M. (2026). Theory of speciation transitions in diffusion models with general class structure.
- Achilli, B., Ventura, E., Silvestri, G., Pham, B., Raya, G., Krotov, D., Lucibello, C., and Ambrogioni, L. (2024). Losing dimensions: geometric memorization in generative diffusion.
- Ackley, D. H., Hinton, G. E., and Sejnowski, T. J. (1985). A learning algorithm for Boltzmann machines. *Cognitive Science*, 9(1):147–169.
- Anderson, B. D. O. (1982). Reverse-time diffusion equation models. *Stochastic Processes and their Applications*, 12(3):313–326.
- Bachtis, D., Biroli, G., Decelle, A., and Seoane, B. (2024). Cascade of phase transitions in the training of energy-based models. In *Advances in Neural Information Processing Systems 37*.
- Besag, J. (1974). Spatial interaction and the statistical analysis of lattice systems. *Journal of the Royal Statistical Society: Series B*, 36(2):192–225.
- Biroli, G., Bonnaire, T., de Bortoli, V., and Mézard, M. (2024). Dynamical regimes of

- diffusion models. *Nature Communications*, 15:9957.
- Biroli, G. and Mézard, M. (2023). Generative diffusion in very large dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2023(9):093402.
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer, New York.
- Bonnaire, T., Urfin, R., Biroli, G., and Mézard, M. (2025). Why diffusion models don’t memorize: the role of implicit dynamical regularization in training. In *Advances in Neural Information Processing Systems*. Oral presentation; arXiv:2505.17638.
- Brockwell, P. J. and Davis, R. A. (1991). *Time Series: Theory and Methods*. Springer, New York, 2 edition.
- Cui, H., Pehlevan, C., and Lu, Y. M. (2025). A solvable model of learning generative diffusion: theory and insights.
- Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society, Series B*, 39(1):1–38.
- Donoho, D. L., Maleki, A., and Montanari, A. (2009). Message-passing algorithms for compressed sensing. *Proceedings of the National Academy of Sciences*, 106(45):18914–18919.
- Efron, B. (2011). Tweedie’s formula and selection bias. *Journal of the American Statistical Association*, 106(496):1602–1614.
- Favero, A., Sclocchi, A., Cagnetta, F., Frossard, P., and Wyart, M. (2025a). How compositional generalization and creativity improve as diffusion models are trained. In *International Conference on Machine Learning*.
- Favero, A., Sclocchi, A., and Wyart, M. (2025b). Bigger isn’t always memorizing: early stopping overparameterized diffusion models.
- Garnier-Brun, J., Biggio, L., Beltrame, D., Mézard, M., and Saglietti, L. (2026). Biased generalization in diffusion models. Spotlight, International Conference on Machine Learning.

- Garnier-Brun, J., Mézard, M., Moscato, E., and Saglietti, L. (2024). How transformers learn structured data: insights from hierarchical filtering.
- Genovese, G. and Piana, A. (2026). Derivation of the AMP equations from belief propagation for the ℓ_2 minimisation problem. arXiv:2602.15191 [math.PR]. <https://arxiv.org/abs/2602.15191> [last access: 14/07/26].
- Ghio, D., Dandi, Y., Krzakala, F., and Zdeborová, L. (2024). Sampling with flows, diffusion, and autoregressive neural networks from a spin-glass perspective. *Proceedings of the National Academy of Sciences*, 121(27):e2311810121.
- Ho, J., Jain, A., and Abbeel, P. (2020). Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 6840–6851.
- Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558.
- Hyvärinen, A. (2005). Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6:695–709.
- Kadkhodaie, Z., Guth, F., Simoncelli, E. P., and Mallat, S. (2024). Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *International Conference on Learning Representations*. Outstanding Paper award.
- Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45.
- Karras, T., Aittala, M., Aila, T., and Laine, S. (2022). Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 26565–26577.
- Koller, D. and Friedman, N. (2009). *Probabilistic Graphical Models: Principles and Techniques*. MIT Press, Cambridge, MA.
- Krizhevsky, A., Sutskever, I., and Hinton, G. E. (2012). ImageNet classification with deep

- convolutional neural networks. In *Advances in Neural Information Processing Systems* 25.
- Krzakala, F. and Zdeborová, L. (2024). Statistical physics methods in optimization and machine learning: An introduction to replica, cavity and message-passing techniques. Lecture notes.
- Kschischang, F. R., Frey, B. J., and Loeliger, H.-A. (2001). Factor graphs and the sum-product algorithm. *IEEE Transactions on Information Theory*, 47(2):498–519.
- Lai, C.-H., Song, Y., Kim, D., and Mitsufuji, Y. (2025). The principles of diffusion models: From origins to advances. arXiv:2510.21890.
- LeCun, Y., Bengio, Y., and Hinton, G. (2015). Deep learning. *Nature*, 521:436–444.
- LeCun, Y., Chopra, S., Hadsell, R., Ranzato, M., and Huang, F. J. (2006). A tutorial on energy-based learning. In Bakır, G., Hofmann, T., Schölkopf, B., Smola, A. J., Taskar, B., and Vishwanathan, S., editors, *Predicting Structured Data*. MIT Press, Cambridge, MA.
- Malioutov, D. M., Johnson, J. K., and Willsky, A. S. (2006). Walk-sums and belief propagation in Gaussian graphical models. *Journal of Machine Learning Research*, 7:2031–2064.
- Mei, S. (2024). U-nets as belief propagation: Efficient classification, denoising, and diffusion in generative hierarchical models. arXiv:2404.18444.
- Meng, X.-L. and Rubin, D. B. (1993). Maximum likelihood estimation via the ECM algorithm: A general framework. *Biometrika*, 80(2):267–278.
- Mézard, M. (2017). Mean-field message-passing equations in the Hopfield model and its generalizations. *Physical Review E*, 95:022117.
- Mézard, M. and Montanari, A. (2009). *Information, Physics, and Computation*. Oxford University Press, Oxford.
- Minka, T. P. (2001). Expectation propagation for approximate Bayesian inference. In *Uncertainty in Artificial Intelligence (UAI)*, pages 362–369.

- Miyasawa, K. (1961). An empirical Bayes estimator of the mean of a normal population. *Bulletin of the International Statistical Institute*, 38(4):181–188.
- Pearl, J. (1988). *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, San Mateo, CA.
- Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.
- Rauch, H. E., Tung, F., and Striebel, C. T. (1965). Maximum likelihood estimates of linear dynamic systems. *AIAA Journal*, 3(8):1445–1450.
- Raya, G. and Ambrogioni, L. (2023). Spontaneous symmetry breaking in generative diffusion models. In *Advances in Neural Information Processing Systems*, volume 36.
- Robbins, H. (1956). An empirical Bayes approach to statistics. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 157–163.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695.
- Ronneberger, O., Fischer, P., and Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241.
- Sclocchi, A., Favero, A., and Wyart, M. (2025). A phase transition in diffusion models reveals the hierarchical nature of data. *Proceedings of the National Academy of Sciences*, 122(1):e2408799121.
- Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., and Ganguli, S. (2015). Deep unsupervised learning using nonequilibrium thermodynamics. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 2256–2265.
- Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. (2021).

- Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations (ICLR)*.
- The Royal Swedish Academy of Sciences (2024). The Nobel Prize in Physics 2024. <https://www.nobelprize.org/prizes/physics/2024/summary/>. Last accessed 19 July 2026.
- Thouless, D. J., Anderson, P. W., and Palmer, R. G. (1977). Solution of ‘Solvable model of a spin glass’. *Philosophical Magazine*, 35(3):593–601.
- Uhlenbeck, G. E. and Ornstein, L. S. (1930). On the theory of the Brownian motion. *Physical Review*, 36(5):823–841.
- Vincent, P. (2011). A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674.
- Weiss, Y. and Freeman, W. T. (2001). Correctness of belief propagation in Gaussian graphical models of arbitrary topology. *Neural Computation*, 13(10):2173–2200.
- Zdeborová, L. and Krzakala, F. (2016). Statistical physics of inference: thresholds and algorithms. *Advances in Physics*, 65(5):453–552.